Scheduled_AC_CLReqControlMode	complexType: Scheduled_AC_CLReqControlModeType refer to 8.3.5.4.4	This element is used by the EVCC for offering and setting parameters for scheduled control mode energy transfer.
BPT_Scheduled_AC_CLReqControlMode	complexType: BPT_Scheduled_AC_CLReqControlModeType refer to 8.3.5.4.7.4	This element is used by the EVCC for offering and setting parameters for scheduled control mode BPT.

[V2G20-1810] The absolute value of the following elements, if provided within any AC energy transfer control loop message, shall be less than or equal to the absolute value of the same element provided earlier in the AC_ChargeParameterDiscovery message pair:

- EVMaximumChargePower
- EVMaximumDischargePower

[V2G20-1812] The absolute value of the following elements, if provided within any AC energy transfer control loop message, shall be higher than or equal to the absolute value of the same element provided earlier in the AC_ChargeParameterDiscovery message pair:

- EVMinimumChargePower
- EVMinimumDischargePower

8.3.4.4.3.3 AC_ChargeLoopRes

After receiving the AC_ChargeLoopReq from the EVCC, the SECC sends the AC_ChargeLoopRes.

[V2G20-1278] The EVCC and the SECC shall implement the message elements as defined in Table 57 and Figure 61.

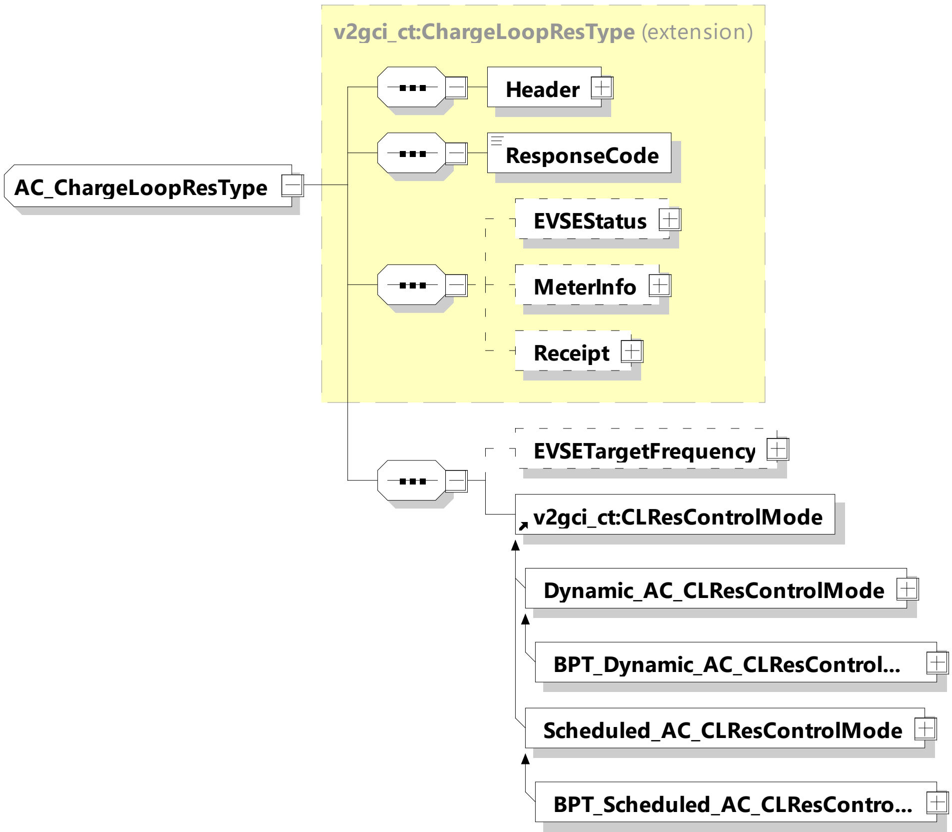

Figure 61 — Schema diagram – AC_ChargeLoopRes

The elements of this message are used according to Table 57.

Table 57 — Semantics and type definition for AC_ChargeLoopRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType Refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.

Element Name	Type	Semantics
EVSEStatus	complexType: EVSEStatusType refer to 8.3.5.3.26	Optional: This element is used by the SECC for indicating the EVSE status and for signaling an event the SECC expects the EVCC to react to.
MeterInfo	complexType: MeterInfoType refer to 8.3.5.3.6	Optional: Includes the energy charged during this service session and other meter relevant data.
Receipt	complexType: ReceiptType Refer to 8.3.5.3.59	Optional: Receipt related information.
EVSETargetFrequency	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: Target frequency requested by the EVSE
Dynamic_AC_CLResControlMode	complexType: Dynamic_AC_CLResControlModeType refer to 8.3.5.4.5	This element is used by the SECC for offering and setting parameters for dynamic control mode energy transfer.
BPT_Dynamic_AC_CLResControlMode	complexType: BPT_Dynamic_AC_CLResControlModeType refer to 8.3.5.4.7.5	This element is used by the SECC for offering and setting parameters for dynamic control mode charging and discharging.
Scheduled_AC_CLResControlMode	complexType: Scheduled_AC_CLResControlModeType refer to 8.3.5.4.6	This element is used by the SECC for offering and setting parameters for scheduled control mode energy transfer.

Element Name	Type	Semantics
BPT_Scheduled_AC_CLResControlMode	complexType: BPT_Scheduled_AC_CLResControlModeType refer to 8.3.5.4.7.5	This element is used by the SECC for offering and setting parameters for scheduled control mode charging and discharging.

[V2G20-1555] The SECC shall send EVSETargetFrequency values only in situations when the EVSENominalFrequency is no longer the desired equilibrium point.

NOTE 1 This might be necessary in situations like the reconnection of residential power island to the main power grid.

[V2G20-1556] The SECC shall not terminate the charging session only for the reason that an EVSETargetFrequency has not been achieved, as the frequency cannot be attributed to the EV's behavior.

NOTE 2 In AC power grids the frequency is the result of the collective behavior of all connected devices. The frequency is always drifting unless perfect equilibrium has been reached. The grid is typically never running at exactly the EVSENominalFrequency. The EVSENominalFrequency only defines a theoretical target frequency and provides a reference point for the definition of technical requirements which define the detection of and the reaction to abnormal situations. However, in some situations (power grid emergencies, small power islands, etc.) the desired frequency for the equilibrium point intentionally gets pushed outside the normal operating bandwidth. Only in those situations the EVSETargetFrequency can be used. The goal of the EVSETargetFrequency is to communicate the equilibrium point and to enable the EV's power inverter to operate in a robust way under extreme circumstances.

[V2G20-1558] In case EVCC selects the dynamic control mode with the service parameter MobilityNeedsMode set to "2", the SECC shall send a departure time in AC_ChargeLoopRes if it intends to update the DepartureTime sent in the latest ScheduleExchangeRes message.

[V2G20-2179] In dynamic control mode the SECC shall provide the target setpoints for the power transfer by the means of EVSETargetActivePower and EVSETargetReactivePower in AC_ChargeLoopRes.

8.3.4.5 DC messages

8.3.4.5.1 Overview

Messages defined as DC-Messages belong to the DC message set(s).

8.3.4.5.2 DC_ChargeParameterDiscoveryReq/Res

8.3.4.5.2.1 DC_ChargeParameterDiscoveryReq/Res handling

After being authorized for charging at the EVSE (SECC) the EVCC and the SECC negotiate the energy transfer parameters with the DC_ChargeParameterDiscovery message pair.

Concepts treated for an optimal supply of energy that corresponds to the customer needs.

When using scheduled control mode, the energy transfer parameters negotiation that precedes the delivery of energy or may be engaged during the energy delivery phase is destined to ensure that the user will be satisfied while at the same time ensuring that the energy will effectively be available and fall within the capacity of power supply grid at the local level (private network) and at the regional level (public network). This required negotiation will become more and more necessary as the number of EVs increase, as well as an increase in volatility of local renewable energy production.

When using the dynamic control mode, the energy transfer parameters can be changed dynamically while in the charging loop, depending on the need of the EV supply equipment.

8.3.4.5.2.2 DC_ChargeParameterDiscoveryReq

By sending the DC_ChargeParameterDiscoveryReq message the EVCC provides its energy transfer parameters to the SECC. This message provides status information about the EV, i.e. the capabilities of the EV charging system.

[V2G20-1268] The EVCC and the SECC shall implement the message elements as defined in Table 58 and Figure 62.

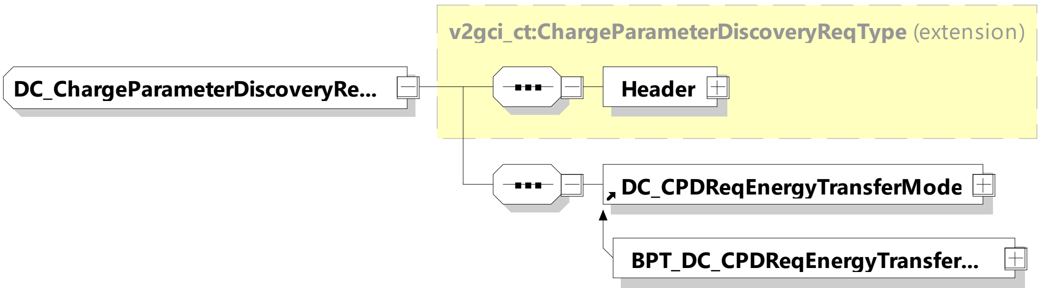

Figure 62 — Schema diagram – DC_ChargeParameterDiscoveryReq

The elements of this message are used according to Table 58.

Table 58 — Semantics and type definition for DC_ChargeParameterDiscoveryReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
DC_CPDReqEnergyTransferMode	complexType: DC_CPDReqEnergyTransferModeType refer to 8.3.5.5.1	This element is used by the EVCC for initiating the target setting process for DC charging.
BPT_DC_CPDReqEnergyTransferMode	complexType: BPT_DC_CPDReqEnergyTransferModeType refer to 8.3.5.5.7.1	This element is used by the EVCC for initiating the target setting process for DC bidirectional charging.

NOTE The parameters related to current, power and voltage provided in DC_ChargeParameterDiscovery message pair can be considered as physical limitations, meaning that they are related to the physical abilities of the system under rated operating conditions. Later during the charging process, depending on the external and environmental conditions (SOC, temperatures, etc.), the values of these limitations can change into more conservative values.

8.3.4.5.2.3 DC_ChargeParameterDiscoveryRes

With the DC_ChargeParameterDiscoveryRes message the SECC provides applicable charge parameters from the grid's perspective.

[V2G20-1207] The EVCC and the SECC shall implement the message elements as defined in Table 59 and Figure 63.

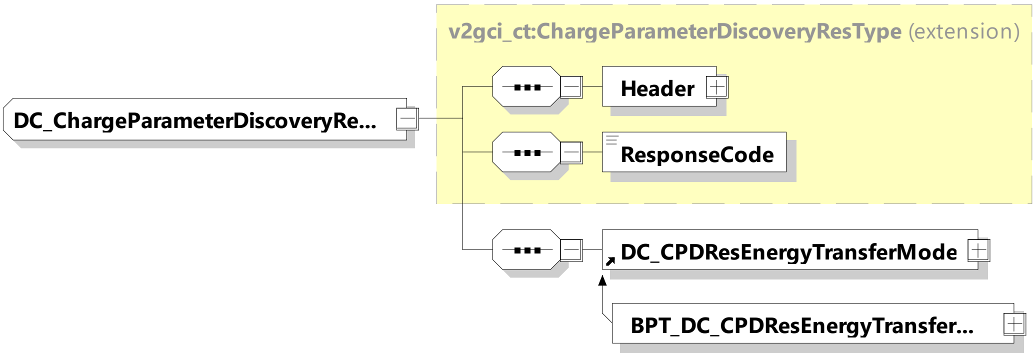

Figure 63 — Schema diagram – DC_ChargeParameterDiscoveryRes

The elements of this message are used according to Table 59.

Table 59 — Semantics and type definition for DC_ChargeParameterDiscoveryRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
DC_CPDResEnergyTransferMode	complexType: DC_CPDResEnergyTransferModeType refer to 8.3.5.5.2	This element is used by the SECC for initiating the target setting process for DC charging.
BPT_DC_CPDResEnergyTransferMode	complexType: BPT_DC_CPDResEnergyTransferModeType refer to 8.3.5.5.7.2	This element is used by the SECC for initiating the target setting process for DC bidirectional charging.

8.3.4.5.3 DC_CableCheckReq/Res

8.3.4.5.3.1 DC_CableCheckReq/Res handling

[V2G20-1557] To ensure safety in DC energy transfer, a cable check shall be performed.

8.3.4.5.3.2 DC_CableCheckReq

The DC_CableCheckReq asks for the cable check status of the EVSE and, for example, tells the EVSE if the connector is locked on EV side and if the EV is ready to transfer energy.

[V2G20-1279] The EVCC and the SECC shall implement the message elements as defined in Table 60 and Figure 64.

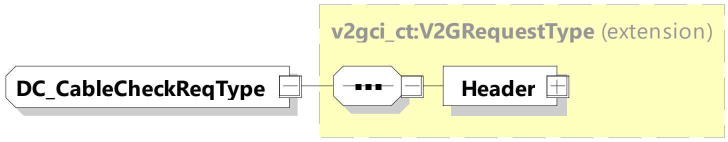

Figure 64 — Schema diagram – DC_CableCheckReq

The element of this message is used according to Table 60.

Table 60 — Semantics and type definition for DC_CableCheckReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.

NOTE DC_CableCheckReq contains no elements

8.3.4.5.3.3 DC_CableCheckRes

After receiving the DC_CableCheckReq from the EVCC the SECC sends the DC_CableCheckRes informing the EVCC about the result of the cable check and EVSE status.

[V2G20-1280] The EVCC and the SECC shall implement the message elements as defined in Table 61 and Figure 65.

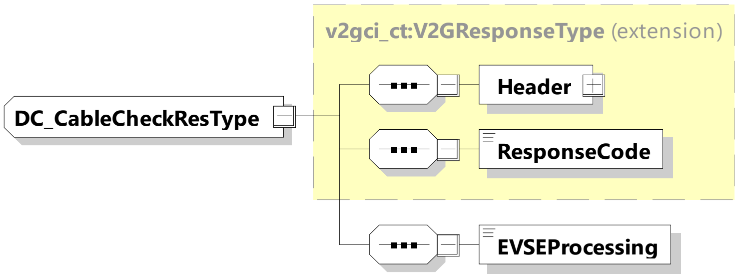

Figure 65 — Schema diagram – DC_CableCheckRes

The elements of this message are used according to Table 61.

Table 61 — Semantics and type definition for DC_CableCheckRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.

EVSEProcessing	simpleType: processingType enumeration refer to Annex A for the type definition	Parameter indicating that the EVSE has finished the processing that was initiated after the DC_CableCheckReq or if the EVSE is still processing at the time, the response message was sent.

NOTE By using the EVSEProcessing parameter, the EVSE can indicate to the EVCC that the processing is not finished but a response message is sent to fulfil the timeout and performance requirements defined in 8.5.4. This allows continuing the communication session while fulfilling the performance and timeout requirements.

8.3.4.5.4 DC_PreChargeReq/Res

8.3.4.5.4.1 DC_PreChargeReq/Res handling

PreCharge is used for adjusting the EVSE output voltage to the EV battery voltage.

8.3.4.5.4.2 DC_PreChargeReq

The DC_PreChargeReq is used to start the Pre Charge process from EV side.

[V2G20-1281] The EVCC and the SECC shall implement the message elements as defined in Table 62 and Figure 66.

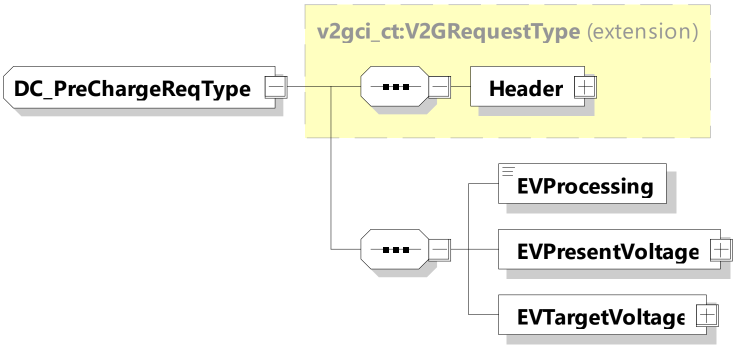

Figure 66 — Schema diagram – DC_PreChargeReq

The elements of this message are used according to Table 62.

Table 62 — Semantics and type definition for DC_PreChargeReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVProcessing	simpleType: processingType enumeration refer to Annex A for the type definition	Parameter that indicates the stop of the DC_PreCharge process by the EV. When the EV wants to continue in the Sequence, the EVProcessing is set to "Finished", while still running it is set to "Ongoing".
EVPresentVoltage	complexType: RationalNumberType refer to 8.3.5.3.8	Present voltage of the EV
EVTargetVoltage	complexType RationalNumberType refer to 8.3.5.3.8	Target voltage of the EV (see 8.3.4.5.5.2 for more background)

The SECC shall adjust the voltage on the EVSE side, which is reported in the EVSEPresentVoltage element within the DC_PreChargeRes, to match the EVTargetVoltage as closely as necessary (for details see IEC 61851-23) in order to allow the contactors to close and to exit the PreCharge phase.

8.3.4.5.4.3 DC_PreChargeRes

After receiving the DC_PreChargeReq from the EVCC the SECC sends the DC_PreChargeRes informing the EV about the EVSE status and the present EVSE output voltage.

[V2G20-1282] The EVCC and the SECC shall implement the message elements as defined in Table 63 and Figure 67.

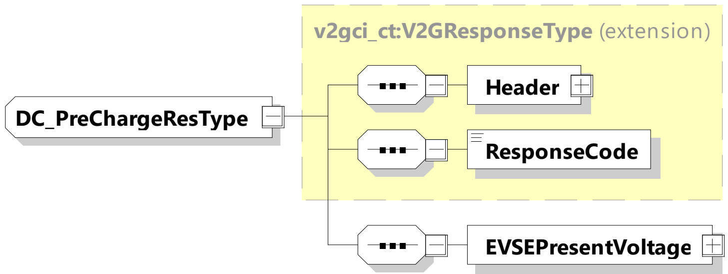

Figure 67 — Schema diagram – DC_PreChargeRes

The elements of this message are used according to Table 63.

Table 63 — Semantics and type definition for DC_PreChargeRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.

ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
EVSEPresentVoltage	complexType RationalNumberType refer to 8.3.5.3.8	Present voltage as measured by the EVSE

8.3.4.5.5 DC_ChargeLoopReq/Res

8.3.4.5.5.1 DC_ChargeLoopReq/Res handling

For DC charging control cyclic exchange of the requested current from EV side is necessary. Also, the target voltage and the difference in current and voltages is transferred.

8.3.4.5.5.2 DC_ChargeLoopReq

By sending the DC_ChargeLoopReq the EV requests a certain current from the EVSE. Also, the target voltage, current and voltage difference are transferred.

[V2G20-1283] The EVCC and the SECC shall implement the message elements as defined in Table 64 and Figure 68.

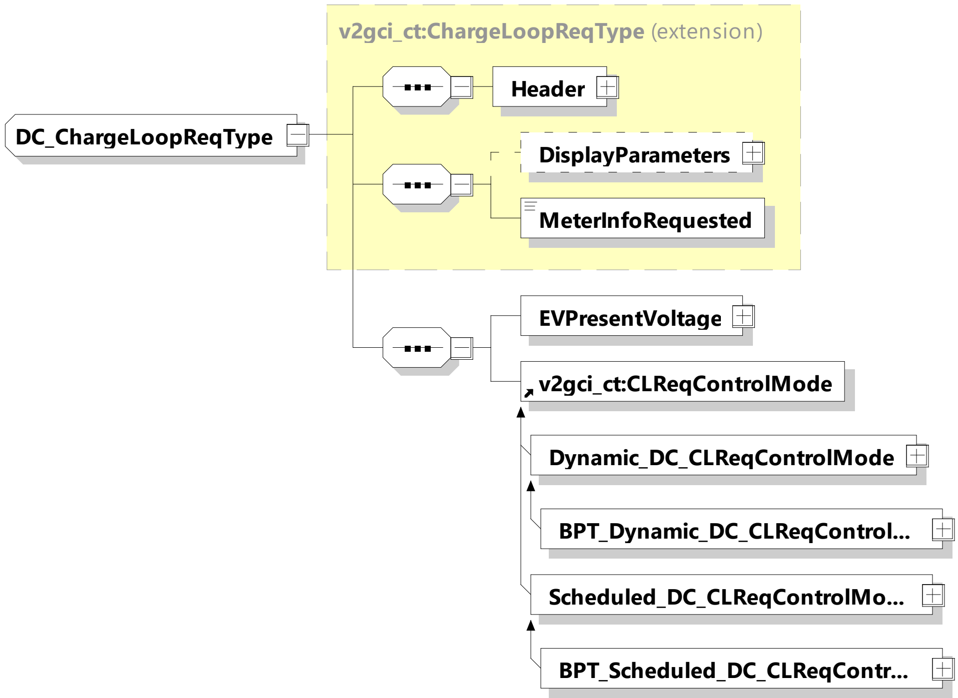

Figure 68 — Schema diagram – DC_ChargeLoopReq

The elements of this message are used according to Table 64.

Table 64 — Semantics and type definition for DC_ChargeLoopReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
DisplayParameters	complexType: DisplayParametersType refer to 8.3.5.3.28	Optional: Parameters that may be displayed on the EVSE or any other user interface which is connected directly or indirectly to the EVSE. They shall under no circumstances influence the V2G communication session.
MeterInfoRequested	simpleType: xs:boolean	This Parameter indicates, if set to "TRUE" that the EV wants to receive the MeterInfo by the EVSE.
EVPresentVoltage	complexType: RationalNumberType refer to 8.3.5.3.8	Present voltage of the EV
Dynamic_DC_CLReqControlMode	complexType: Dynamic_DC_CLReqControlModeType refer to 8.3.5.5.3	This element is used by the EVCC for offering and setting parameters for dynamic control mode energy transfer.
BPT_Dynamic_DC_CLReqControlMode	complexType: BPT_Dynamic_DC_CLReqControlModeType refer to 8.3.5.5.7.3	This element is used by the EVCC for offering and setting parameters for dynamic control mode BPT.
Scheduled_DC_CLReqControlMode	complexType: Scheduled_DC_CLReqControlModeType refer to 8.3.5.5.4	This element is used by the EVCC for offering and setting parameters for Scheduled control mode energy transfer.
BPT_Scheduled_DC_CLReqControlMode	complexType: BPT_Scheduled_DC_CLReqControlModeType refer to 8.3.5.5.7.4	This element is used by the EVCC for offering and setting parameters for Scheduled control mode BPT.

NOTE 1 EVTargetVoltage and EVTargetCurrent are used by the EV in the scheduled control mode to request a certain voltage or current value from the EVSE. In dynamic control mode, where the EVSE is in charge of choosing the energy transfer plan and the related power setpoints, these elements are not available. In all DC_ChargeLoopReq variations, the EVs limitations are provided by means of EVMaximumChargeCurrent, EVMaximumChargePower, EVMaximumDischargeCurrent, EVMaximumDischargePower, EVMaximumVoltage and EVMinimumVoltage (for example, see [V2G20-2180]).

[V2G20-2180] EVMaximumVoltage and EVMinimumVoltage shall express limits for EVPresentVoltage and shall represent the state at the EV inlet. If a different physical measurement point is used inside the EV than the EVCC shall apply the necessary correction factors before communicating the EVPresentVoltage.

NOTE 2 If the SECC sees, that the EVPresentVoltage exceeds one of the limits then there is the risk of a termination of the charging session due to the EV enforcing safety protection measures.

If, for example, the EV is measuring voltage via the BMS inside the battery and is also using a DC-DC converter between the battery and the DC inlet then the EVCC shall apply a meaningful mathematical transformation to the actual measured voltage values. It is important that the EVPresentVoltage multiplied with EVSEPresentCurrent does not contradict any power limitations communicated by the EV because the SECC might use that as a plausibility check to validate its charging strategy.

[V2G20-2198] The absolute value of the following elements, if provided within any DC energy transfer control loop message, shall be less than or equal to the absolute value of the same element provided earlier in the DC_ChargeParameterDiscovery message pair:

– EVMaximumChargeCurrent
– EVMaximumDischargeCurrent
– EVMaximumChargePower
– EVMaximumDischargePower
– EVMaximumVoltage

[V2G20-2199] The absolute value of the following elements, if provided within any DC energy transfer control loop message, shall be higher than or equal to the absolute value of the same element provided earlier in the DC_ChargeParameterDiscovery message pair:

– EVMinimumChargeCurrent
– EVMinimumDischargeCurrent
– EVMinimumChargePower
– EVMinimumDischargePower
– EVMinimumVoltage

8.3.4.5.5.3 DC_ChargeLoopRes

After receiving the DC_ChargeLoopReq from the EVCC the SECC sends the DC_ChargeLoopRes informing the EV about the EVSE status and the present EVSE output voltage and current.

[V2G20-1284] The EVCC and the SECC shall implement the message elements as defined in Table 65 and Figure 69.

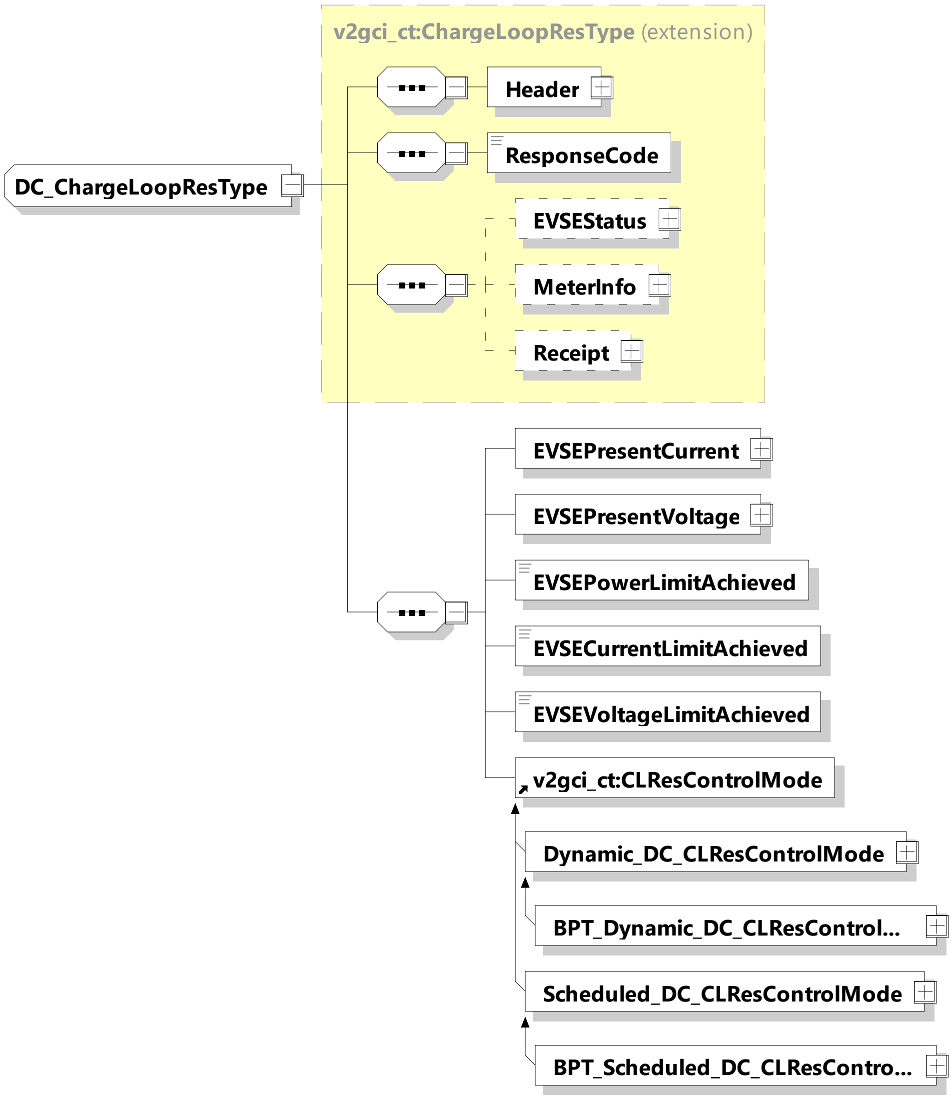

Figure 69 — Schema diagram – DC_ChargeLoopRes

The elements of this message are used according to Table 65.

Table 65 — Semantics and type definition for DC_ChargeLoopRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.

Element Name	Type	Semantics
EVSEStatus	complexType: EVSEStatusType refer to 8.3.5.3.26	Optional: This element is used by the SECC for indicating the EVSE status and for signalling an event the SECC expects the EVCC to react to.
MeterInfo	complexType: MeterInfoType refer to 8.3.5.3.6	Optional: Includes the energy charged during this service session and other meter relevant data.
Receipt	complexType: ReceiptType refer to 8.3.5.3.59	Optional: Receipt related information.
EVSEPresentCurrent	complexType: RationalNumberType refer to 8.3.5.3.8	Present current as measured by the EVSE.
EVSEPresentVoltage	complexType: RationalNumberType refer to 8.3.5.3.8	Present voltage as measured by the EVSE.
EVSECurrentLimitAchieved	simpleType: xs:boolean	If set to TRUE, the EVSE has reached its current limit.
EVSEVoltageLimitAchieved	simpleType: xs:boolean	If set to TRUE, the EVSE has reached its voltage limit.
EVSEPowerLimitAchieved	simpleType: xs:boolean	If set to TRUE, the EVSE has reached its power limit.
Dynamic_DC_CLResControlModeType	complexType: Dynamic_DC_CLResControlModeType refer to 8.3.5.5.5	This element is used by the SECC for offering and setting parameters for dynamic control mode energy transfer.
BPT_Dynamic_DC_CLResControlModeType	complexType: BPT_Dynamic_DC_CLResControlModeType refer to 8.3.5.5.7.5	This element is used by the SECC for offering and setting parameters for dynamic control mode BPT.
Scheduled_DC_CLResControlModeType	complexType: Scheduled_DC_CLResControlModeType refer to 8.3.5.5.6	This element is used by the SECC for offering and setting parameters for scheduled control mode energy transfer.
BPT_Scheduled_DC_CLResControlModeType	complexType: BPT_Scheduled_DC_CLResControlModeType refer to 8.3.5.5.7.6	This element is used by the SECC for offering and setting parameters for scheduled control mode BPT.

NOTE EVSEPresentVoltage and EVSEPresentCurrent implicitly communicate the present power that is applied by the EVSE on the charging cable. This information can be used by the EVCC, for example, to validate if the target setpoints in scheduled control mode have been respected by the EVSE or how the active PriceAlgorithm needs to be applied (see 8.3.5.3.49.1).

[V2G20-1275] In case EVCC selects the dynamic control mode with the service parameter MobilityNeedsMode set to "2", the SECC shall send a departure time in DC_ChargeLoopRes if it intends to update the DepartureTime sent in the latest ScheduleExchangeRes message.

[V2G20-2654] The absolute value of the following elements, if provided within any DC energy transfer control loop message, shall be less than or equal to the absolute value of the same element provided earlier in the DC_ChargeParameterDiscovery message pair:

- EVSEMaximumChargeCurrent
- EVSEMaximumDischargeCurrent
- EVSEMaximumVoltage
- EVSEMaximumChargePower
- EVSEMaximumDishargePower

[V2G20-2655] The absolute value of the following elements, if provided within any DC energy transfer control loop message, shall be higher than or equal to the absolute value of the same element provided earlier in the DC_ChargeParameterDiscovery message pair:

- EVSEMinimumChargeCurrent
- EVSEMinimumDischargeCurrent
- EVSEMinimumVoltage
- EVSEMinimumChargePower
- EVSEMinimumDischargePower

8.3.4.5.6 DC_WeldingDetectionReq/Res

8.3.4.5.6.1 DC_WeldingDetectionReq/Res handling

The Welding Detection messages described in this subclause allow to implement the Welding Detection mechanism according to IEC 61851-23.

8.3.4.5.6.2 DC_WeldingDetectionReq

By sending the DC_WeldingDetectionReq the EV requests welding detection on EVSE side.

[V2G20-1285] The EVCC and the SECC shall implement the message elements as defined in Table 66 and Figure 70.

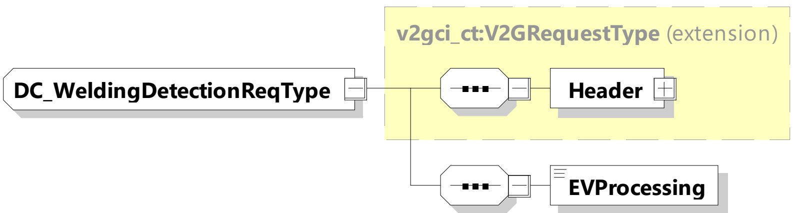

Figure 70 — Schema diagram – DC_WeldingDetectionReq

The elements of this message are used according to Table 66.

Table 66 — Semantics and type definition for DC_WeldingDetectionReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVProcessing	simpleType: processingType enumeration refer to Annex A for the type definition	Parameter that indicates the stop of the DC_WeldingDetection process by the EV. When the EV wants to continue in the Sequence, the EVProcessing is set to "Finished", while still running it is set to "Ongoing".

8.3.4.5.6.3 DC_WeldingDetectionRes

After receiving the DC_WeldingDetectionReq from the EVCC, the SECC sends the Welding Detection Response informing the EV about the EVSE status and the present EVSE output voltage.

[V2G20-1286] The EVCC and the SECC shall implement the message elements as defined in Table 67 and Figure 71.

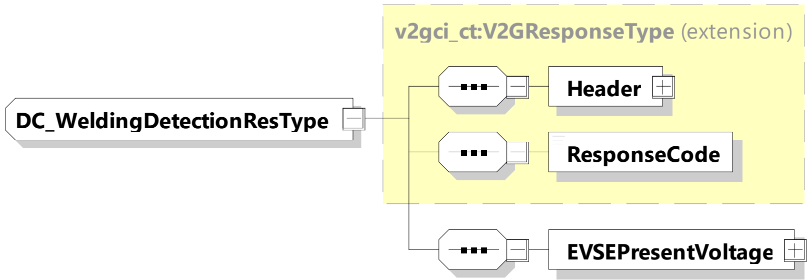

Figure 71 — Schema diagram – DC_WeldingDetectionRes

The elements of this message are used according to Table 67.

Table 67 — Semantics and type definition for DC_WeldingDetectionRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType Refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
EVSEPresentVoltage	complexType RationalNumberType refer to 8.3.5.3.8	Present voltage as measured by the EVSE

8.3.4.6 WPT messages

8.3.4.6.1 Overview

Messages defined as WPT-Messages belong to the WPT message set(s).

8.3.4.6.2 WPT_FinePositioningSetupReq/Res

WPT_FinePositioningSetupReq/Res are used to exchange system setup data between the EVCC and SECC describing the individual positioning system characteristics at the EV device and at the supply device. Additionally, possibilities for the pairing and alignment check methods are exchanged and determined.

8.3.4.6.2.1 WPT_FinePositioningSetupReq/Res handling

After SessionSetup has been done the EVCC sends a WPT_FinePositioningSetupReq in order to determine the possible options of the SECC for fine positioning, pairing and alignment check support. With the WPT_FinePositioningReq the EVCC will inform the SECC about its possible options.

The SECC responds to the request with a WPT_FinePositioningSetupRes containing information about available options with respect to fine positioning, pairing, and alignment check.

After analyzing the available options, the EVCC sends a WPT_FinePositioningSetupReq containing information about the choice for fine position, pairing and alignment check made by the EV.

The SECC responds with a confirmation of the EV's choice in the WPT_FinePositioningSetupRes.

8.3.4.6.2.2 WPT_FinePositioningSetupReq

By sending the WPT_FinePositioningSetupReq message the EVCC provides its WPT_Fine positioning setup parameters to the SECC.

[V2G20-5009] The EVCC and the SECC shall implement the message elements as defined in Figure 72 and Table 68.

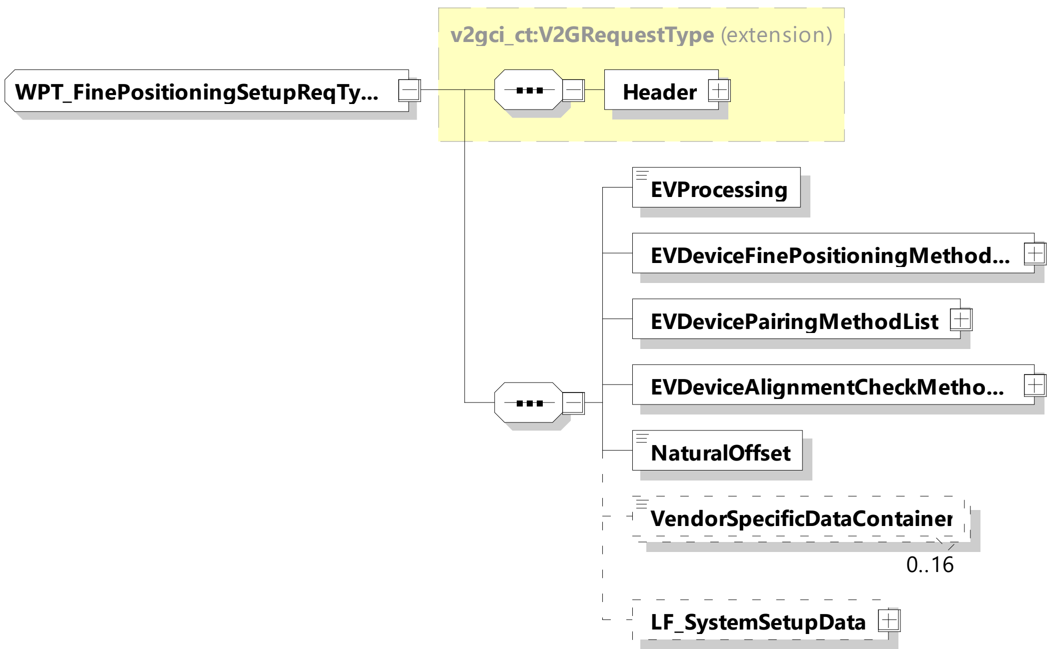

Figure 72 — Schema diagram – WPT_FinePositioningSetupReq

The elements of this message are used according to Table 68.

Table 68 — Semantics and type definition for WPT_FinePositioningSetupReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVProcessing	simpleType: processingType refer to Annex A for the type definition	Indicates the progress of the selection progress for the methods of Fine positioning, pairing and alignment check. The setting of this parameter impacts the usage of the elements "EVDeviceFinePositioningMethodList", "EVDevicePairingMethodList", "EVDeviceAlignmentCheckMethodLists"
EVDeviceFinePositioningMethodList	simpleType: WPT_FinePositioningMethodListType refer to Annex A for the type definition	Fine positioning method intended to be used by the EV: Multiple options are possible, ordered by priority (See IEC 61980-2). – Manual (default): Any fine position method which does not depend on technical data exchange during the positioning process, e.g. positioning by visually detecting specified marking or characteristics – LF_TxEV: LF positioning with transmitter at EV – LF_TxPrimaryDevice: LF positioning with transmitter at primary device – LPE: Low power excitation according – Proprietary: Any proprietary positioning method with individualized data exchange over element VendorSpecificDataContainer in FinePositioningReq/Res messages

Element Name	Type	Semantics
EVDevicePairingMethodList	simpleType: WPT_PairingMethodListType refer to Annex A for the type definition	Pairing method intended to be used by the EV: Multiple options are possible, ordered by priority (See IEC 61980-2). – External confirmation (default) – LPE: application of low power excitation for pairing – LF_TxEV: application of LF signal transmitted by EV for pairing – LF_TxPrimaryDevice: application of LF signal transmitted by primary device for pairing – Optical: Pairing by means of machine vision, e.g. QR code at primary device which can be read by EV and sent to the SECC – Proprietary: Any proprietary positioning method with individualized data exchange over data block in pairingReq/Res messages
EVDeviceAlignmentCheckMethodList	simpleType: WPT_AlignmentCheckMethodListType refer to Annex A for the type definition	Alignment check method intended to be used by the EV: Multiple options are possible, ordered by priority (See IEC 61980-2). – LPE: application of low power excitation for alignment check – Power check (default): utilization of minimal power transfer and anomaly detection for reasonable confidence of alignment – Proprietary: Anything with individualized message structure
NaturalOffset	simpleType: xs:unsignedShort	Offset of the center alignment point in mm (in x-direction) which results from the coil design of EV and infrastructure.
VendorSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Data container for proprietary information
LF_SystemSetupData	simpleType: WPT_LF_SystemSetupDataType refer to Annex A for the type definition	Optional: Additional data needed to describe a LF positioning system. This information is needed, when LF_TxEV or LF_TxPrimaryDevice is given in the list EVDevicePositioningMethods.

[V2G20-5001] As long as EVProcessing is set to "Ongoing" the lists "EVDeviceFinePositioningMethodList", "EVDevicePairingMethodList", "EVDeviceAlignmentCheckMethodList" shall contain more than one element.

[V2G20-5002] If the lists "EVDeviceFinePositioningMethodList", "EVDevicePairingMethodList", "EVDeviceAlignmentCheckMethodList" contain more than one element, the default values shall be part of the list, if the device is a compatibility class A device (see IEC 61980-2 and IEC 61980-3).

[V2G20-5003] If the EVProcessing is set to "Finished", the lists shall contain only one element which represents the EV's choice.

NOTE 1 The value of this element does not need to be the default value.

8.3.4.6.2.3 WPT_FinePositioningSetupRes

With the WPT_FinePositioningSetupRes message the SECC provides information about possible methods for fine positioning, pairing and alignment check together with setup parameters, accordingly.

[V2G20-5010] The EVCC and the SECC shall implement the message elements as defined in Figure 73 and Table 69.

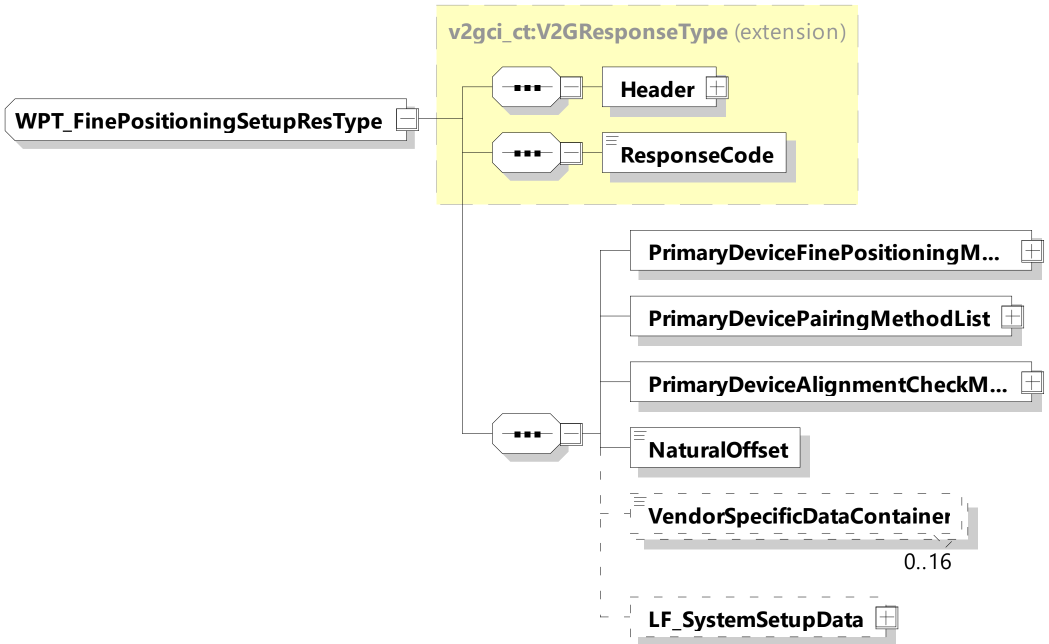

Figure 73 — Schema diagram – WPT_FinePositioningSetupRes

The elements of this message are used according to Table 69.

Table 69 — Semantics and type definition for WPT_FinePositioningSetupRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.

Element Name	Type	Semantics
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
PrimaryDeviceFinePositioningMethodList	simpleType: WPT_FinePositioningMethodListType refer to Annex A for the type definition	Fine positioning method intended to be used by the primary device (confirmation of EVs choice) (See IEC 61980-2). The default value is "Manual".
PrimaryDevicePairingMethodList	simpleType: WPT_PairingMethodListType refer to Annex A for the type definition	Pairing method intended to be used by the primary device (confirmation of EVs choice) (See IEC 61980-2) The default value is "External confirmation".
PrimaryDeviceAlignmentCheckMethodList	simpleType: WPT_AlignmentCheckMethodListType refer to Annex A for the type definition	Alignment check method intended to be used by the primary device (confirmation of EVs choice) (See IEC 61980-2) The default value is "PowerCheck".
NaturalOffset	simpleType: xs:unsignedShort	Offset of the center alignment point in mm (in x-direction) which results from the coil design of EV and infrastructure.
VendorSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Data container for proprietary information.
LF_SystemSetupData	simpleType: WPT_LF_SystemSetupDataType refer to Annex A for the type definition	Optional: Additional data needed to describe a LF positioning system. This information is needed, when LF_TxEV or LF_TxPrimaryDevice is given in the list EVPrimaryDevicePositioningMethodList.

[V2G20-5004] When the SECC has received a FinePositioningSetupReq with EVProcessing is set to "Ongoing" the lists "PrimaryDeviceFinePositioningMethodList", "PrimaryDevicePairingMethodList", "PrimaryDeviceAlignmentCheckMethodList" shall contain those elements, which represent the technical possibilities of the installation on infrastructure side.

[V2G20-5005] If the lists "PrimaryDeviceFinePositioningMethodList", "PrimaryDevicePairingMethodList", "PrimaryDeviceAlignmentCheckMethodList" contain more than one element, the default values shall be part of the list, if the device is a compatibility class A device (See IEC 61980-2 and IEC 61980-3).

[V2G20-5006] When the SECC has received a FinePositioningSetupReq with the EVProcessing set to "Finished", the lists "PrimaryDeviceFinePositioningMethodList", "PrimaryDevicePairingMethodList", "PrimaryDeviceAlignmentCheckMethodList" shall contain only one element which confirm the EV's choice.

NOTE The value of this element does not need to be the default value.

8.3.4.6.3 WPT_FinePositioning

8.3.4.6.3.1 WPT_FinePositioningReq/Res handling

The fine positioning process is started by the EVCC by sending the WPT_FinePositioningSetupReq message. The SECC responds by sending the WPT_FinePositioningSetupRes message.

After this initialization the positioning procedure is continued by looping the WPT_FinePositioningReq/Res message. As long as the vehicle positioning process is not finished the SECC sets the parameter "EVSEProcessing" in the WPT_FinePositioningRes message to "Processing" and the EVCC repeatedly sends the WPT_FinePositioningReq message. The EVCC stops sending WPT_FinePositioningReq when the parameter "EVSEProcessing" in the WPT_FinePositioningRes message is set to "Finished".

8.3.4.6.3.2 WPT_FinePositioningReq

[V2G20-5011] The EVCC and the SECC shall implement the message elements as defined in Figure 74 and Table 70.

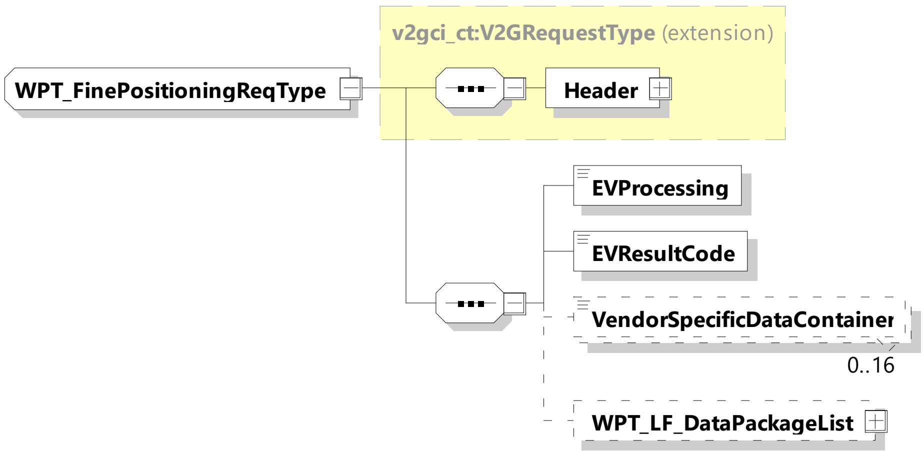

Figure 74 — Schema diagram – WPT_FinePositioningReq

The elements of this message are used according to Table 70.

Table 70 — Semantics and type definition for WPT_FinePositioningReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType Refer to 8.3.3	Contains general information, used for all messages.

Element Name	Type	Semantics
EVProcessing	simpleType: processingType enumeration refer to Annex A for the type definition	Parameter that indicates the stop of the Fine positioning process by the EV. When the EV stops the fine positioning process, the EVProcessing is set to "Finished", while still running it is set to "Ongoing".
EVResultCode	simpleType: WPT_EVResultType refer to Annex A for the type definition	Parameter with which the EV can inform the SECC about the result of the fine positioning process: – EVResultUnknown, – EVResultSuccess, – EVResultFailed.
VendorSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Data container for additional vendor specific information.
WPT_LF_DataPackageList	simpleType: WPT_LF_DataPackageListType refer to Annex A for the type definition	Optional: Used when LF_TxPrimaryDevice of LF_TxEV is chosen as FinePositioning method. The element includes information about received or transmitted LF signal data.

[V2G20-5051] As long as the EVProcessing Element in WPT_FinePositioningReq is "Ongoing" the EVResultCode shall be set to "EVResultUnknown".

[V2G20-5052] If the EVProcessing element in WPT_FinePositioningReq is set to "Finished" the EVResultCode shall not be set to "EVResultUnknown".

8.3.4.6.3.3 WPT_FinePositioningRes

[V2G20-5013] The EVCC and the SECC shall implement the message elements as defined in Figure 75 and Table 71.

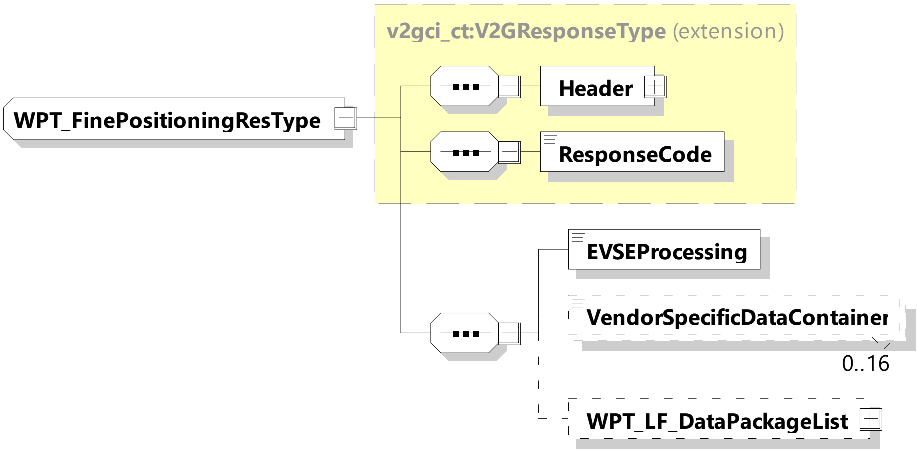

Figure 75 — Schema diagram – WPT_FinePositioningRes

The elements of this message are used according to Table 71.

Table 71 — Semantics and type definition for WPT_FinePositioningRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
EVSEProcessing	simpleType: processingType enumeration refer to Annex A for the type definition	Parameter that indicates the stop of the Fine positioning process by the EV. When the EVSE stops the fine positioning process, the EVSEProcessing is set to "Finished", while still running it is set to "Ongoing".
VendorSpecificDataContainer	simpleType: WPT_DataContainerType, refer to Annex A for the type definition	Optional: Data container for additional vendor specific information.
WPT_LF_DataPackageList	simpleType: WPT_LF_DataPackageListType refer to Annex A for the type definition	Optional: Used when LF_TxPrimaryDevice of LF_TxEV is chosen as FinePositioning method. The element includes information about received or transmitted LF signal data.

8.3.4.6.4 WPT_Pairing

The application of pairing for WPT and the sequence of parameter exchange is described in IEC 61980-2. It describes the exchange of parameters between EVCC and SECC for the different possible methods, which are considered in combination with WPT.

After the SDP process, SessionSetup and vehicle positioning have been finished the pairing shall be applied in order to identify the EVSE and the primary device, over which the EV has been positioned.

NOTE The identifier of the EV and the EVSE which are typically determined during pairing might already be exchanged during the SDP process. This, however, does not replace the exchange of the pairing messages together with the pairing process as described in IEC 61980-2.

8.3.4.6.4.1 WPT_PairingReq/Res handling

The EV will start the pairing process by applying the WPT_PairingReq (see 8.3.4.6.4.2). The SECC answers with the WPT_PairingRes (see 8.3.4.6.4.3). The parameters of the messages and their values are used according to the selected pairing method. The selection of the pairing method takes place during WPT_FinePositioningSetup. Refer to IEC 61980-2 for the sequence and application of parameter exchange.

8.3.4.6.4.2 WPT_PairingReq

[V2G20-5100] The EVCC and the SECC shall implement the message elements as defined in Figure 76 and Table 72.

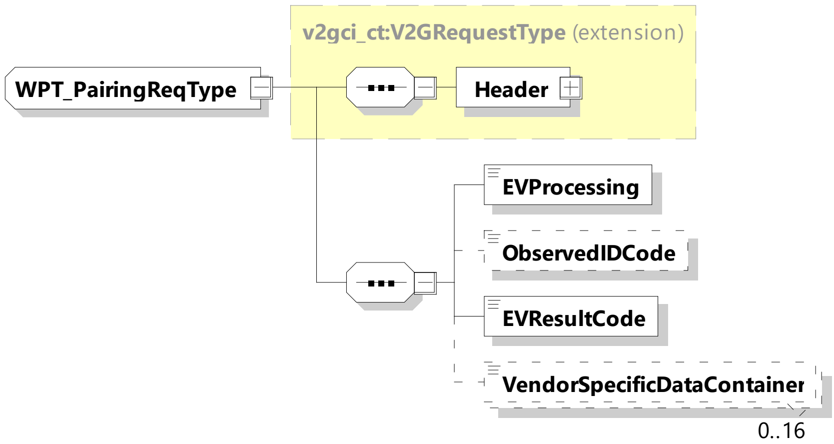

Figure 76 — Schema diagram – WPT_PairingReq

The message elements of this message are used as defined in Table 72.

Table 72 — Semantics and type definition for WPT_PairingReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVProcessing	simpleType: processingType enumeration refer to Annex A for the type definition	Parameter that indicates the stop of the Pairing process by the EV. When the EV stops the pairing process, the EVProcessing is set to "Finished", while still running it is set to "Ongoing".

ObservedIDCode	simpleType: numericIDType refer to Annex A for the type definition	Optional: The identifier observed by the EV through P2PS signaling according to the pairing method applied. (omitted if not required for the pairing method).
EVResultCode	simpleType: WPT_EVResultType refer to Annex A for the type definition	Parameter with which the EV can inform the SECC about the result of the pairing process: – EVResultUnknown, – EVResultSuccess, – EVResultFailed .
VendorSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Data container for additional vendor specific information.

8.3.4.6.4.3 WPT_PairingRes

[V2G20-5101] The EVCC and the SECC shall implement the message elements as defined in Figure 77 and Table 73.

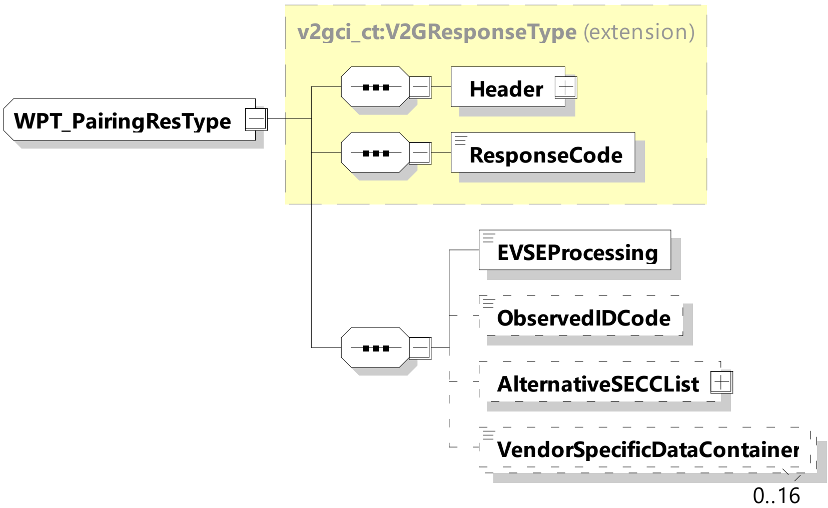

Figure 77 — Schema diagram – WPT_PairingRes

The message elements of this message are used as defined in Table 73.

Table 73 — Semantics and type definition for WPT_PairingRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType Refer to 8.3.3	Contains general information, used for all messages.

Element Name	Type	Semantics
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
EVSEProcessing	simpleType: processingType enumeration refer to Annex A for the type definition	Parameter indicating that the EVSE has finished the processing that was initiated after the WPT_PairingReq or if the EVSE is still processing at the time, the response message was sent.
ObservedIDCode	simpleType: numericIDType refer to Annex A for the type definition	Optional: The identifier observed by the supply device through P2PS signaling according to the pairing method applied (omitted if not required for the pairing method).
AlternativeSECCList	complexType: AlternativeSECCListType refer to 8.3.5.6.1	Optional: List of connection information to alternative SECCs in the order of highest probability in case the pairing failed.
VendorSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Data container for additional vendor specific information.

8.3.4.6.4.4 General WPT_Pairing Requirements

[V2G20-5007] Both EVCC and SECC shall support the set of WPT pairing messages as described in 8.3.4.6.4, which are required to process the WPT pairing.

8.3.4.6.4.4.1 Requirements for EVCC

[V2G20-5008] To start the pairing process the EVCC shall send a WPT_PairingReq message.

[V2G20-5012] If the EVCC has identified a pairing code it shall send a WPT_PairingReq message with the parameter ObservedIDCode set to the pairing identification code and the EVProcessing parameter set to "Finished".

[V2G20-5049] As long as the EVProcessing Element in WPT_PairingReq is "Ongoing" the EVResultCode shall be set to "EVResultUnknown".

[V2G20-5050] If the EVProcessing element in WPT_PairingReq is set to "Finished" the EVResultCode shall not be set to "EVResultUnknown".

8.3.4.6.4.4.2 Requirements for SECC

[V2G20-5016] If the SECC has received the WPT_PairingReq message with the parameter ObservedIDCode set to a pairing identification code and the validation of the identifier used for pairing fails it shall respond with a WPT_PairingRes message with the parameter ResponseCode equal to FAILED or WARNING_WPT.

NOTE 1 It is up to the implementation whether to give a response code FAILED or WARNING_WPT. In case of FAILED, the communication is terminated. In case of WARNING_WPT, the system allows to repeat the positioning and pairing procedure.

[V2G20-5065] If the SECC has received the WPT_PairingReq message with the parameter EVResultCode set to "EVResultFailed", then the SECC shall respond with a WPT_PairingRes message with the parameter ResponseCode equal to FAILED or WARNING_WPT.

[V2G20-5017] If the SECC has received the WPT_PairingReq message with the parameter ObservedIDCode set to a pairing identification code and the validation of the ObservedIDCode succeeds it shall respond with a WPT_PairingRes message with the parameter ResponseCode equal to "OK".

[V2G20-5018] If the SECC has identified a pairing code it shall respond to a WPT_PairingReq with a WPT_PairingRes message with the parameter ObservedIDCode set to the pairing identification code and the EVSEProcessing parameter set to "Finished".

[V2G20-5053] If [V2G20-5016] applies or the EVCC sent a WPT_PairingReq with the parameter EVResultCode set to "EVResultFailed, then the SECC should include AlternativeSECCList parameter in the PairingRes message when such information is available.

NOTE 2 The AlternativeSECCList provides a list of SECCs that the EVCC is recommended to connect to, in the order of priority, when the pairing process with the current SECC fails. Given the list, the EVCC can connect to those SECCs in the given order. For example, the EVCC connects to the wireless network with the provided SSID and start SDP protocol, or connect to the SECC by provided IP address and port number directly. How the list of alternative SECCs is obtained by the current SECC is not in the scope of this document.

8.3.4.6.5 WPT_ChargeParameterDiscoveryReq/Res

The WPT_ChargeParameterDiscoveryReq and WPT_ChargeParameterDiscoveryRes messages are used to exchange information about the technical characteristics of the power transfer devices in order to tune the system components, accordingly.

By exchanging this information the final compatibility check of EV device and the supply device is performed (see IEC 61980-2).

8.3.4.6.5.1 WPT_ChargeParameterDiscoveryReq/Res handling

After being authorized for charging at the EVSE (SECC) the EVCC and the SECC negotiate the energy transfer parameters with the WPT_ChargeParameterDiscovery message pair.

Concepts treated for an optimal supply of energy that corresponds to the customer needs below.

When using scheduled control mode, the energy transfer parameters negotiation that precedes the delivery of energy or may be engaged during the energy delivery phase is destined to ensure that the user will be satisfied while at the same time ensuring that the energy will effectively be available and fall within the capacity of power supply grid at the local level (private network) and at the regional level (public network). This required negotiation will become more and more necessary as the number of EVs increase, as well as an increase in volatility of local renewable energy production.

When using the dynamic control mode, the energy transfer parameters can be changed dynamically while in the Charging Loop, depending on the need of the EVSE.

8.3.4.6.5.2 WPT_ChargeParameterDiscoveryReq

By sending the WPT_ChargeParameterDiscoveryReq message the EVCC provides its energy transfer parameters to the SECC. This message provides status information about the EV, i.e. the capabilities of the EV charging system.

[V2G20-5019] The EVCC shall send the WPT_ChargeParameterDiscoveryReq message with the elements of the message set according to the technical characteristics of the EV device.

[V2G20-5128] The EVCC and the SECC shall implement the message elements as defined in Figure 78 and Table 74.

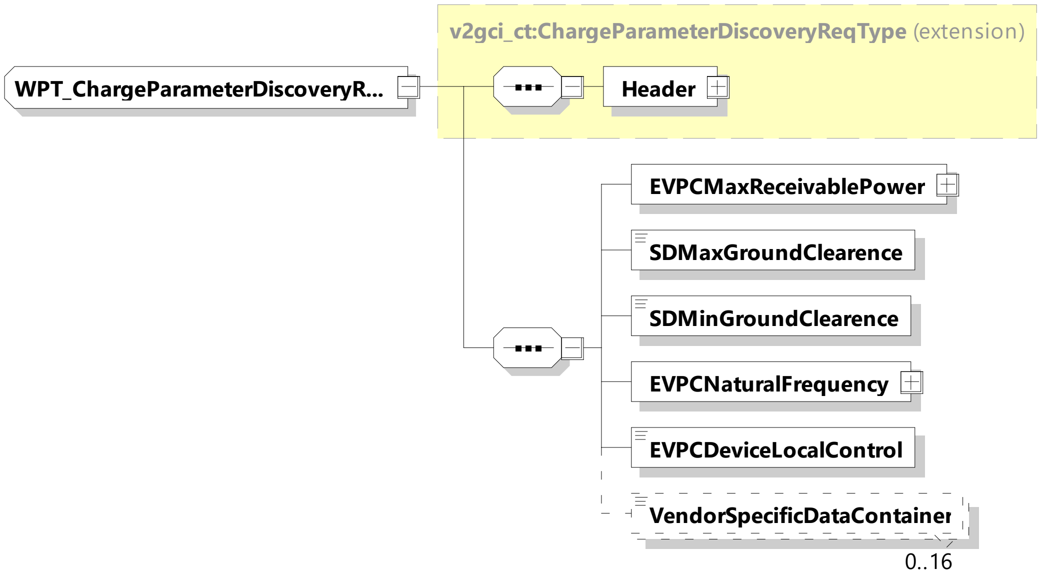

Figure 78 — Schema diagram – WPT_ChargeParameterDiscoveryReq

The elements of this message are used according to Table 74.

Table 74 — Semantics and type definition for WPT_ChargeParameterDiscoveryReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVPCMaxReceivablePower	complexType RationalNumberType refer to 8.3.5.3.8	Maximum power in watts which EVPC can receive
SDMaxGroundClearence	simpleType: xs:unsignedShort	Maximum value of secondary device ground clearance in mm
SDMinGroundClearence	simpleType: xs:unsignedShort	Minimum value of secondary device ground clearance in mm

Element Name	Type	Semantics
EVPCNaturalFrequency	complexType RationalNumberType refer to 8.3.5.3.8	Self-resonance frequency of the EV device under fully loaded condition (primary device not present)
EVPCDeviceLocalControl	simpleType: xs:boolean	True, if EV device has a local control loop, which modifies reactance or power, false otherwise.
VendorSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Manufacturer specific data container, containing, e.g. manufacturer ID and Model ID

NOTE The parameters related to current, power and voltage provided in WPT_ChargeParameterDiscovery message pair can be considered as physical limitations, meaning that they are related to the physical abilities of the system under rated operating conditions. Later during the charging process, depending on the external and environmental conditions (SOC, temperatures, etc.), the values of these limitations can change into more conservative values.

8.3.4.6.5.3 WPT_ChargeParameterDiscoveryRes

With the WPT_ChargeParameterDiscoveryRes message the SECC provides applicable charge parameters from the grid’s perspective.

[V2G20-5021] The SECC shall respond with a WPT_ChargeParameterDiscoveryRes with the ResponseCode = OK when the element values received from the EVCC allow a system setup for successful power transfer at the supply device.

[V2G20-5022] The SECC shall respond with a WPT_ChargeParameterDiscoveryRes with the ResponseCode = FAILED when the element values received from the EVCC do not allow a system setup for any power transfer at the supply device.

[V2G20-5066] The SECC shall respond with a WPT_ChargeParameterDiscoveryRes with the ResponseCode = WARNING_WPT when the element values received from the EVCC only allow a system setup for limited (significantly lower) power transfer at the supply device.

NOTE It is up to the implementation of the SECC when to response with WARNING_WPT or FAILED based on the system behavior.

[V2G20-5127] The EVCC and the SECC shall implement the message elements as defined in Figure 79 and Table 75.

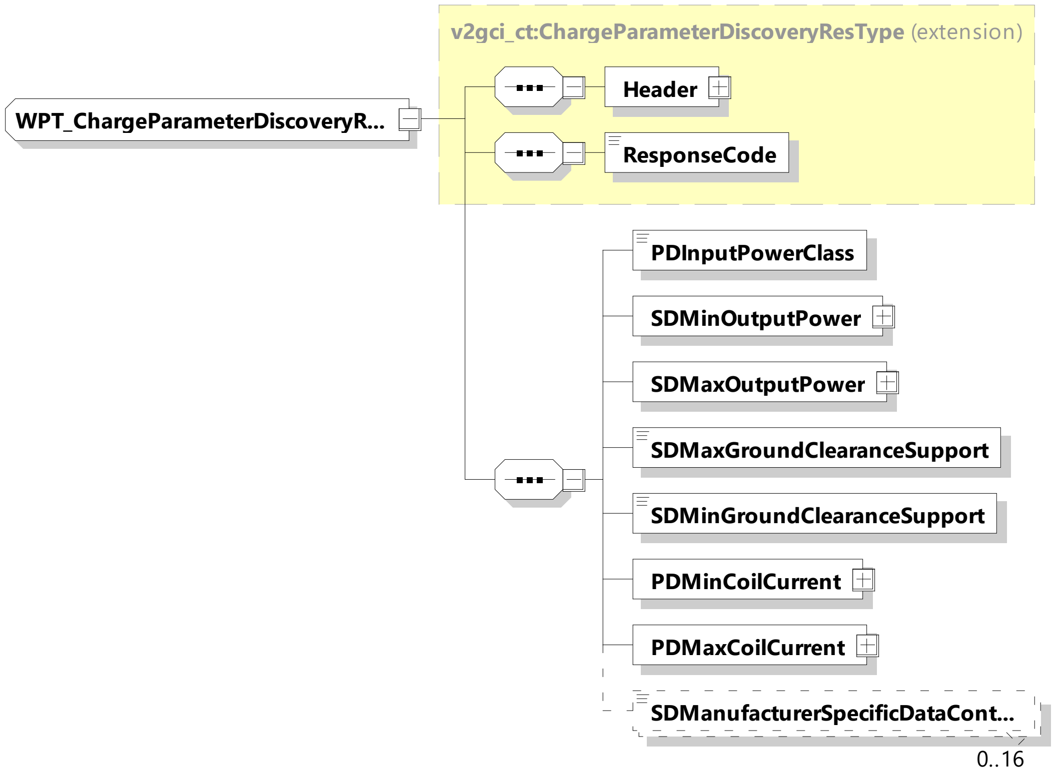

Figure 79 — Schema diagram – WPT_ChargeParameterDiscoveryRes

The elements of this message are used according to Table 75.

Table 75 — Semantics and type definition for WPT_ChargeParameterDiscoveryRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
PDInputPowerClass	simpleType: WPT_PowerClassType refer to Annex A for the type definition	Power class supported by the primary device. Possible values for MF-WPT input power classes are: MF-WPT1, MF-WPT2, MF-WPT3 and MF-WPT4 according to IEC TS 61980-3
SDMinOutputPower	complexType: RationalNumberType refer to 8.3.5.3.8	Minimum power in watts drawn by the secondary device
SDMaxOutputPower	complexType: RationalNumberType refer to 8.3.5.3.8	Maximum power in watts drawn by the secondary device

Element Name	Type	Semantics
SDMaxGroundClearanceSupport	simpleType: xs:unsignedShort	Maximum value of supported secondary device ground clearance in mm
SDMinGroundClearanceSupport	simpleType: xs:unsignedShort	Minimum value of the secondary device ground clearance in mm
PDMinCoilCurrent	complexType: RationalNumberType refer to 8.3.5.3.8	Minimum current through the primary coil, which can be controlled by the primary device
PDMaxCoilCurrent	complexType: RationalNumberType refer to 8.3.5.3.8	Maximum current through the primary coil, which can be controlled by the primary device
SDManufacturerSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Manufacturer specific data container, containing, e.g. manufacturer ID and Model ID

8.3.4.6.6 WPT_AlignmentCheckReq/Res

The WPT_AlignmentCheckReq/Res messages are used to exchange information needed to perform the alignment check, with which the proper position of the EV device with the primary device is checked and confirmed (See IEC 61980-2).

8.3.4.6.6.1 WPT_AlignmentCheckReq/Res handling

8.3.4.6.6.2 WPT_AlignmentCheckReq

[V2G20-5015] The EVCC and the SECC shall implement the message elements as defined in Figure 80 and Table 76.

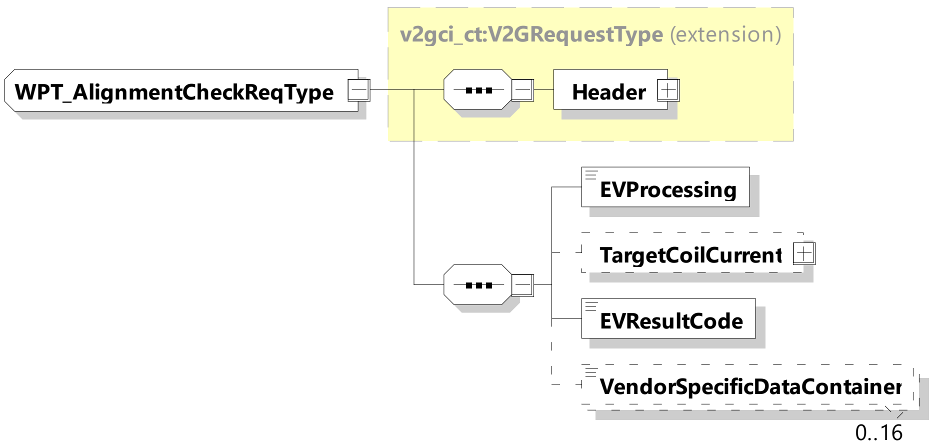

Figure 80 — Schema diagram – WPT_AlignmentCheckReq

The elements of this message are used according to Table 76.

Table 76 — Semantics and type definition for WPT_AlignmentCheckReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType Refer to 8.3.3	Contains general information, used for all messages.
EVProcessing	simpleType: processingType enumeration refer to Annex A for the type definition	Parameter that indicates the stop of the alignment check process by the EV. When the EV stops the pairing process, the EVProcessing is set to "Finished", while still running it is set to "Ongoing".
TargetCoilCurrent	complexType RationalNumberType refer to 8.3.5.3.8	Optional: Used for Power check and LPE: – Current in amperes, the EVSE shall feed the primary coil with in order to enable the secondary device to perform the alignment check.
EVResultCode	simpleType: WPT_EVResultType refer to Annex A for the type definition	Parameter with which the EV can inform the SECC about the result of the alignment check: – EVResultUnknown, – EVResultSuccess, – EVResultFailed.
VendorSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Data container for additional vendor specific information. This element is used primarily when a proprietary alignment check method was selected with the FinePositioningSetup.

[V2G20-5047] As long as the EVProcessing Element in WPT_AlignmentCheckReq is "Ongoing" the EVResultCode shall be set to "EVResultUnknown".

[V2G20-5048] If the EVProcessing element in WPT_AlignmentCheckReq is set to "Finished" the EVResultCode shall not be set to "EVResultUnknown".

8.3.4.6.6.3 WPT_AlignmentCheckRes

[V2G20-5014] The EVCC and the SECC shall implement the message elements as defined in Figure 81 and Table 77.

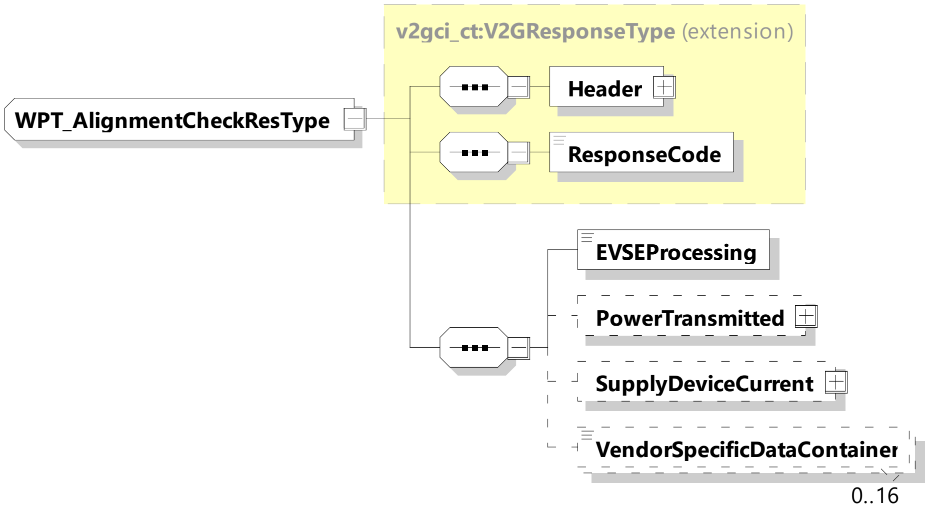

Figure 81 — Schema diagram – WPT_AlignmentCheckRes

The elements of this message are used according to Table 77.

Table 77 — Semantics and type definition for WPT_AlignmentCheckRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
EVSEProcessing	simpleType: processingType refer to Annex A for the type definition	Parameter indicating that the EVSE has finished the processing that was initiated after the WPT_AlignmentCheckReq or if the EVSE is still processing at the time, the response message was sent.
PowerTransmitted	complexType RationalNumberType refer to 8.3.5.3.8	Optional: The amount of power in watts transmitted by the supply device
SupplyDeviceCurrent	complexType RationalNumberType refer to 8.3.5.3.8	Optional: The amount of current in amperes into the supply device coil
VendorSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Data container for additional vendor specific information. This element is used primarily when a proprietary alignment check method was selected with the FinePositioningSetup

8.3.4.6.7 WPT_ChargeLoopReq/Res

For WPT periodic exchange of process control data is requested from EV side, see IEC 61980-2.

8.3.4.6.7.1 WPT_ChargeLoopReq/Res handling

The WPT_ChargeLoopReq/Res messages are used for the periodic exchange of process control data. The messages are derived from the ChargeLoopReq/Res messages defined in the common module (see ChargeLoopReqType) and extended by the parameters necessary for the handling of wireless power transfer according to IEC 61980-2.

8.3.4.6.7.2 WPT_ChargeLoopReq

By sending the WPT_ChargeLoopReq the EV requests a certain power from the EVSE.

[V2G20-1296] The EVCC and the SECC shall implement the message elements as defined Table 78 and Figure 82.

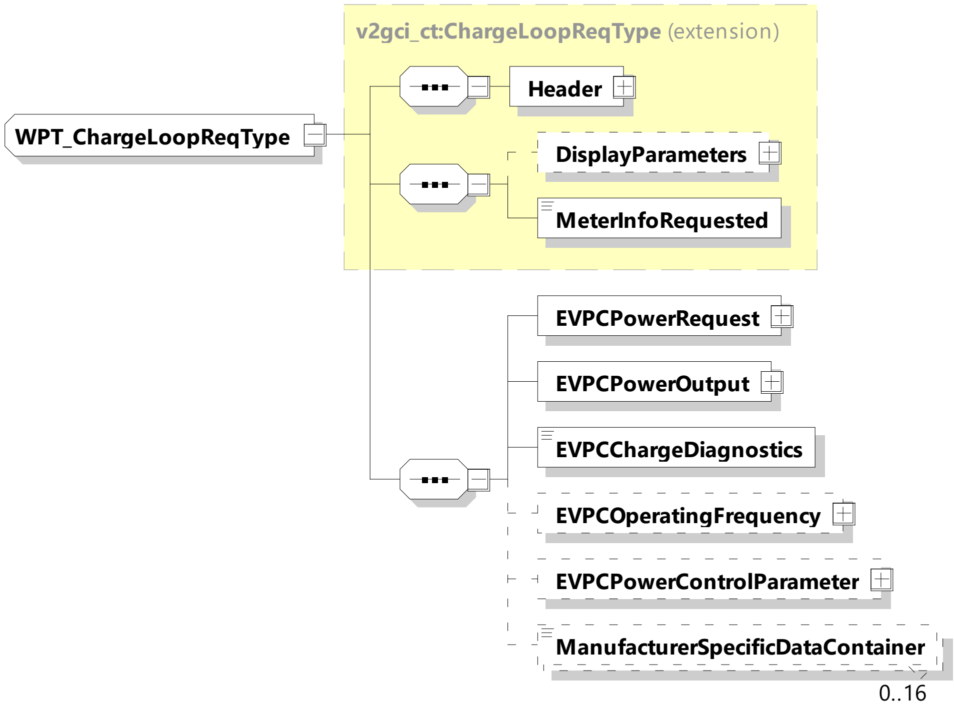

Figure 82 — Schema diagram – WPT_ChargeLoopReq

The elements of this message are used according to Table 78.

Table 78 — Semantics and type definition for WPT_ChargeLoopReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVPCPowerRequest	complexType RationalNumberType refer to 8.3.5.3.8	Power the EVPC would like to have as output in Watt. Note that setting this value to zero is valid.

Element Name	Type	Semantics
EVPCPowerOutput	complexType RationalNumberType refer to 8.3.5.3.8	Power measured at the output of the EVPC electronics in Watt
EVPCChargeDiagnostics	simpleType: WPT_ChargeDiagnosticsType refer to Annex A for the type definition	Diagnostics information, can have the following values: – EVPCNoIssue: no anomaly detected, Power Control is running normally; – EVPCTempOverheatDetected: EVPC has detected a temperature anomaly; – EVPCPowerTransferAnomalyDetected: EVPC has detected an unexpected behavior of the magnetic power transfer; – EVPCAnomalyDetected: EVPC has detected an unexpected system behavior.
EVPCOperatingFrequency	complexType RationalNumberType refer to 8.3.5.3.8	Optional: EVPC measured MF-WPT operating frequency
EVPCPowerControlParameter	simpleType: WPT_EVPCPowerControlParameterType refer to Annex A for the type definition	Optional: additional parameters for the applied wireless power transfer method: – EVPC coil current request: EVPC wants the primary device to set a specific (preferred) coil current value. NOTE The EVPCPowerRequest is overriding the value given here in case of mismatch: – EVPC coil current information: information about secondary device coil current (AC); – EVPC current output information: information about DC current supplied to the EV; – EVPC voltage output information: information about DC bus or battery voltage.
ManufacturerSpecificDataContainer	simpleType: WPT_DataContainerType refer to Annex A for the type definition	Optional: Manufacturer specific parameters: additional field for proprietary manufacturer specific information during power transfer

8.3.4.6.7.3 WPT_ChargeLoopRes

After receiving the WPT_ChargeLoopReq from the EVCC the SECC sends the WPT_ChargeLoopRes informing the EV about the EVSE status.

[V2G20-5028] The EVCC and the SECC shall implement the message elements as defined in Table 79 and Figure 83.

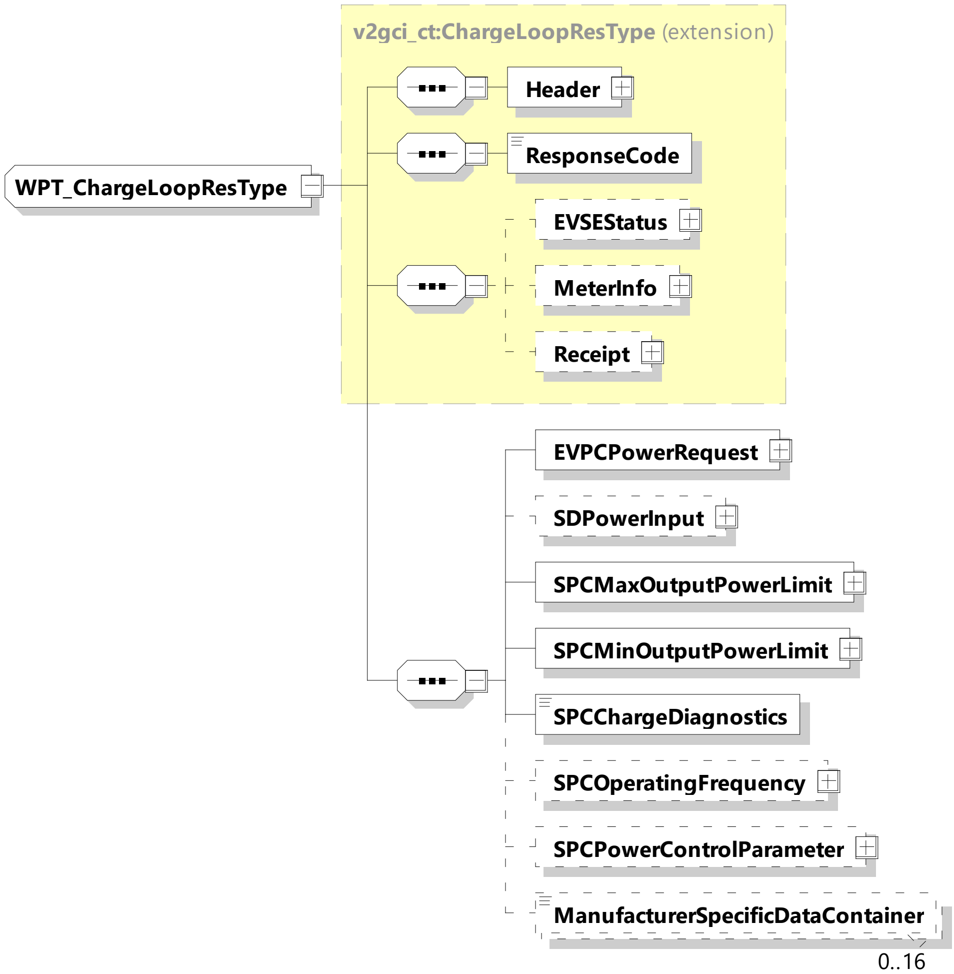

Figure 83 — Schema diagram – WPT_ChargeLoopRes

The elements of this message are used according to Table 79.

Table 79 — Semantics and type definition for WPT_ChargeLoopRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVPCPowerRequest	complexType RationalNumberType refer to 8.3.5.3.8	Power value which was requested by the EVCC
SDPowerInput	complexType RationalNumberType refer to 8.3.5.3.8	Optional: DC power consumed from the MF-WPT system input by the supply device in Watt. The EVCC compares the value with the received EVPCPowerOutput in order to detect possible misbehavior. This value is for information only without any impact on safety relative actions.

The wireless communication for ACDP with infrastructure mounted pantograph requires early association of the EVCC with the associated SECC before the V2G communication can be started. This is needed to be able to control the EV positioning and ACDP activation processes correctly. For this the early association provides the required information as the EVID and EV position information.

For ACDP the multi SECC communication architecture according Figure 29 applies for multiple SECC architecture as described in 7.10.1.2.2.

For ACDP a pairing and positioning device (PPD) mounted on the infrastructure applies as described in EN 50696:2021, A.2 and A.3.

NOTE 1 An architecture using a single AP with single SECC, and single PPD can be viewed as a special version of the Multiple SECC architecture and is therefore also a permitted communication architecture.

NOTE 2 For ACDP mounted pantograph on the infrastructure the pairing and positioning device can be located on the infrastructure side. The identification object which holds the EVID is located on the EVs ACDP counterpart side. In this case the PPD can be able to read an identification mark positioned on or in the EV. The PPD identification method is used to ensure that the EV's EVID is detected only in close proximity of the EV to the EVSE so crosstalk to another EV is not possible (Partitioning). RFID identification tags are recommended to be present on the EV to ensure a basic level of interoperability. These tags are specified in Annex E.2.

NOTE 3 The EVID can be any unique identifier number or string. For electric bus applications it can comprise a readable name or, e.g. the EV number plate. For any other application it can be any random number which is unique at that time, thus privacy is not concerned.

The ACDP V2G communication consists of the following messages:

- common messages;
- DC messages;
- ACDP specific messages.

8.3.4.7.2.1 ACDP terminology

There are three distinct roles for the pairing and positioning device:

Pairing

The EV connects to the right supply equipment communication controller (SECC) using the wireless SDP protocol in installations where there are several ACDP EVSEs in close proximity. This is a required role.

Partitioning

The area around the ACDP is partitioned into a properly sized acceptancezone, outside which a charging session cannot be started. The zone should not be so small as to restrict the positioning leeway, but small enough that a nearby EV cannot inadvertently start a session and lower the ACDP such that a dangerous situation as a collision between the ACDP and another EV can occur. This is a required role for safety.

Positioning

The driver can position the bus with enough precision that the ACDP contacts will connect with the ACDP counterpart contacts on the EV correctly so that correct charging is assured. This is an optional role for operational efficiency.

The pairing and positioning device provides a spatial selective orthogonal communication channel to the WLAN communication when the EV is in close proximity to the ACDP in order to resolve the undetermined association of the EV communication to the ACDP’s EVSE. The PPD consists of a marker located on the EV and a receiver located in the ACDP infrastructure.

A recommended, basic RFID PPD system with sufficient positional accuracy for Partitioning is defined in E.2.1.3.

8.3.4.7.3 ACDP message overview

ACDP with infrastructure mounted pantograph will use the common message set, the DC V2G message set and will have ACDP specific messages to handle the status information, the EV positioning process and the ACDP control. Table 80 provides an overview of the ACDP communication messages.

Table 80 — ACDP message communication overview

Message/Action	Abbr	Com	DC	ACDP	Msg Looped	Next message(s) on success	Exception handling on app level failure	Exception handling on msg timeout
SupportedAppProtocol	SAP	x			N	SSU	SDP-Restart	SDP-Restart
SessionSetup	SSU	x			N	APOS	SDP-Sestart	SDP-Restart
ACDP_VehiclePositioning	APOS			x	Y	APOS, ASUP	SSP	SDP-Restart
AuthorizationSetup	ASUP	x			N	AUTH, CINS	SSP	SDP-Restart
Authorization	AUTH	x			Y	AUTH, SDI	SSP	SDP-Restart
CertificateInstallation	CINS	x			Y	CINS, AUTH	SSP	SDP-Restart
ServiceDiscovery	SDI	x			N	SDE	SSP	SDP-Restart
ServiceDetail	SDE	x			N	SSEL	SSP	SDP-Restart
ServiceSelection	SSEL	x			N	CPDA	SSP	SDP-Restart
DC_ChargeParameter Discovery	CPDA		x		N	SE	SSP	SDP-Restart
ScheduleExchange	SE	x			Y	SE, CONA	SSP	SDP-Restart
ACDP_Connect	CONA			x	Y	CONA, CC	DISA	ESDW
DC_CableCheck	CC		x		Y	CC, PC	DISA	ESDW
DC_PreCharge	PC		x		N	PD(Start)	DISA	ESDW
PowerDelivery(Start)	PD(Start)		x		N	DCCL	DISA	ESDW
DC_ChargeLoop	DCCL		x		Y	DCCL, MCON, PD(Stop)	PD(Stop)	ESDW
PowerDelivery(Stop)	PD(Stop)		x		N	DISA	DISA	ESDW
ACDP_Disconnect	DISA			x	Y	DISA, SSP	ESDW	ESDW

Message/Action	Abbr	Com	DC	ACDP	Msg Looped	Next message(s) on success	Exception handling on app level failure	Exception handling on msg timeout
SessionStop	SSP	x			N	SSP	SDP-Restart	ESDW
Wait for new session						SDP-Restart	SDP-Restart	
Multiplexed communication								
MeteringConfirmation	MCON		x		N	MCON		ESDW
ACDP_SystemStatus	SYSS			x	Y	SYSS		SDP-Restart

NOTE 1 Emergency shut down (ESDW) according to IEC 61851-23:2014, Annex CC.5 and wait for change of control pilot to State A.

NOTE 2 ACDP_SystemStatus has a separate payload type, see Table 14.

8.3.4.7.4 SDP parameters for ACDP

For ACDP use case the SDP client needs to select the appropriate configuration compliant with the following requirement:

[V2G20-4003] For ACDP an SDP client shall send the SDP request message with the elements "PPD/P2PS" (Bit 0) set to 1 and "Coupling Type" (Bit 0/1) set to 1 accordingly.

8.3.4.7.5 ACDP_VehiclePositioning

8.3.4.7.5.1 ACDP_VehiclePositioningReq/Res handling

During the EV approaching process the ACDP_VehiclePositioningReq/Res messages may be used to guide the EV to the ACDP infrastructure charging area with defined x and y tolerances of the contact interface according to EN 50696.

The definition of the coordinate system for ACDP for infrastructure mounted pantograph is based on ISO 4130:1978, Annex D.

For ACDP the actual position of the EV contact interface in relation to the ACDP contact interface is determined by the pairing and positioning device.

The position information provided during the EV approaching process as well as the effective tolerance dimensions can be communicated via the ACDP_VehiclePositioningRes message. This information may be forwarded to the EV driver as guiding information.

There are three cases for positioning support between EV and EVSE. Either both the EV and the EVSE support positioning support, the EV does not, or the EV does and the EVSE does not. The sequence is slightly different in these three cases.

Case 1: Positioning support from both sides. The EVCC will start sending ACDP_VehiclePositioningReq with EVPositioningSupport=True, and EVMobilityStatus=Mobilized since it is asking the SECC to help it position the EV. The SECC will reply ACDP_VehiclePositioningRes with EVSEPositioningSupport=True and EVSEProcessing=Ongoing and provide positioning information to assist the EV in moving to the correct charging position. When the EVCC receives this information, it can decide whether it is sufficiently in position to initiate charging or send another ACDP_VehiclePositioningReq. This looping continues until the EVCC decides that it is in position and is also immobilized. Then, it will then send a ACDP_VehiclePositioningReq with EVMobilityStatus=Immobilized. The SECC will reply with a ACDP_VehiclePositioningRes with EVSEProcessing=Finished and the sequence will continue with AuthorizationSetupReq.

Case 2: the EVCC does not support SECC supported positioning. The EVCC will send ACDP_VehiclePositioningReq with EVPositioningSupport=False, and EVMobilityStatus=Mobilized. The SECC will reply with ACDP_VehiclePositioningRes with EVSEPositioningSupport=False/True (do not care) and EVSEProcessing=Ongoing. When the EVCC receives this information, the EV can decide whether it is sufficiently in position (using other means, such as the driver using visual aids to park, or advanced EV sensors) to stop the positioning process, or to send another ACDP_VehiclePositioningReq in a loop. This looping continues until the EV decides that the EV is in position, and is also immobilized. Then it will then send a ACDP_VehiclePositioningReq with EVMobilityStatus=Immobilized. The SECC will reply with a ACDP_VehiclePositioningRes with EVSEProcessing=Finished, and the sequence will continue with AuthorizationSetupReq.

Case 3: the EVCC supports positioning, but the SECC does not. In this case, the EVCC will send ACDP_VehiclePositioningReq with EVPositioningSupport=True, and EVMobilityStatus=Mobilized. The SECC will reply ACDP_VehiclePositioningRes with EVSEPositioningSupport=False and EVSEProcessing=Ongoing. At this point, the EVCC may allow for manual positioning of the EV as described in the second case. If the EVCC decides that it requires SECC positioning support, it may stop the charging process by sending SessionStop.

In all cases where there is an ongoing positioning loop, the SECC monitors its PPD to verify that pairing is still correct. If the EV moves out of the pairing zone, then the SECC will send ACDP_VehiclePositioningRes with ResponseCode=FAILED_AssociationError, the EVCC will send SessionStop and the SDP process will restart.

NOTE 1 If the EV already knows it is in position and is also immobilized, then no loop is required, and the EVCC can directly send a ACDP_VehiclePositioningReq with EVMobilityStatus=Immobilized. The SECC will then reply with a ACDP_VehiclePositioningRes with EVSEProcessing=Finished, and the sequence will continue with AuthorizationSetupReq.

NOTE 2 Positioning support with the RFID PPD is very basic. The only information that is returned by the SECC is EV_InChargePosition = True/False. This information indicates that the EV is within/not within the partition zone for charging.

NOTE 3 It is possible that more than one PPD technology is available. For example, the EV and EVSE have both the Basic RFID PPD as well as an advanced optical PPD. Since the active part of the PPD is always on the EVSE, the EVSE can determine the presence of the advanced optical PPD during the SDP pairing for WLAN process and continue using this during the positioning process.

8.3.4.7.5.2 ACDP_VehiclePositioningReq

[V2G20-4004] The EVCC and the SECC shall implement the message elements as defined in Figure 84 and Table 81.

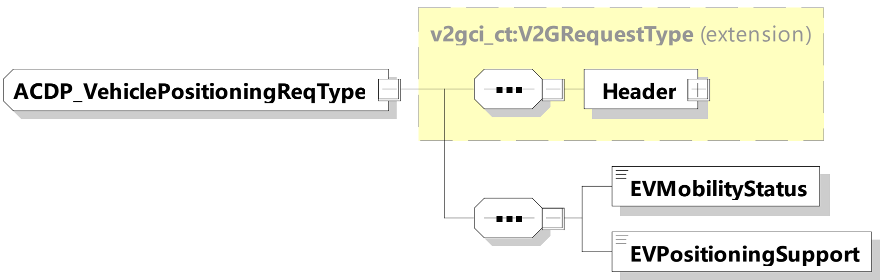

Figure 84— Schema diagram ACDP_VehiclePositioningReq

The elements of this message are used according to Table 81.

Table 81 — Semantics and type definition for ACDP_VehiclePositioningReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType Refer to 8.3.3	Contains general information, used for all messages.
EVMobilityStatus	simpleType: xs:boolean	– EV is mobilized = 0 – EV is immobilized = 1
EVPositioningSupport	simpleType: xs:boolean	Indicates whether positioning is supported by the EV. – No EV Position Support = 0 – EV Position Support = 1

8.3.4.7.5.3 ACDP_VehiclePositioningRes

[V2G20-4005] The EVCC and the SECC shall implement the message elements as defined in Figure 85 and Table 82.

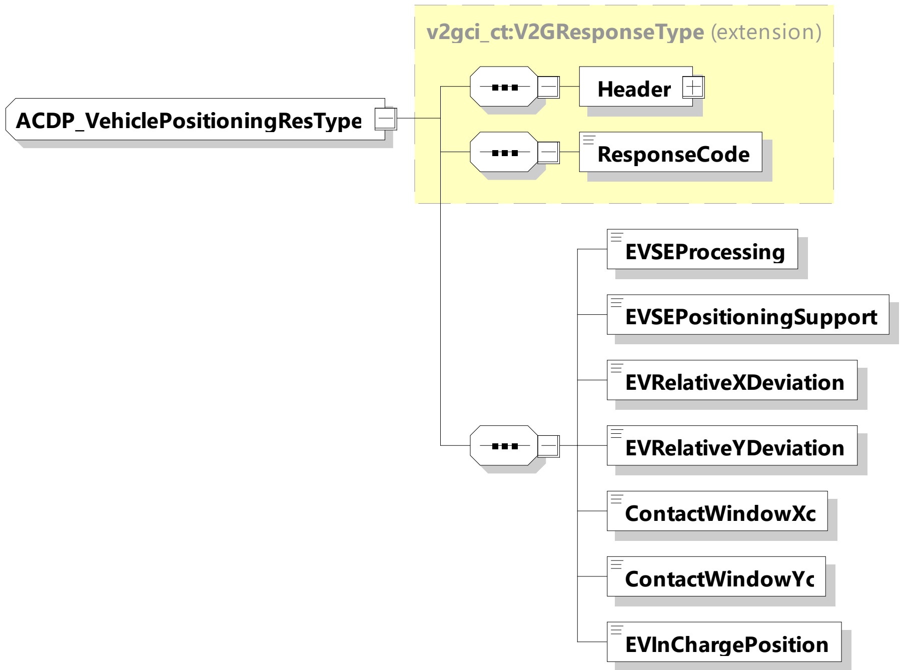

Figure 85— Schema diagram ACDP_VehiclePositioningRes

The elements of this message are used according to Table 82.

Table 82 — Semantics and type definition for ACDP_VehiclePositioningRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
EVSEProcessing	simpleType: processingType refer to Annex A for the type definition	Parameter indicating that the EVSE has finished the processing that was initiated after the ACDP_VehiclePositioningReq or that the EVSE is still processing at the time, the response message was sent.
EVSEPositioningSupport	simpleType: xs:boolean	Indicates whether positioning is supported by the EVSE.
EVRelativeXDeviation	simpleType: xs:short	Option in case of positioning supported: EV relative ACDP X (counter longitudinal driving direction) deviation in mm.
EVRelativeYDeviation	simpleType: xs:short	Option in case of positioning supported: EV relative ACDP Y (lateral) deviation in mm.
ContactWindowXc	simpleType: xs:short	Option in case of positioning supported: Is used to provide driver information about usable contact position. ContactWindow_Xc defines half of the useable ACDP contact area in driving direction in mm for consideration of a tolerance margin.
ContactWindowYc	simpleType: xs:short	Option in case of positioning supported: Is used to provide driver information about usable contact position. ContactWindow_Yc defines half of the useable contact area lateral to the driving direction in mm for consideration of a tolerance margin.

Element Name	Type	Semantics
EVInChargePosition	simpleType: xs:boolean	Option in case of positioning supported: Indicates whether the EV is positioned within allowed tolerances to ACDP. States: • EVNotInChargePosition = 0 • EVInChargePosition = 1

8.3.4.7.6 ACDP_ConnectReq/Res

8.3.4.7.6.1 ACDP_ConnectReq/Res handling

This message pattern is used to initiate the connecting process of the ACDP.

8.3.4.7.6.2 ACDP_ConnectReq

[V2G20-4006] The EVCC and the SECC shall implement the message elements as defined in Figure 86 and Table 83.

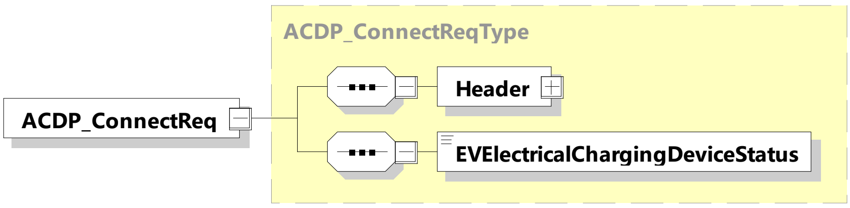

Figure 86 — Schema diagram – ACDP_ConnectReq

The elements of this message are used according to Table 83.

Table 83 — Semantics and type definition for ACDP_ConnectReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVElectricalChargingDeviceStatus	simpleType: electricalChargingDeviceStatusType refer to Annex A for the type definition	This element provides the information if the conductive connection between the EV and EVSE is in state "connected" (CP State = B, C or D) or "disconnected" (CP State = A).

8.3.4.7.6.3 ACDP_ConnectRes

[V2G20-4007] The EVCC and the SECC shall implement the message elements as defined in Figure 87 and Table 84.

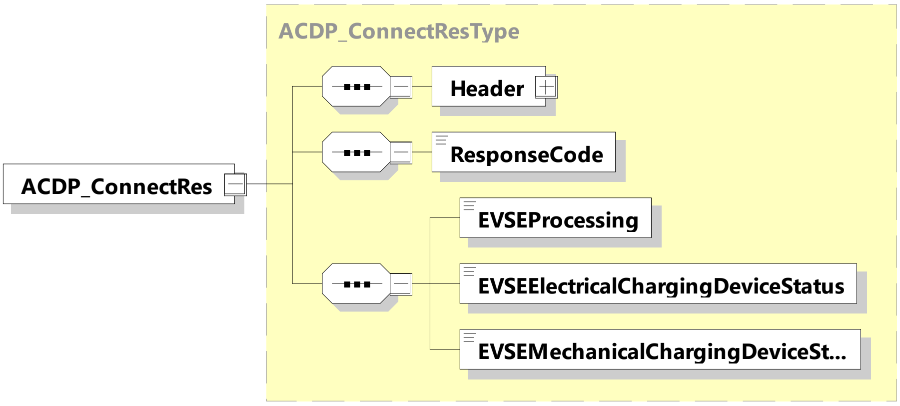

Figure 87 — Schema diagram – ACDP_ConnectRes

The elements of this message are used according to Table 84.

Table 84 — Semantics and type definition for ACDP_ConnectRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
EVSEProcessing	simpleType: processingType refer to Annex A for the type definition	Parameter indicating that the EVSE has finished the processing that was initiated after the ACDP_ConnectReq or if the EVSE is still processing at the time, the response message was sent.
EVSEElectricalChargingDeviceStatus	simpleType: electricalChargingDeviceStatusType refer to Annex A for the type definition	This element provides the information if the conductive connection between the EV and EVSE is in state "connected" (CP State = B, C or D) or "disconnected" (CP State = A).

Element Name	Type	Semantics
EVSEMechanicalChargingDeviceStatus	simpleType: mechanicalChargingDeviceStatusType refer to Annex A for the type definition	This element provides the information if the ACDP status of the EVSE is in home, or end position or if it is in a moving state.

8.3.4.7.7 ACDP_DisconnectReq/Res

8.3.4.7.7.1 ACDP_DisconnectReq

[V2G20-4008] The EVCC and the SECC shall implement the message elements as defined in Figure 88 and
Table 85.

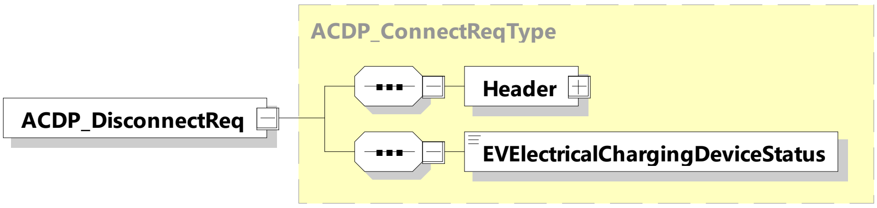

Figure 88 — Schema diagram – ACDP_DisconnectReq

The elements of this message are used according to Table 85.

Table 85 — Semantics and type definition for ACDP_DisconnectReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVElectricalChargingDeviceStatus	simpleType: electricalChargingDeviceStatusType refer to Annex A for the type definition	This element provides the information if the conductive connection between the EV and EVSE is in state "connected" (CP State = B, C or D) or "disconnected" (CP State = A).

8.3.4.7.7.2 ACDP_DisconnectRes

[V2G20-4009] The EVCC and the SECC shall implement the message elements as defined in Figure 89 and Table 86.

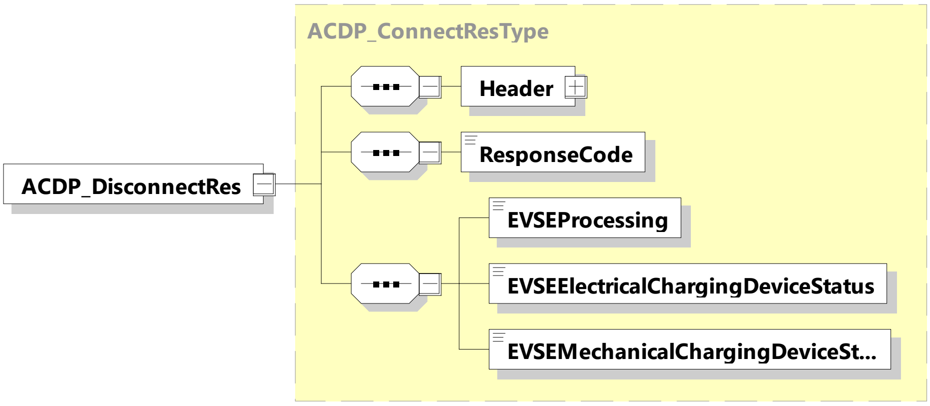

Figure 89 — Schema diagram – ACDP_DisconnectRes

The elements of this message are used according to Table 86.

Table 86 — Semantics and type definition for ACDP_DisconnectRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G Messages received by the SECC.
EVSEProcessing	simpleType: processingType refer to Annex A for the type definition	Parameter indicating that the EVSE has finished the processing that was initiated after the ACDP_ConnectReq or if the EVSE is still processing at the time, the response message was sent.

Element Name	Type	Semantics
EVSEElectricalChargingDeviceStatus	simpleType: electricalChargingDeviceStatusType refer to Annex A for the type definition	This element provides the information if the conductive connection between the EV and EVSE is in state "connected" (CP State = B, C or D) or "disconnected" (CP State = A).
EVSEMechanicalChargingDeviceStatus	simpleType: mechanicalChargingDeviceStatusType refer to Annex A for the type definition	This element provides the information if the ACDP status of the EVSE is in home, or end position or if it is in a moving state.

8.3.4.7.8 ACDP_SystemStatusReq/Res

8.3.4.7.8.1 ACDP_SystemStatusReq/Res handling

This message pattern is used to get current status information of involved charging equipment. The ACDP_SystemStatus message pair provides important diagnostic information as well.

The ACDP_SystemStatus messages have a separate payload type thus they execute in parallel and independent to the V2G messages.

NOTE 1 Safety relevant control with respect to the charging equipment and charging process is not in the scope of this document.

NOTE 2 The V2G communication is kept until the end of the charging session as well in the case of an error until the system is in a safe status after the ACDP has been moved into the home position.

The intended application of ACDP_SystemStatus is:

- first time after successfully processing of SessionSetupRes;
- at any time within the V2G communication process, asynchronous of the primary communication;
- in case of any relevant internal EV status change;
- in case of any EV error after processing of the last EVSE response message with “ResponseCode = OK”;
- in case of any EVSE error indicated by “ResponseCode = FAILED”, or “FAILED_xxx”;
- ACDP_SystemStatus may possibly be omitted in loop messages with “ResponseCode = OK”.

NOTE 3 The ACDP_SystemStatus message pair can also be applied in a synchronous manner in between the charging control V2G messages.

For the usage of ACDP_SystemStatus in error cases see 8.3.4.7.8.5 and 8.3.4.7.8.7.

8.3.4.7.8.2 ACDP_SystemStatusReq

[V2G20-4010] The EVCC and the SECC shall implement the message elements as defined in Figure 90 and Table 87.

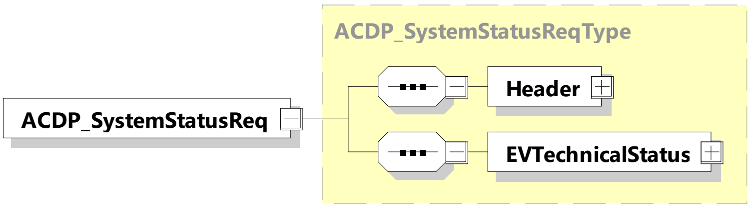

Figure 90 — Schema diagram – ACDP_SystemStatusReq

The elements of this message are used according to Table 87.

Table 87 — Semantics and type definition for ACDP_SystemStatusReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVTechnicalStatus	complexType: EVTechnicalStatusType refer to 8.3.5.7.1	This element provides detailed EV technical status

8.3.4.7.8.3 ACDP_SystemStatusRes

[V2G20-4012] The EVCC and the SECC shall implement the message elements as defined in Figure 91 and Table 88.

For ACDP charging method the EV positions itself correctly relative to the position of the ACDP. Before the ACDP can be activated the EV becomes immobilized. At the end of the charging process the EV is kept immobilized as long as the ACDP is not at its home position. The driver can control the immobilization request by the EV hand brake or any other technical means. On the other hand, the driver can end the charging process at any time by releasing the hand brake. In this case the mobilization of the EV waits until the ACDP is in home position.

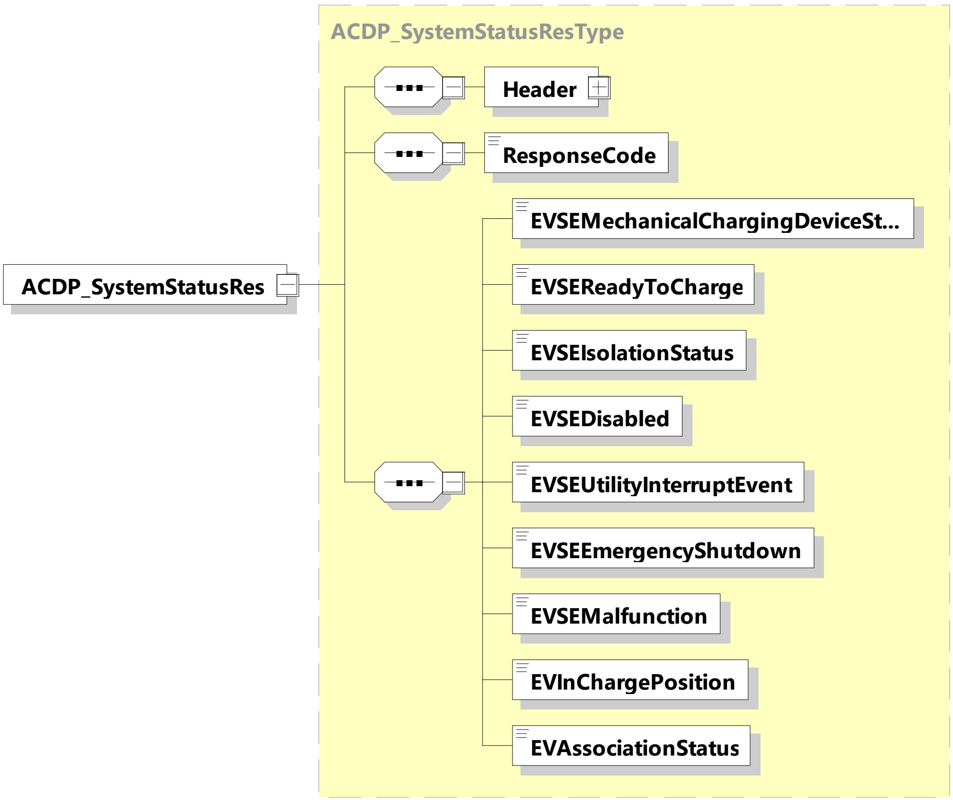

Figure 91 — Schema diagram – ACDP_SystemStatusRes

The elements of this message are used according to Table 88.

Table 88 — Semantics and type definition for ACDP_SystemStatusRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
EVSEMechanicalChargingDeviceStatus	simpleType: mechanicalChargingDeviceStatusType refer to Annex A for the type definition	This element provides the information if the ACDP status of the EVSE is in home, or end position or if it is in a moving state.
EVSEReadyToCharge	simpleType: xs:boolean	Element signalizes when the EVSE is READY or NOT READY to charge. This comprises all required EVSE sub-systems.
EVSEIsolationStatus	simpleType: isolationStatusType refer to Annex A for the type definition	This element provides the information about the isolation status of the EVSE. Possible states are: invalid, safe, warning, fault.

Element Name	Type	Semantics
EVSEDisabled	simpleType: xs:boolean	This element indicates when the EVSE is disabled by the operator or for maintenance. Possible states are: true, false.
EVSEUtilityInterruptEvent	simpleType: xs:boolean	This element indicates a utility power interrupt event. The EVSE control and communication functions may be backed up by an UPS.
EVSEEmergencyShutdown	simpleType: xs:boolean	This element indicates a charging system emergency shutdown or "E-Stop" button pressed at EVSE.
EVSEMalfunction	simpleType: xs:boolean	This element indicates a non-recoverable EVSE fault has occurred (e.g. breaker failure, ...).
EVInChargePosition	simpleType: xs:boolean	Option, only when position detection is available: This element indicates whether the EV is positioned within allowed tolerances to the ACDP.
EVAssociationStatus	simpleType: xs:boolean	Option, only when PPD is available: Indicates whether EV and EVSE are associated correctly.

NOTE 1 For ACDP it is recommended to provide the optional parameters to enable diagnoses of the charge process.

8.3.4.7.8.4 ACDP_SystemStatus EVCC requirements

[V2G20-4013] In case of an EV error the EVCC shall send a ACDP_SystemStatusReq message with Parameter "EV_Status_ReadyToCharge = FALSE" and all EV parameters representing the current EV (error) status and the EVCC shall execute normal shut down of the charging process according to Figure 92.

[V2G20-4014] In case [V2G20-4013] applies and the EVCC receives a ACDP_SystemStatusRes with parameter ResponseCode = OK the EVCC may process the received EVSE status parameters and shall continue to execute the normal shutdown process.

[V2G20-4015] The EVCC shall ignore to process the parameters provided by the ACDP_SystemStatusRes message when the "ResponseCode = FAILED".

8.3.4.7.8.5 Usage of ACDP_SystemStatusReq/Res on EVCC error

Figure 92 depicts an example EV error case during the processing of the DC_ChargeLoop message pair while charging. In this example the EVCC will execute the normal shutdown procedure as defined in IEC 61851-23:2014, Annex CC.3.3 starting with ramping down the target current to zero amp in a ramp down phase. After the present current is zero the EVCC will issue the PowerDeliveryReq with 'ChargeProgress' equal to 'STOP'. At this time the value of the ACDP_SystemStatus element EVImmobilizationRequest will be True, meaning the driver did not initiate to stop the charging process by releasing the hand brake. In that way the ACDP_SystemStatus will provide concrete information about the root cause of a normal shut down process.

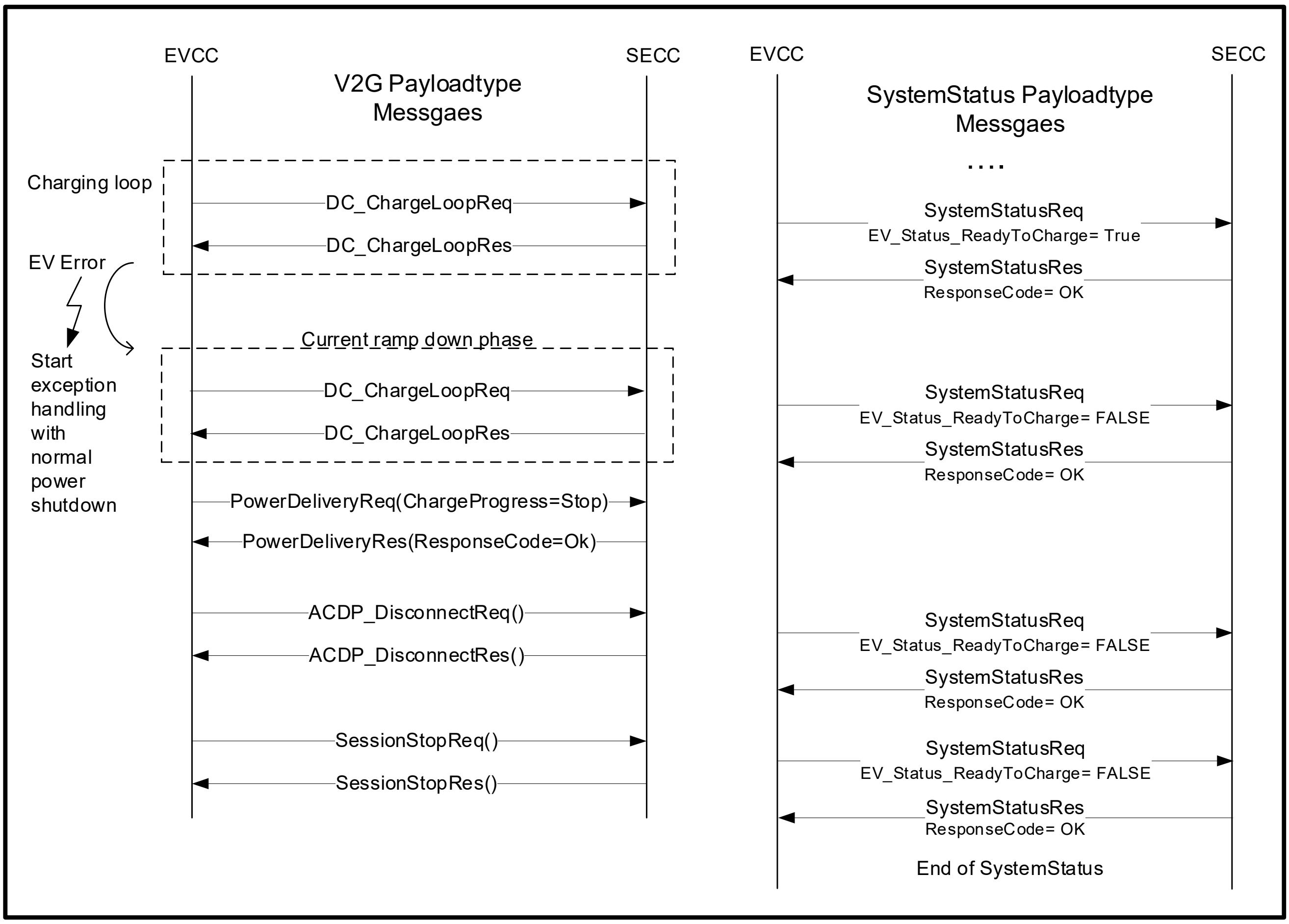

Figure 92 — ACDP exception handling example on EVCC error

In this sample EV initiated error case is due to RESS overtemperature therefore the

EVTechnicalStatus of the ACDP_SystemStatusReq will show the following contents:

- EVReadyToCharge = False;
- EVImmobilizationRequest = True;
- EVWLANStrength = - 66 dBm;
- EVCPStatus = C;
- EVSOC = 74 %;
- EVErrorCode = 4 (RESS overtemperature);
- EVTimeout = False.

The ACDP_SystemStatusRes will provide the following contents:

- ResponseCode = OK;
- EVSEMechanicalChargingDeviceStatus = EndPosition;
- EVSEReadyToCharge = True;
- EVSEIsolationStatus = Valid;
- EVSEDisabled = False;
- EVSEUtilityInterruptEvent = False;

– EVSEEmergencyShutdown = False;
– EVSEMalfunction = False;
– EVInChargePosition = True;
– EVAssociationStatus = True.

In the usual case of a normal shut down due to the driver's-initiated end of the charging by releasing the hand brake the EVTechnicalStatus will look like:

– EVReadyToCharge = False;
– EVImmobilizationRequest = False;
– EVWLANStrength = - 66 dBm;
– EVCPStatus = C;
– EVSOC = 74 %;
– EVErrorCode = 0;
– EVTimeout = False.

In this case the status of the element EVImmobilizationRequest = False indicates that the release of the hand brake was the cause to end the charging process.

8.3.4.7.8.6 ACDP_SystemStatus EVSE requirements

[V2G20-4016] After receiving the SessionSetupRes with parameter "ResponseCode = OK" the EVCC shall start to send ACDP_SystemStatusReq messages.

[V2G20-4017] The SECC shall respond to ACDP_SystemStatusReq message with "ResponseCode = OK" when it is able to represent the current EVSE status with the assigned parameters.

[V2G20-4018] The SECC shall respond to ACDP_SystemStatusReq message with "ResponseCode = FAILED" when it is not able to represent the current EVSE status with the assigned parameters.

[V2G20-4019] In case the SECC receives a ACDP_SystemStatusReq message with Parameter "EV_Status_ReadyToCharge = FALSE" it shall respond with ACDP_SystemStatusRes with "ResponseCode = OK" and all other EVSE ACDP_SystemStatus parameters representing the current EVSE status.

The timeout for System_Status_EndTimer is defined in Table 89.

NOTE The System_Status service is stopped after a SessionStopRes with parameter ResposeCode = OK was received.

Table 89 — System_Status service timer timeout value

Element Name	Value[S]	Semantics
System_Status_EndTimer	System_Status_End_Timeout	Duration of system status service
System_Status_End_Timeout	60	Timeout value of system status duration

8.3.4.7.8.7 Usage of ACDP_SystemStatusReq/Res on EVSE error

Figure 93 depicts an example EVSE error case during the processing of the DC_ChargeLoop message pair. The SECC will respond with DC_ChargeLoopRes with ResponseCode = FAILED to initiate the EV exception handling by sending a ACDP_DisconnectReq. In parallel the EV will send ACDP_SystemStatusReq in order to receive diagnostic information of the EVSE error cause.

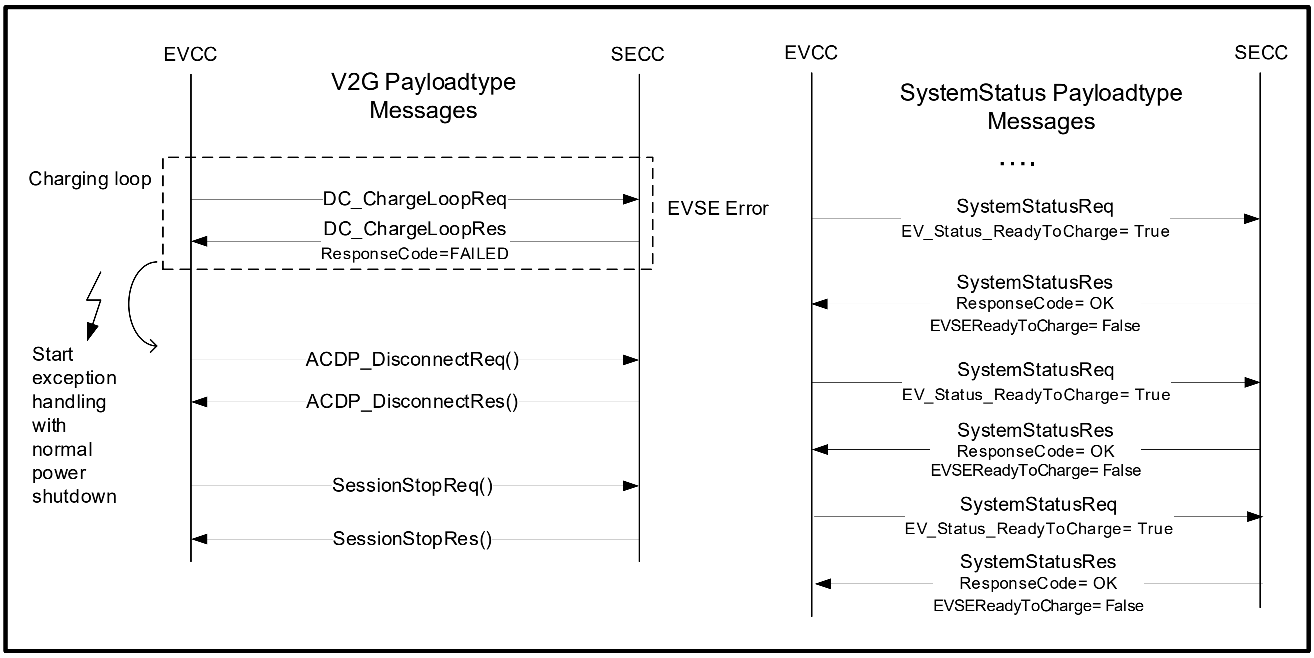

Figure 93 — ACDP Error handling on EVSE error

8.3.4.8 VAS messages

8.3.4.8.1 Parking Status service

8.3.4.8.1.1 VehicleCheckInReq/Res

8.3.4.8.1.1.1 VehicleCheckInReq/Res handling

This message is used by a parking status service for approaching EV to get information about an EVSE. A particular application of this service is an automated valid parking system (AVPS). The main purpose of EV check in message is that EV gets information of the EV target position for optical auto parking. The reference point marked by parking guideline is recognized by optical sensor. Target offset, 3 dimensional distances between pad target point and reference point is provided by this message response. The EV will be able to know the target position by adding the target offset to the reference point. When EVs complete parking to the target space, the EVCC informs the SECC and will exit parking status.

NOTE These messages are part of the parking status service.

Further description about parking status service is provided in 8.4.3.2.7.

8.3.4.8.1.1.2 VehicleCheckInReq

[V2G20-1298] The EVCC and the SECC shall implement the message elements as defined in Table 90 and Figure 94.

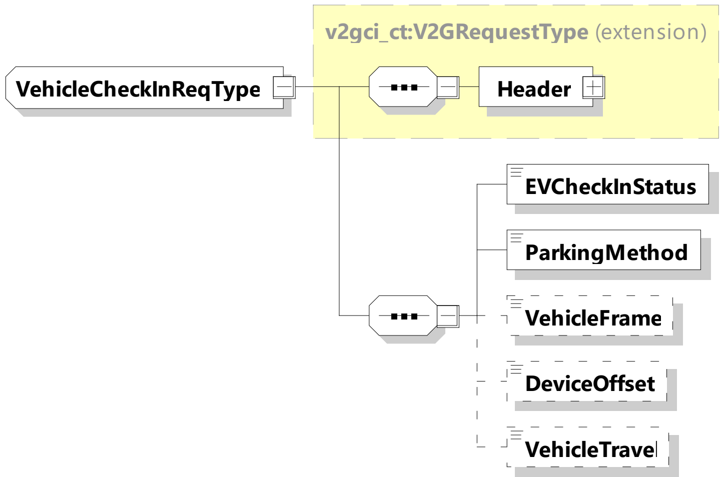

Figure 94 — Schema diagram – VehicleCheckInReq

The elements of this message are used according to Table 90.

Table 90 — Semantics and type definition for VehicleCheckInReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVCheckInStatus	simpleType: EVCheckInStatusType enumeration refer to Annex A for the type definition	Set "check in", "processing" or "completed" This element is used to inform to the SECC the check in status.
ParkingMethod	simpleType: parkingMethodType enumeration refer to Annex A for the type definition	Set "auto parking" or "manual" This element is used to inform to the SECC the parking method.
VehicleFrame	simpleType: xs:short	Optional: This element provides the information about 2 dimensional EV enveloping square, given by (VehicleWidth, VehicleLength).
DeviceOffset	simpleType: xs:short	Optional: This element provides information of EV WPT device offset from EV frame, given by (OffsetWidth, OffsetLength).
VehicleTravel	simpleType: xs:short	Optional: This element provides the distance information between WPT parking device and EV device measured by EV, given by (TravelX, TravelY).

Actually, positioning control in parking only requires information about device locations, such as "DeviceOffset" and "DeviceLocation" (see 8.3.4.8.1.1.3). The information about parking lot and EV outline, such as "VehicleFrame" and "ParkingSpace" will have a chance to be applied to the EV geometry and direction check within a parking lot in some cases.

8.3.4.8.1.1.3 VehicleCheckInRes

[V2G20-1299] The EVCC and the SECC shall implement the message elements as defined in Table 91 and Figure 95.

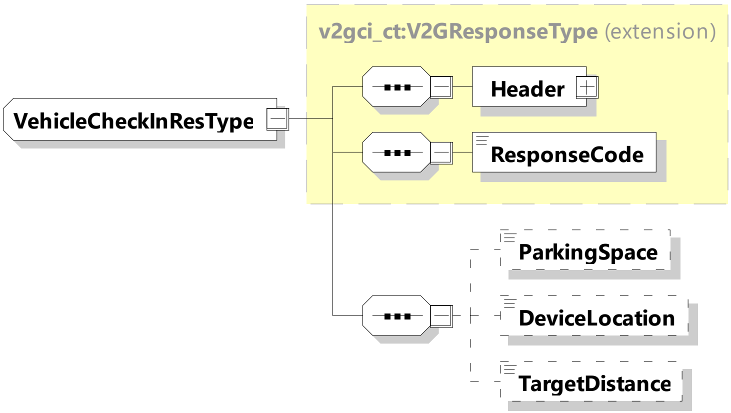

Figure 95 — Schema diagram – VehicleCheckInRes

The elements of this message are used according to Table 91.

Table 91 — Semantics and type definition for VehicleCheckInRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
ParkingSpace	simpleType: xs:short	Optional: This element provides the information about 2 dimensional parking space, given by (ParkingWidth, ParkingLength).
DeviceLocation	simpleType: xs:short	Optional: This element provides information of parking WPT device location in parking space, given by (LocationWidth, LocationLength).
TargetDistance	simpleType: xs:short	Optional: This element provides the distance information between WPT parking device and EV device measured by EVSE, given by (DistanceX, DistanceY).

8.3.4.8.1.2 VehicleCheckOutReq/Res

8.3.4.8.1.2.1 VehicleCheckOutReq/Res handling

This message is used by the EVCC to inform the SECC about the departure of the EV. The definition of a charging session includes EV departure, so it is necessary for the SECC to know it. This message is applied to parking status services.

NOTE These messages are part of the parking status service.

8.3.4.8.1.2.2 VehicleCheckOutReq

[V2G20-1300] The EVCC and the SECC shall implement the message elements as defined in Table 92 and Figure 96.

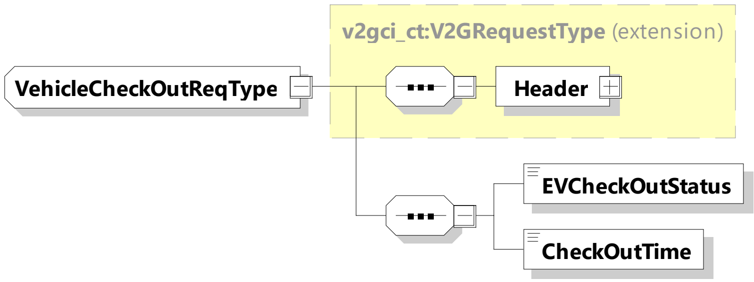

Figure 96 — Schema diagram – VehicleCheckOutReq

The elements of this message are used according to Table 92.

Table 92 — Semantics and type definition for VehicleCheckOutReq

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
EVCheckOutStatus	simpleType: EVCheckOutStatusType enumeration refer to Annex A for the type definition	Set "check out", "processing" or "completed" This element is used to inform to the SECC the CheckOut status.
CheckOutTime	simpleType: xs:unsignedLong	This element is used to inform to the SECC about the planned/estimated/expected departure time of the EV expressed as an absolute time stamp in SECC Time [in seconds].

8.3.4.8.1.2.3 VehicleCheckOutRes

[V2G20-1301] The EVCC and the SECC shall implement the message elements as defined in Table 93 and Figure 97.

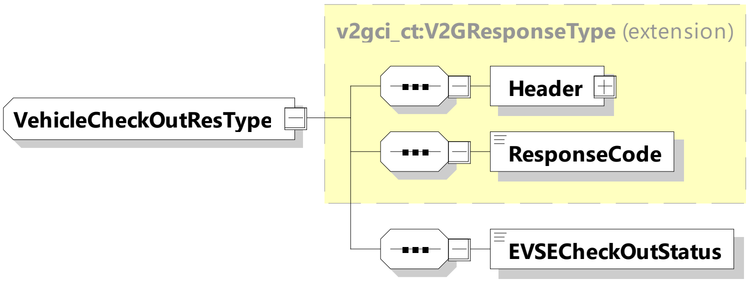

Figure 97 — Schema diagram – VehicleCheckOutRes

Table 93 — Semantics and type definition for VehicleCheckOutRes

Element Name	Type	Semantics
Header	complexType: MessageHeaderType refer to 8.3.3	Contains general information, used for all messages.
ResponseCode	simpleType: responseCodeType enumeration refer to Annex A for the type definition	ResponseCode indicating the acknowledgment status of any of the V2G messages received by the SECC.
EVSECheckOutStatus	simpleType: evseCheckOutStatusType enumeration refer to Annex A for the type definition	Set "scheduled" or "completed". This element is used to inform to the EVCC the CheckOut status.

8.3.5 Complex data types

8.3.5.1 Overview

This subclause defines complex data types (complexType), which are used in the messages. Complex data types are composed of several elements which themselves are based on simple data types.

NOTE Types starting with xs: are defined by W3C XML Schema 2 documents.

8.3.5.2 Physical Values

[V2G20-1813] The absolute value of the element defining the maximum shall always be greater than the absolute value defining the matching minimum, e.g. EVMaximumChargePower and EVMinimumChargeCurrent. This relationship shall also be maintained if the values of an element pair get communicated in two different messages.

[V2G20-1204] For all message elements of type RationalNumberType the SECC and the EVCC shall apply the value range and unit definition as according to Table 94.

Table 94 — Value range and unit definition for message elements using RationalNumberType

Element Name	Physical unit
BatteryEnergyCapacity	Wh
EVEnergyRequest	Wh
EVMaximumChargeCurrent	A

Element Name	Physical unit
EVMaximumChargePower	W
EVMaximumCurrent	A
EVMaximumCurrentLimit	A
EVMaximumDischargeCurrent	A
EVMaximumDischargePower	W
EVMaximumEnergyRequest	Wh
EVMaximumPowerLimit	W
EVMaximumVoltage	V
EVMinimumChargeCurrent	A
EVMinimumCurrent	A
EVMinimumDischargeCurrent	A
EVMinimumEnergyRequest	Wh
EVPresentActivePower	W
EVPresentReactivePower[_L2,_L3]	W
EVPresentVoltage	Volt
EVSEMaximumChargeCurrent	A
EVSEMaximumChargePower	W
EVSEMaximumCurrent	A
EVSEMaximumCurrentLimit	A
EVSEMaximumDischargeCurrent	A
EVSEMaximumDischargePower	W
EVSEMaximumPowerLimit	W
EVSEMinimumChargePower	W
EVSEMinimumCurrentLimit	A
EVSEMinimumDischargePower	W
EVSENominalFrequency	Hz
EVSENominalVoltage	V
EVSEPresentActivePower [_L2,_L3]	W
EVSEPresentCurrent	A
EVSEPresentVoltage	V
EVSETargetActivePower[_L2,_L3]	W
EVSETargetFrequency	Hz
EVSETargetReactivePower[_L2,_L3]	VA
EVTargetCurrent	A
EVTargetEnergyRequest	Wh
EVTargetVoltage	V
Power	W

Element Name	Physical unit
PowerTolerance	W
RemainingTimeToFullSOC	s

NOTE 1 The maximum and minimum transmittable values are calculated using the boundaries of the RationalNumberType. They are given through the signed 16-bit Integer (Range: -32 768 to +32 767) of the value and the exponent.

NOTE 2 Values from -32 768 – +32 767 can be changed till the last digit, but when the exponent is used the changeable amount changes. For example if an exponent of 1 is used the changes can only be made until the 10 digit range or if the exponent of 2 is used the number can only be changed till 100 digit range.

NOTE 3 The transmittable range is not the range of allowed values for the used system. It is important that these ranges are implemented according to the limits of the implementation and the corresponding IEC documents, to follow safety regulations.

8.3.5.2.1 Common rules for physical values

[V2G20-1034] For all applications a positive value for elements with the physical unit "W", "Wh" and "A" shall represent a power transfer from the EVSE to the EV (charging mode).

[V2G20-1035] Negative values shall be used to define energy transfer from the EV to the EVSE (discharging mode).

[V2G20-1783] For all applications in AC power transfer the elements with the physical unit:

- "V" and "A" shall be interpreted as root mean square values;
- "V" shall always be measured between one phase and neutral, unless stated otherwise;
- "W" shall be based on the assumption of a purely resistive AC circuit with a power factor of 1.0;
- for the "VA reactive" unit positive values shall represent inductive and negative values shall represent capacitive reactive power.

Equipment with other characteristics shall apply proper adjustments during related value calculations.

In the case of AC (dis)charging the EVSETargetActivePower and EVSETargetReactive elements serve two purposes. In dynamic control mode they are the normal means of operation while in Scheduled control mode they should only be used as a technique to guarantee that the mandatory local technical requirements from the grid code or urgent local requirements can be satisfied.

[V2G20-1823] EVSETargetActivePower and EVSETargetReactivePower shall always be based on EVSENominalVoltage.

NOTE From the perspective of the EV the EVSETargetActivePower and EVSETargetReactivePower elements therefore primarily communicate target setpoints for the active and reactive currents and the resulting phi angle ("power factor") as the grids voltage can and will be fluctuating, even during the delay resulting from the communication processing.

[V2G20-1824] For the EVSETargetReactivePower positive values shall represent inductive and negative values shall represent capacitive reactive power.

[V2G20-1825] The EV shall adjust its energy transfer behavior to the new setpoints, that have been communicated by the EVSETargetActivePower and/or the EVSETargetReactive elements, as fast as technically feasible.

8.3.5.2.2 Rules related to asymmetric polyphaser values

In order to support asymmetric polyphase energy transfers many elements which are related to power in the context of AC messages are provided in sets with phase specific variants. The following requirements apply to all such element sets unless explicitly stated otherwise in different requirements.

[V2G20-1814] For AC power phase specific element sets the line names L1, L2 and L3 shall identify the associated pins on the charging connector (e.g. as defined in IEC 62196).

NOTE This document only controls the energy exchange between the EV and the EVSE. In that context L1, L2 and L3 are clearly defined by the connector. A mapping to the associated lines at the grid connection point is only possible in the context of a specific installation within the premises and therefore out of the scope for this document.

[V2G20-1815] In the context of AC services with the connector choice SinglePhase only the power phase specific base elements (e.g. EVMaximumChargePower) shall be used. Elements with the suffix "_L2" and "_L3" shall not be used.

[V2G20-1816] In the context of AC services with the connector choice ThreePhase power phase specific elements with the suffix "_L2" and "_L3" shall be allowed.

[V2G20-1817] In the context of AC services with the connector choice ThreePhase the power phase specific base elements (e.g. EVMaximumChargePower) shall represent the sum of all three lines if neither the element sets peer with the suffix "_L2" nor "_L3" is present in the same message. If a sum value is communicated an even distribution across all three power phases shall be applied.

[V2G20-1818] In the context of AC services with the connector choice ThreePhase the power phase specific base elements (e.g. EVMaximumChargePower) shall identify the value associated with the L1 line if either the element sets peer with the suffix "_L2" or "_L3" is present in the same message.

[V2G20-1819] All requirements which apply to the base element (e.g. EVMaximumChargePower) of a power phase specific element set shall also apply to the peers with the suffix "_L2" or "_L3" in an equivalent way.

8.3.5.2.3 Rules related to EV energy request values

The EV energy request parameters defined below are used by the EV to inform the EVSE of its energy demands and limitations. They are instantaneous values that will change over time during the energy transfer loops. They are calculated, updated and sent during the energy transfer loop by the EV and they form the foundation for the DynamicControlMode.

[V2G20-1217] The EV shall ensure that at any time: EV Minimum Energy Level ≤ EV Target Energy Level ≤ EV Maximum Energy Level and therefore the resulting energy requests satisfy the following relationship: EVMinimumEnergyRequest ≤ EVTargetEnergyRequest ≤ EVMaximumEnergyRequest.

[V2G20-2191] In dynamic control mode the objective of the energy transfer plan shall be that at the end of the service session the EVTargetEnergyRequest will be less than or equal to zero.

NOTE 1 This means that the SECC is allowed to reach a target range defined between EV maximum energy level and EV target energy level.

NOTE 2 When the EVMaximumEnergyRequest value is negative then an immediate discharge is requested by the EV.

EVMinimumEnergyRequest is used to indicate the minimum amount of energy requested by the EV at any given time during the energy transfer loop. It is the difference between the minimum level of energy requested by the EV to be transferred as soon as possible and the present level of energy of the EV battery.

NOTE 3 When the EVMinimumEnergyRequest value is positive then an immediate charge is requested by the EV.

NOTE 4 When the EVMinimumEnergyRequest value is equal to zero or negative then charging can be delayed or discharging can be applied.

EVMaximumEnergyRequest is used to indicate the maximum energy level accepted by the EV at any given time during the energy transfer loop. It is the difference between the maximum level of energy accepted by the EV and the present level of energy of the EV battery.

EVTargetEnergyRequest is used to indicate the amount of energy requested by the EV before departure time. It is the difference between the amount of energy required by the EV at departure time and the present level of energy in the EV battery.

NOTE 5 When the EVMinimumEnergyRequest value is equal to zero or negative and the EVMaximumEnergyRequest value is equal to zero or positive then charging or discharging is possible.

Since the energy request values get constantly updated by the EV to the actual conditions this mechanism allows to compensate for auxiliary energy demand (e.g. due to cabin heating or cooling) that over time will drain the storage system. Such type of energy "leakage" would be reflected in a reduced EV present energy level, which then would be reflected by adjusted EV[*]EnergyRequest parameters. The SECC gets a chance to derive educated guesses on leakage behavior and can include such assumptions in its energy transfer strategy. On the other hand the EVCC should include a known leakage energy demand in its energy request description as early as possible to give the SECC a better foundation for planning its strategy.

8.3.5.3 Common

8.3.5.3.1 ServiceType

This type represents a tag for a specific service. It gives a short definition and identification of a specific service.

[V2G20-1308] The SECC and the EVCC shall implement this type as defined in Table 95 and Figure 98.

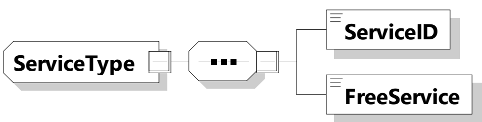

Figure 98 — Schema diagram – ServiceType

The elements of this message are used according to Table 95.

Table 95 — Semantics and type definition for ServiceType

Element Name	Type	Semantics
ServiceID	simpleType: xs:unsignedShort	Unique identifier of the service
FreeService	simpleType: xs:boolean	This element is used by the SECC to indicate if a service can be used by the EVCC free of charge or not. If FreeService is equal to true, the EV can use the offered service without an additional cost. If FreeService is equal to false, the service, if used by the EV, will be billed using the method negotiated during the authorization phase.

8.3.5.3.2 ServiceListType

[V2G20-1309] The SECC and the EVCC shall implement this type as defined in Table 96 and Figure 99.

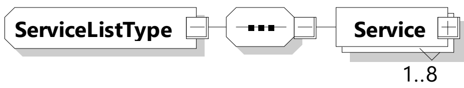

Figure 99 — Schema diagram – ServiceListType

The elements of this message are used according to Table 96.

Table 96 — Semantics and type definition for ServiceListType

Element Name	Type	Semantics
Service	complexType: serviceType refer to 8.3.5.3.1	Contains all information for identifying a service. The ServiceList includes at least one service and may include up to 8 services.

8.3.5.3.3 CertificateChainType

This data type stores the leaf certificate, and all certificates in the chain up to the root. The root certificate is not included in this data type. In the special (but unlikely) case, that a leaf certificate is directly signed by the root, the "SubCertificates" field is empty. In all other cases, this field contains an ordered list of sub-CA certificates to follow the trust-path from the leaf certificate up to the root.

[V2G20-1312] The SECC and the EVCC shall implement this type as defined in Table 97 and Figure 100.

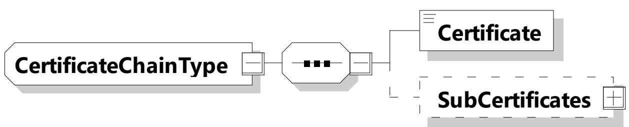

Figure 100 — Schema diagram – CertificateChainType

The elements of this message are used according to Table 97.

Table 97 — Semantics and type definition for CertificateChainType

Element Name	Type	Semantics
Certificate	simpleType: certificateType refer to Annex A for the type definition	A x.509v3 leaf certificate (the "client" certificate). The certificate is DER encoded. Refer to ITU-T X.690 for details of DER encoding.
SubCertificates	complexType: SubCertificatesType refer to 8.3.5.3.6	Optional: The certificate chain containing all sub-certificates leading up to the CA certificate, but not including the root CA certificate

[V2G20-2683] If the certificate provided in element "Certificate" included in the parameter of "CertificateChainType" is not directly signed by a root CA certificate, element "SubCertificates" shall be included in the parameter of "CertificateChainType".

NOTE If the certificate included in element "Certificate" is directly signed by a root CA certificate, element "SubCertificates" is not included in the parameter of "CertificateChainType".

8.3.5.3.4 SignedCertificateChainType

This data type stores the leaf certificate, and all certificates in the chain up to the root. The root certificate is not included in this data type. In the special (but unlikely) case, that a leaf certificate is directly signed by the root, the "SubCertificates" field is empty. In all other cases, this field contains an ordered list of sub-CA-certificates to follow the trust-path from the leaf certificate up to the root. In addition to 8.3.5.3.3, this element may be signed, since an Id attribute is included.

[V2G20-1312] The SECC and the EVCC shall implement this type as defined in Table 98 and Figure 101.

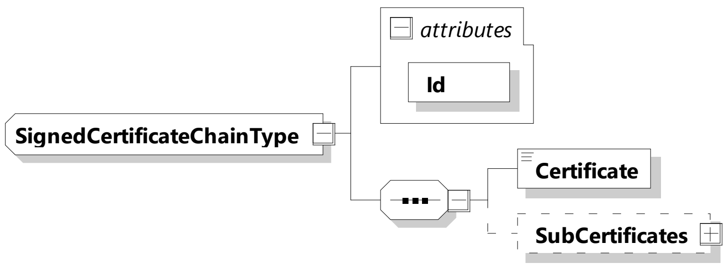

Figure 101 — Schema diagram – SignedCertificateChainType

The elements of this message are used according to Table 98.

Table 98 — Semantics and type definition for SignedCertificateChainType

Element Name	Type	Semantics
Id	Attribute	Allows inclusion of this element into a signature.
Certificate	simpleType: certificateType refer to Annex A for the type definition	A x.509v3 leaf certificate (the "client" certificate). The certificate is DER encoded. Refer to ITU-T X690 for details of DER encoding.

SubCertificates	complexType: SubCertificatesType refer to 8.3.5.3.6	Optional: The chain with all subcertificates to the root certificate (not including root certificate).

8.3.5.3.5 ContractCertificateChainType

[V2G20-1859] The SECC and the EVCC shall implement this type as defined in Figure 102 and Table 99.

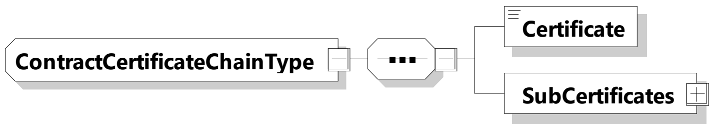

Figure 102 — Schema diagram – ContractCertificateChainType

The elements of this message are used according to Table 99.

Table 99 — Semantics and type definition for ContractCertificateChainType

Element Name	Type	Semantics
Certificate	simpleType: certificateType refer to Annex A for the type definition	An x.509v3 certificate (the "client" certificate). The certificate is DER encoded.
SubCertificates	complexType: SubCertificatesType refer to 8.3.5.3.6	The chain with all subcertificates to the root-certificate (not including root certificate).

8.3.5.3.6 SubCertificatesType

[V2G20-1336] The SECC and the EVCC shall implement this type as defined in Table 100 and Figure 103.

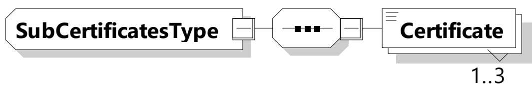

Figure 103 — Schema diagram – SubCertificatesType

The elements of this message are used according to Table 100.

Table 100 — Semantics and type definition for SubCertificatesType

Element Name	Type	Semantics
Certificate	simpleType: certificateType refer to Annex A for the type definition	An x.509v3 certificate. The certificate is DER encoded. Number of occurrences is limited to three.

8.3.5.3.7 MeterInfoType

[V2G20-1313] The SECC and the EVCC shall implement this type as defined in Table 101 and Figure 104.

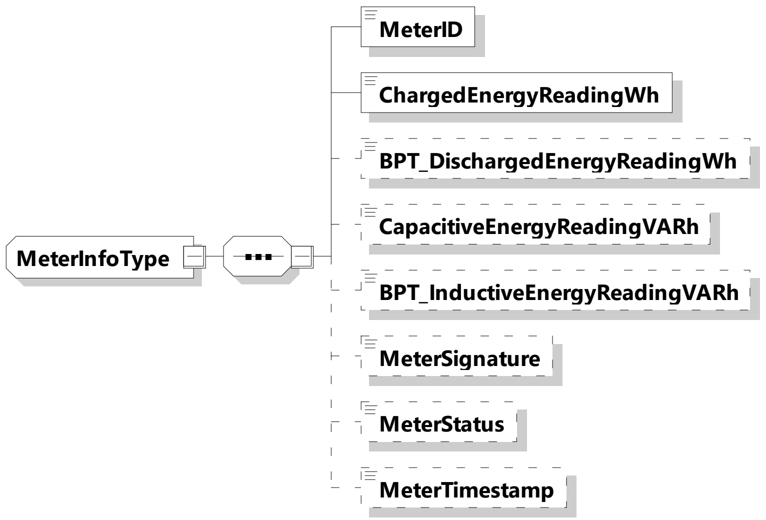

Figure 104 — Schema diagram – MeterInfoType

The elements of this message are used according to Table 101.

Table 101 — Semantics and type definition for MeterInfoType

Element Name	Type	Semantics
MeterID	simpleType: meterIDType refer to Annex A for the type definition	Unique identifier of the EVSE meter.
ChargedEnergyReadingWh	simpleType: xs:unsignedLong	The meter reading in Watthours that counts the energy flow from the EVSE to the EV, which is related to the normal charging process. The value is not guaranteed to start at zero during every energy transfer session.
BPT_DischargedEnergyReadingWh	simpleType: xs:unsignedLong	Optional for BPT, not applicable for other energy transfer services: The meter reading in Watthours that counts the energy flow from the EV to the EVSE, which is related to the discharging process. The value is not guaranteed to start at zero during every energy transfer session.
CapacitiveEnergyReadingVARh	simpleType: xs:unsignedLong	Optional: The meter reading in volt-ampere-reactive-hour (varh) which reflects the amount of capacitive reactive energy exchanged with the grid. The value is not guaranteed to start at zero during every energy transfer session.

Element Name	Type	Semantics
BPT_InductiveEnergyReading VARh	simpleType: xs:unsignedLong	Optional for BPT, not applicable for other energy transfer services: The meter reading in volt-ampere-reactive-hour (varh) which reflects the amount of inductive reactive energy exchanged with the grid. The value is not guaranteed to start at zero during every energy transfer session.
MeterSignature	simpleType: meterSignatureType base64Binary (max length: 64) refer to Annex A for the type definition	Optional: Signature of the meter reading This signature is generated by the EVSE meter. It is not verified at the EVCC. It might be used by a SA system for billing purposes if local regulations on metering permit it
MeterStatus	simpleType: meterStatusType short refer to Annex A for the type definition	Optional: Present status of the meter. The definition of the content of the MeterStatus is not in the scope of this document. The content may be defined by the EVSE operator or utility.
MeterTimeStamp	simpleType xs:unsignedLong	Optional: Timestamp of the time when the last element of the provided meter readings has been registered. The value is encoded at microseconds resolution in SECC time, a concept that is defined in 8.3.3.3. For use cases that require metering SECC time would be aligned with UTC.

[V2G20-1415] As exception to [V2G20-1034] and [V2G20-1035], the values of the element DischargedEnergyReadingWh shall always be positive numbers.

Due to legal metrology requirements in some countries it cannot be guaranteed that the elements ChargedEnergyReadingWh, DischargedEnergyReadingWh, MeterReadingVARhCapacitive and MeterReadingVARhInductive will always start at zero when a new charging session begins. Therefore, it is necessary to compute deltas in order to derive energy amounts for a particular charging session. Requirement [V2G20-1833] guarantees that a MeterInfoType element is transmitted prior to the beginning of the actual energy flow. This is the base line which allows the EV to compute proper deltas for an individual charging session.

NOTE The definition of value ranges for legal metrology is out of the scope for this document. As general guidance one can assume that values for elements ChargedEnergyReadingWh, DischargedEnergyReadingWh, MeterReadingVARhCapacitive and MeterReadingVARhInductive, at a household level, commonly can reach up to 999,999,999 (Wh, VARh). But at industrial level (e.g. for electric truck charging) even higher value ranges are likely. Implementations, therefore, can support the full range of the underlying unsignedLong value type.

8.3.5.3.8 RationalNumberType

[V2G20-1314] The SECC and the EVCC shall implement this type as defined in Table 102 and Figure 105.

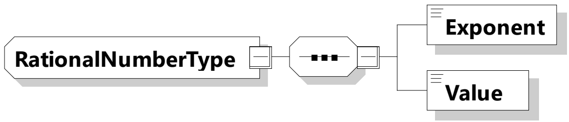

Figure 105 — Schema diagram – RationalNumberType

The elements of this message are used according to Table 102.

Table 102 — Semantics and type definition for RationalNumberType

Element Name	Type	Semantics
Exponent (exp)	simpleType: ExponentType xs:byte	The exponent to base 10 (dec). The final physical value is determined by: $x_{\text{RationalNumberType}} = y \times 10^{\text{exp}}$
Value (y)	simpleType xs:short	Value which shall be multiplied

NOTE For more information on the aspect of measurement tolerances of RationalNumberType elements see Annex I.

8.3.5.3.9 EVPowerProfileType

[V2G20-1316] The SECC and the EVCC shall implement this type as defined in Table 103 and Figure 106.

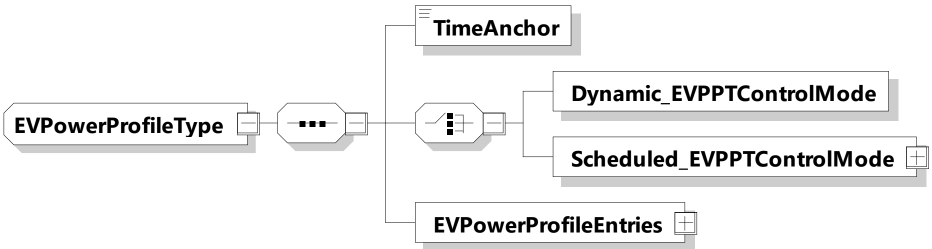

Figure 106 — Schema diagram – EVPowerProfileType

The elements of this message are used according to Table 103.

Table 103 — Semantics and type definition for EVPowerProfileType

Element Name	Type	Semantics
TimeAnchor	simpleType: xs:unsignedLong	The time that defines the starting point for the first EVPowerProfileEntry in this power profile. The value is encoded at microseconds resolution in SECC time, a concept that is defined in 8.3.3.3. Commonly the TimeAnchor will define a time that either is the present or some point in the near future.

Element Name	Type	Semantics
Dynamic_EVPPTControlMode	complexType: Dynamic_EVPPTControlModeType refer to 8.3.5.3.11	Contains all elements that are only required in case the control mode "Dynamic" is chosen.
Scheduled_EVPPTControlMode	complexType: Scheduled_EVPPTControlModeType refer to 8.3.5.3.11	Contains all elements that are only required in case the control mode "Scheduled" is chosen.
EVPowerProfileEntries	complexType: EVPowerProfileEntryListType refer to 8.3.5.3.10	Element used for encapsulating an individual power profile entry of the charge schedule. The number of EVPowerProfileEntry elements is limited to 2048.

[V2G20-1861] The TimeAnchor element shall defined the point in time when the first EVPowerProfileEntry element starts to be active.

[V2G20-1559] The power values of the EVPowerProfileEntry elements at no time shall exceed the corresponding hard limits of the power boundaries defined by the PowerSchedule and PowerDischargeSchedule of the last correctly negotiated ScheduleTuple which was selected by the EV during the related ScheduleExchangeReq/Res phase.

[V2G20-1010] The power values of the EVPowerProfileEntry elements shall represent the projected average target power levels.

NOTE 1 An average target power value allows the SECC to guess the required energy per EVPowerProfileEntry.

NOTE 2 As the EVPowerProfile can communicate charging and discharging phases at the same time the power values can be either positive or negative.

8.3.5.3.10 EVPowerProfileEntryListType

[V2G20-1867] The SECC and the EVCC shall implement this type as defined in Figure 107 and Table 104.

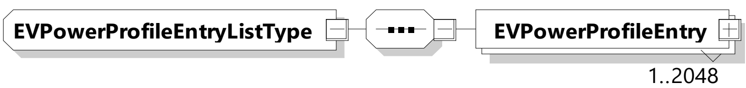

Figure 107 — Schema diagram – EVPowerProfileEntryListType

The elements of this message are used according to Table 104.

Table 104 — Semantics and type definition for EVPowerProfileEntryListType

Element Name	Type	Semantics
EVPowerProfileEntry	complexType: PowerScheduleEntryType refer to 8.3.5.3.20	List of EVPowerProfileEntry elements. The maximum amount of elements shall be 2 048.

8.3.5.3.11 Dynamic_EVPPTControlModeType

This type contains all elements of the EVPowerProfile that are only required in case the control mode "Dynamic" is chosen.

[V2G20-1875] The SECC and the EVCC shall implement this type as defined in Table 105 and Figure 108.

Figure 108 — Schema diagram – Dynamic_EVPPTControlModeType

Table 105 — Semantics and type definition for Dynamic_EVPPTControlModeType

Element Name	Type	Semantics
n/a	n/a	n/a

NOTE This parameter does not contain any elements.

8.3.5.3.12 Scheduled_EVPPTControlModeType

This type contains all elements of the EVPowerProfile that are only required in case the control mode "Scheduled" is chosen.

[V2G20-1782] The SECC and the EVCC shall implement this type as defined in Table 106 and Figure 109.

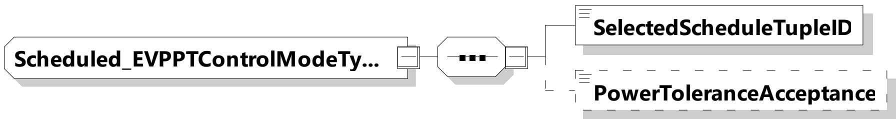

Figure 109 — Schema diagram – Scheduled_EVPPTControlModeType

The elements of this message are used according to Table 106.

Table 106 — Semantics and type definition for Scheduled_EVPPTControlModeType

Element Name	Type	Semantics
SelectedScheduleTupleID	simpleType: numericIDType refer to Annex A for the type definition	Unique identifier of the selectedScheduleTuple
PowerToleranceAcceptance	simpleType: powerToleranceAcceptanceType refer to Annex A for the type definition	Optional: This value is used to indicate if the EVCC acknowledges the PowerTolerance(s) sent by the SECC in ScheduleExchangeRes that was selected by the EVCC with the SelectedScheduleTupleID. This value can be set to "PowerToleranceConfirmed" for a positive reply or set to "PowerToleranceNotConfirmed" for a negative reply. This Element shall always be sent if a PowerTolerance has been provided by the SECC in the latest ScheduleExchangeRes.

8.3.5.3.13 Dynamic_SEReqControlModeType

This type contains all elements of the ScheduleExchangeReq message that are only required in case the control mode "Dynamic" is chosen.

[V2G20-1755] The SECC and the EVCC shall implement this type as defined in Table 107 and Figure 110.

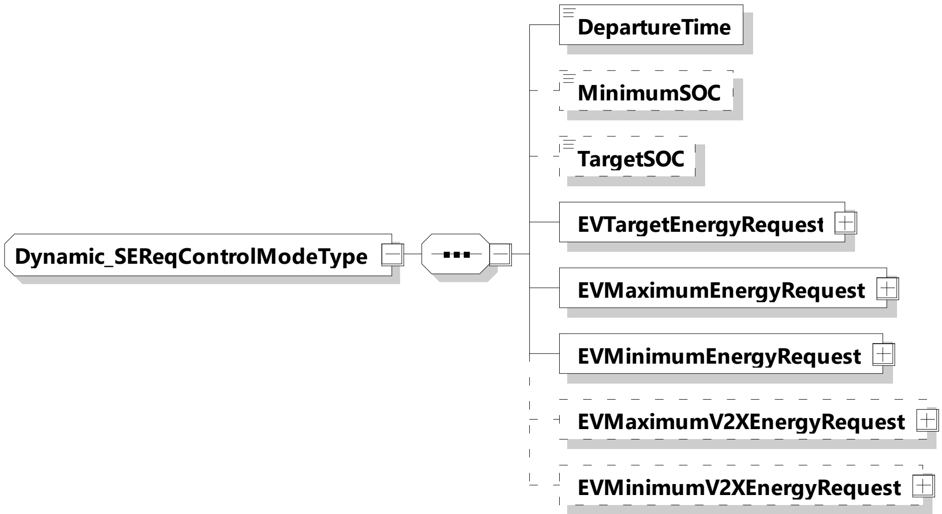

Figure 110 — Schema diagram – Dynamic_SEReqControlModeType

The elements of this message are used according to Table 107.

Table 107 — Semantics and type definition for Dynamic_SEReqControlModeType

Element Name	Type	Semantics
DepartureTime	simpleType: xs:unsignedInt	This element is used to indicate when the EV intends to finish the energy transfer process. The value is encoded in seconds since the TimeStamp of the message header (see [V2G20-2104]).
MinimumSOC	simpleType: percentValueType refer to Annex A for the type definition	Optional: Minimum State of Charge EV needs to keep throughout the charging session. Mandatory when service parameter MobilityNeedsMode was set to 2.
TargetSOC	simpleType: percentValueType refer to Annex A for the type definition	Optional: Target State of Charge of the EV Battery EV needs after charging. Mandatory when service parameter MobilityNeedsMode was set to 2.
EVTargetEnergyRequest	complexType: RationalNumberType refer to 8.3.5.3.8	The energy request of the EV it needs to fulfil the target SOC as specified by the owner.

Element Name	Type	Semantics
EVMaximumEnergyRequest	complexType: RationalNumberType refer to 8.3.5.3.8	Maximum acceptable energy level of the EV.
EVMinimumEnergyRequest	complexType: RationalNumberType refer to 8.3.5.3.8	The energy request of the EV it needs to fulfil the minimum SOC as specified by the owner. NOTE This value is not to be understood as a guaranteed minimal amount of energy that is to be consumed by or factured to the EV.
EVMaximumV2XEnergyRequest	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: Energy which may be charged until the PresentSOC has left the range dedicated for cycling activity. A negative value indicates that PresentSOC is above the V2X range.
EVMinimumV2XenergyRequest	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: Energy which needs to be charged until the PresentSOC enters the range dedicated for cycling activity. A positive value indicates that PresentSOC is below the V2X range.

NOTE If the EVCC sends the parameters EVMaximumV2XEnergyRequest and EVMinimumV2XEnergyRequest, it is indicating a preferred operational V2X range (i.e. for bidirectional cycling) and the SECC can attempt to stay within these limits as long as possible.

[V2G20-2103] Unless defined otherwise DepartureTime in all messages shall always be larger than zero.

[V2G20-2104] Unless defined otherwise DepartureTime in all messages shall express the offset in seconds from the point in time of sending this message, which is marked in the TimeStamp element of the common V2G message header.

[V2G20-2681] If EVMaximumV2XEnergyRequest and EVMinimumV2XEnergyRequest are sent by the EVCC, they always need to be sent together.

[V2G20-2682] If the EVCC provides EVMaximumV2XEnergyRequest and EVMinimumV2XEnergyRequest, EVMinimumEnergyRequest shall be less than or equal to EVMinimumV2XEnergyRequest, which shall be less than or equal to EVMaximumV2XEnergyRequest, which shall be less than or equal to EVMaximumEnergyRequest.

8.3.5.3.14 Scheduled_SEReqControlModeType

This type contains all elements of the ScheduleExchangeReq message that are only required in case the control mode Scheduled is chosen.

[V2G20-1753] The SECC and the EVCC shall implement this type as defined in Table 108 and Figure 111.

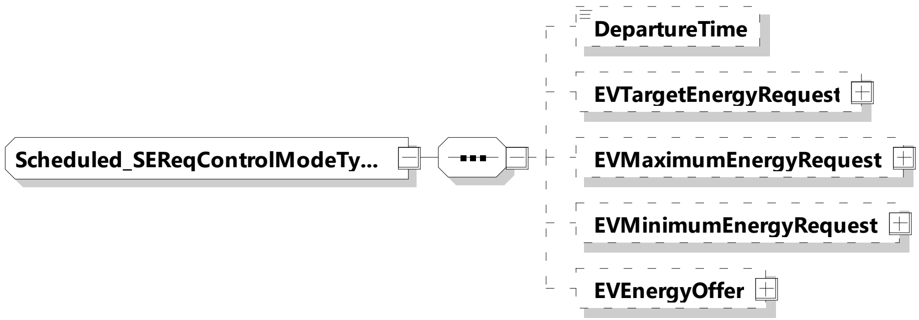

Figure 111 — Schema diagram – Scheduled_SEReqControlModeType

The elements of this message are used according to Table 108.

Table 108 — Semantics and type definition for scheduled_SEReqControlModeType

Element Name	Type	Semantics
DepartureTime	simpleType: xs:unsignedInt	Optional: This element is used to indicate when the EV intends to finish the energy transfer process. The value is encoded in seconds since the TimeStamp of the message header (see [V2G20-2104]).
EVTargetEnergyRequest	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: The energy request of the EV it needs to fulfil the target SOC as specified by the owner.
EVMaximumEnergyRequest	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: Maximum acceptable energy level of the EV.
EVMinimumEnergyRequest	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: The energy request of the EV it needs to fulfil the minimum SOC as specified by the owner.
EVEnergyOffer	complexType: EVEnergyOfferType refer to 8.3.5.3.41	Optional: Includes several tuples of schedules from the EV.

NOTE If the EVTargetEnergyRequest has a negative value and the EVCC communicates an EVEnergyOffer, it is to be set as the amount of Energy the EV offers in the EVEnergyOffer at a certain price per kWh at a certain power-over-time.

8.3.5.3.15 Dynamic_SEResControlModeType

This type contains all elements of the ScheduleExchangeRes message that are only required in case the control mode "Dynamic" is chosen.

[V2G20-1868] The SECC and the EVCC shall implement this type as defined in Table 109 and Figure 112.

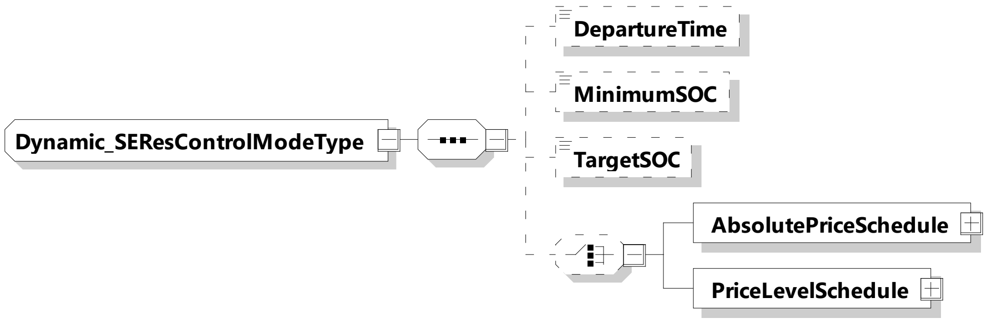

Figure 112 — Schema diagram – Dynamic_SEResControlModeType

The elements of this message are used according to Table 109.

Table 109 — Semantics and type definition for Dynamic_SEResControlModeType

Element Name	Type	Semantics
DepartureTime	simpleType: xs:unsignedInt	Optional: This element is used to indicate when the EV intends to finish the charging process Only used when service parameter MobilityNeedsMode was set to 2. The value is encoded in seconds since the TimeStamp of the message header (see [V2G20-2104]).
TargetSOC	simpleType: percentValueType refer to Annex A for the type definition	Optional: New TargetSOC value updated by the user from EVSE's side Only used when service parameter MobilityNeedsMode was set to 2.
MinimumSOC	simpleType: percentValueType refer to Annex A for the type definition	Optional: New MinimumSOC value updated by the user from EVSE's side Only used when service parameter MobilityNeedsMode was set to 2.
AbsolutePriceSchedule	complexType: AbsolutePriceScheduleType refer to 8.3.5.3.49	Optional: Absolute price schedule for this session.
PriceLevelSchedule	complexType: PriceLevelScheduleType refer to 8.3.5.3.62	Optional: PriceLevel schedule for this session.

[V2G20-1640] The SECC shall only send MinimumSOC that is less than or equal to TargetSOC.

[V2G20-2660] If the EVCC selected the service parameter "MobilityNeedsMode" set to 2 and the SECC received new values for departure time, target SOC or minimum SOC from a secondary actor, the SECC shall include those updated values in DepartureTime, TargetSOC or MinimumSOC, respectively.

[V2G20-1648] If the EVCC selected the service parameter MobilityNeedsMode set to 2 and the EVCC receives a ScheduleExchangeRes with DepartureTime, TargetSOC or MinimumSOC in Dynamic_SEResControlMode, and if the EVCC accepts this DepartureTime, TargetSOC or MinimumSOC as new Mobility Need information, the EVCC shall send updated DepartureTime, TargetSOC or MinimumSOC in the subsequent ChargeLoopReq.

[V2G20-1649] If the EVCC selected the service parameter MobilityNeedsMode set to 2 and the EVCC receives a ScheduleExchangeRes with DepartureTime, TargetSOC or MinimumSOC in Dynamic_SEResControlMode, and if the EV cannot accept this DepartureTime, TargetSOC or MinimumSOC, e.g. for internal reasons, the EVCC shall send different values in the subsequent ChargeLoopReq than the values in the ScheduleExchangeRes.

NOTE 1 In this case, the behavior of the SECC is not in scope.

NOTE 2 Both EVCC and SECC are expected to adapt the respective valid parameter and, if applicable, update their system, user interface, etc. with the information that is used for the session to ensure a high level of transparency and to avoid conflicting information to be displayed.

8.3.5.3.16 Scheduled_SEResControlModeType

[V2G20-1318] The SECC and the EVCC shall implement this type as defined in Table 110 and Figure 113.

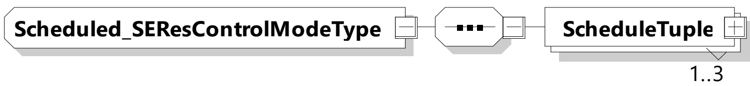

Figure 113 — Schema diagram – Scheduled_SEResControlModeType

The elements of this message are used according to Table 110.

Table 110 — Semantics and type definition for Scheduled_SEResControlModeType

Element Name	Type	Semantics
ScheduleTuple	complexType: ScheduleTupleType refer to 8.3.5.3.17	Includes several tuples of schedules from secondary actors. The number of ScheduleTuple elements is limited to 3. If this element includes discharging service, relevant parameter especially power in PowerSchedule will be used with negative value.

NOTE 1 The EVCC can implement a mechanism to compare different ScheduleTuple elements in order to optimize the charge schedule considering any given kind of cost.

[V2G20-297] The first ScheduleTuple element shall be defined as the default schedule.

[V2G20-298] If the EVCC is not capable of comparing different ScheduleTuple elements or comparison fails, the EVCC shall choose the default ScheduleTuple according to [V2G20-297].

NOTE 2 If the SECC (or a secondary actor) is interested in the energy offered by the EVCC, it can use the information provided in the EVEnergyOffer and integrate it into one of the provided ScheduleTuples.

NOTE 3 The schedule to which an EVCC refers to via the ScheduleTupleID in the PowerDeliveryReq, always originates from the SECC.

[V2G20-1835] The currency which is used within one ScheduleTupleType with one ScheduleTupleID in the charge and the discharge schedule shall be the same.

8.3.5.3.17 ScheduleTupleType

[V2G20-1319] The SECC and the EVCC shall implement this type as defined in Table 111 and Figure 114.

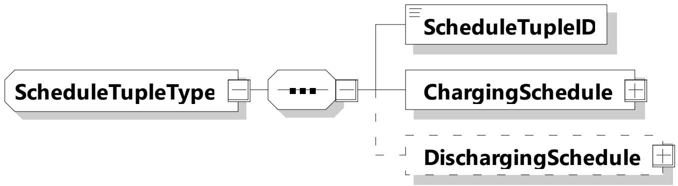

Figure 114 — Schema diagram – ScheduleTupleType

The elements of this message are used according to Table 111.

Table 111 — Semantics and type definition for ScheduleTupleType

Element Name	Type	Semantics
ScheduleTupleID	simpleType: numericIDType refer to Annex A for the type definition	Mandatory for scheduled control mode, not applicable for dynamic control mode: Unique identifier within an energy transfer session for a ScheduleTuple element. An SAID remains a unique identifier for one schedule throughout an energy transfer session.
ChargingSchedule	complexType: ChargingScheduleType refer to 8.3.5.3.40	Encapsulating element describing all relevant details for the charging schedule.
DischargingSchedule	complexType: ChargingScheduleType refer to 8.3.5.3.40	Optional: Encapsulating element describing all relevant details for the discharging schedule.

NOTE 1 A ScheduleTuple contains different elements, two schedules (ChargingSchedule, DischargingSchedule) which all can originate from different secondary actors and therefore can communicate contradicting incentives. While the EV is required to operate within the given physical limits it is free to choose which incentive it wants to follow when deciding on the planned EVPowerProfile.

[V2G20-773] The SECC shall use the values 1 to 255 for the parameter ScheduleTupleID.

[V2G20-1560] The ScheduleTupleID element shall be unique for each ScheduleTuple during the entire charging session.

[V2G20-1561] The SECC shall provide a PowerSchedule element based upon the limits of the local installation if no secondary actor provides a PowerSchedule.

[V2G20-1011] If a secondary actor provided a PowerSchedule, the SECC shall ensure that no power value within the PowerSchedule exceeds the matching EVSEMaximumChargePower limit.

NOTE 2 The EVCC assumes that the hard capability limits communicated in an EVSEMaximumChargePower[_L2,L3] value always will overrule any matching values provided within a PowerSchedule.

[V2G20-1012] If a secondary actor provided a PowerSchedule for discharging, the SECC shall ensure that no Power value within the PowerSchedule exceeds the matching EVSEMaximumDischargePower limit.

NOTE 3 The EVCC assumes that the hard capability limits communicated in an EVSEMaximumDischargePower[_L2,L3] value always will overrule any matching values provided within a PowerDischargeSchedule.

[V2G20-1562] The sum of the individual time intervals described in the PowerSchedule and PowerSchedule for Discharge (refer to 8.3.5.3.19) and PriceSchedule provided in the ScheduleExchangeRes message should match the period of time indicated by the EVCC in the message element DepartureTime of the ScheduleExchangeReq message.

[V2G20-1563] If the EVCC did not provide a DepartureTime Target Setting (refer to 8.3.4.4.2.2 and 8.3.5.4.1) in the ScheduleExchangeReq message, the sum of the individual time intervals described in the PowerSchedule and PriceSchedule provided in the ScheduleExchangeRes message, shall be greater or equal to 24 h.

[V2G20-1564] If the PriceSchedule or PowerSchedule provided by the secondary actor(s) are not covering the entire period of time until DepartureTime, the target setting EVTargetEnergyRequest (refer to 8.3.4.4.2.2 and 8.3.5.4.1) has not been met and the communication session has not been finished, it is the responsibility of the EVCC to request a new ScheduleTuple as soon as the last element of PowerSchedule or PriceSchedule becomes active by triggering a schedule renegotiation.

NOTE 4 The EVCC can request an update of tariff information by applying the schedule renegotiation process at any time during the charging loop.

[V2G20-1565] If the number of PriceSchedule elements provided by the secondary actor is not covering the entire time period until the EV has reached its mobility needs, it is in the responsibility of the EVCC to optimize the schedule based on the available information.

NOTE 5 The algorithm for optimizing the energy transfer profile is out of the scope of this document.

NOTE 6 Independent of the selected authorization mode of PnC or EIM, the secondary actor can decide to sign the PriceSchedule. This is not mandatory, as security is expected to provide through the CSMS, CSO, and EVSE trust chain. It is not in the scope of this document whether signing a PriceSchedule and/or validating an attached signature is mandatory or not. This can be defined by the legislative or other entities.

[V2G20-1567] The SECC shall not change the signature attached to the PriceSchedule, when receiving a signed PriceSchedule from a secondary actor.

NOTE 7 This presumes that the data structure used by the secondary actor for the provisioning of the PriceSchedule is identical with the data structure defined in this document.

[V2G20-2130] If the SA provided a PriceSchedule along with a signature, the SECC shall include the PriceSchedule along with the signature in the ScheduleExchangeRes.

[V2G20-1568] If the PriceSchedule is signed, it shall be signed by the same private key that was used to issue the leaf contract certificate that the EVCC used during this V2G communication session during the Authorization phase.

NOTE 8 This allows the EVCC to verify the integrity of the sales tariff by performing unique SECC verification. SECC/SA implementation is supposed to use the EVCC contract certificate chain which was transmitted in the element ContractCertificateChain of the message AuthorizationReq to identify the private key which can be used for signing the PriceSchedule. This requirement could be fulfilled by cooperation between SECC and SA including online communication; or the signed PriceSchedule can be cached.

[V2G20-907] In case of PnC, the EVCC, after receiving the PriceSchedule, should verify the signature using the same sub-CA certificate that was used to issue the contract certificate that the EVCC previously used during authorization. If this verification fails, the EVCC may treat the PriceSchedule as invalid. See [V2G20-1569].

NOTE 9 In case of EIM, the EVCC can ignore the signature (if it exists).

[V2G20-1569] If the EVCC treats the PriceSchedule as invalid, it shall ignore the PriceSchedule, i.e. the behavior of the EVCC shall be the same as if this invalid PriceSchedule was not received. Furthermore, the EVCC may close the connection. It then may reopen the connection again.

[V2G20-1869] The pricing within one ScheduleTupleID shall be consistent.

8.3.5.3.18 PowerScheduleType

[V2G20-310] The SECC and the EVCC shall implement this type as defined in Table 112 and Figure 115.

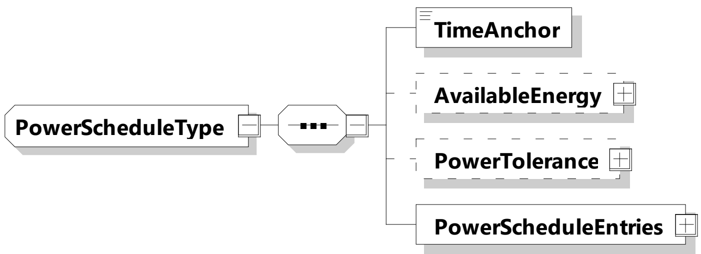

Figure 115 — Schema diagram – PowerScheduleType

The elements of this message are used according to Table 112.

Table 112 — Semantics and type definition for PowerScheduleType

Element Name	Type	Semantics
TimeAnchor	simpleType: xs:unsignedLong	The time that defines the starting point for the first PowerScheduleEntry in this power schedule. The value is encoded at ms resolution in SECC time, a concept that is defined in 8.3.3.2. For use cases that involve external secondary actors SECC time would be aligned with UTC. Commonly the TimeAnchor will define some point in time that lies in the past.
AvailableEnergy	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: Gives the Available Energy the EVSE can offer with this Schedule.

Element Name	Type	Semantics
PowerTolerance	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: A value greater than zero which defines the highest acceptable deviation from the power values communicated via the EVPowerProfile. The SECC forms each PowerTolerance band by applying the PowerTolerance symmetrically to each EVPowerProfile entry. The SECC shall only send the parameter PowerTolerance if it requires the EVCC to stay within a certain tolerance band.
PowerScheduleEntries	complexType: PowerScheduleEntryListType refer to 8.3.5.3.19	List of PowerScheduleEntry elements. The number of PowerScheduleEntry elements is limited by the parameter MaximumSupportingPoints (according to [V2G20-1544]).

[V2G20-1016] The TimeAnchor of a PowerScheduleType element shall define the point in time when the first PowerScheduleEntry element starts to be active.

[V2G20-1863] In case the applied power profile violates the accepted band provided by the parameter PowerTolerance and exceeds the boundaries referred to by [V2G20-1547] the SECC may respond with the response code "WARNING_EVPowerProfileViolation" and trigger a schedule renegotiation. In case the EVCC fails to comply with the renegotiated EVPowerProfile, the SECC may stop the charging session and respond with the response code "FAILED_EVPowerProfileViolation".

[V2G20-1864] If a PowerTolerance band was defined (see [V2G20-1870]), then the SECC shall issue a "WARNING_EVPowerProfileViolation", as defined in [V2G20-1947], [V2G20-1948] and [V2G20-5078], whenever the applied power is outside the presently valid boundaries of the PowerTolerance band. This shall not apply if any of the exception conditions of [V2G20-1865] are met.

[V2G20-1865] If a PowerTolerance band was defined (see [V2G20-1870]), then the SECC shall issue a "FAILED_EVPowerProfileViolation", as defined in [V2G20-1950], [V2G20-1951] and [V2G20-5079], whenever the arithmetic average of the applied power over a time period of the last 30 s is outside the presently valid boundaries of the PowerTolerance band. This shall not apply if the violation of the PowerTolerance band is the result of other instructions issued by the SECC which have a higher priority (e.g. EVTargetActivePower goals).

[V2G20-1866] The EVCC shall always send the parameter PowerToleranceAcceptance in PowerDeliveryReq if the parameter PowerTolerance has been provided by the SECC in the schedule that was selected by the EVCC with the SelectedScheduleTupleID.

[V2G20-1870] The parameter PowerTolerance defines the absolute allowed deviation in relation to the power values communicated via the EVPowerProfile in PowerDeliveryReq message. As the EVPowerProfile is defining the average power over time ([V2G20-1010]), the PowerTolerance parameter is applied symmetrically (+/-) in order to define what is called the PowerTolerance band. The PowerTolerance band defines the boundaries of the acceptable true power flow. This does not apply, if the

PowerTolerance band at any point in time, is in conflict with the maximum power value provided by the selected PowerSchedule(s) in the latest ScheduleExchangeRes message [V2G20-1559].

NOTE 1 For example, if the requested charge power for a time segment within the EVPowerProfile is 20 kW and PowerTolerance is set to 1,5 kW, the acceptable PowerTolerance band for charging during that segment is (18,5 – 21,5) kW.

Since the EV energy transfer process is impacted by external factors (e.g. environmental conditions) the predictions of the EVPowerProfile are already subject to uncertainty. Therefore, it is not recommended for the SECC to set a too tight PowerTolerance parameter below 5 % of the EVSEMaximumChargePower because it makes unintended violations of the PowerTolerance band more likely.

[V2G20-1871] The parameter PowerTolerance shall be a value greater than zero.

[V2G20-2189] The AvailableEnergy element shall only be included in instances of PowerScheduleType which are describing charging limits within a ChargingSchedule element.

[V2G20-2190] If used in the context of a ChargingSchedule element then the AvailableEnergy element shall only contain values greater than zero.

NOTE 2 For example, the AvailableEnergy information can be used by EVSEs which are part of a micro grid system. It would allow the EVCC to detect very early in the charging process that the desired mobility needs (energy demand) cannot be met.

8.3.5.3.19 PowerScheduleEntryListType

[V2G20-1873] The SECC and the EVCC shall implement this type as defined in Figure 116 and Table 113.

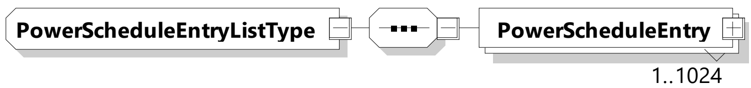

Figure 116 — Schema diagram – PowerScheduleEntryListType

The elements of this message are used according to Table 113.

Table 113 — Semantics and type definition for PowerScheduleEntryListType

Element Name	Type	Semantics
PowerScheduleEntry	complexType: PowerScheduleEntryType refer to 8.3.5.3.208.3.5.3.19	List of PowerScheduleEntry elements. The number of PowerScheduleEntry elements is limited by the parameter MaximumSupportingPoints (according to [V2G20-1056]).

8.3.5.3.20 PowerScheduleEntryType

[V2G20-1320] The SECC and the EVCC shall implement this type as defined in Table 114 and Figure 117.

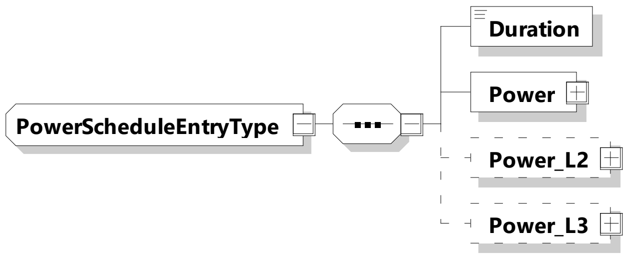

Figure 117 — Schema diagram – PowerScheduleEntryType

The elements of this message are used according to Table 114.

Table 114 — Semantics and type definition for PowerScheduleEntryType

Element Name	Type	Semantics
Duration	complexType: xs:unsignedInt	The amount of seconds that define the duration of the given power schedule entry.
Power	complexType: RationalNumberType refer to 8.3.5.3.8	Defines the amount of power (as defined in 8.3.5.3.8) for the given duration.
Power_L2	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: Defines the amount of power on phase L2 (as defined in 8.3.5.3.8) for the given duration.
Power_L3	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: Defines the amount of power on phase L3 (as defined in 8.3.5.3.8) for the given duration.

NOTE 1 The power values of the PowerScheduleEntryType follow the common semantic rules as defined in the clause on the RationalNumberType (8.3.5.3.8), where also the rules for asymmetric polyphase values are provided (8.3.5.2.2). Additional semantic requirements depend on the parent elements which utilize the PowerScheduleEntryType. For more details see the definitions related to: EVPowerProfile, PowerSchedule and PowerDischargeSchedule.

NOTE 2 The precise physical meaning of power is related to the chosen energy transfer technology. Refer to 8.3.5.3.8.

NOTE 3 According to [V2G20-1035] all power values in a DischargingSchedule need to be negative or zero.

[V2G20-1872] The duration element shall define the active period of time (in seconds) for the respective parent element of type PowerScheduleEntryType.

NOTE 4 In a list of consecutive elements the start time of the first PowerScheduleEntryType element is defined by the parent elements which utilize the PowerScheduleEntryType with their TimeAnchor element. See the definitions related to: EVPowerProfile, PowerSchedule and PowerDischargeSchedule.

[V2G20-1874] The start time of each element in a list of consecutive PowerScheduleEntryType elements shall be defined as the point in time when the previous element becomes inactive (see [V2G20-1872]).

[V2G20-1811] Before the start time of the first element and after the last element in a list of elements of type PowerScheduleEntryType becomes inactive the Power[_L2,L3] value shall be defined as the value "zero".

8.3.5.3.21 ServiceParameterListType

[V2G20-1331] The EVCC and the SECC shall implement this type as defined in Table 115 and Figure 118.

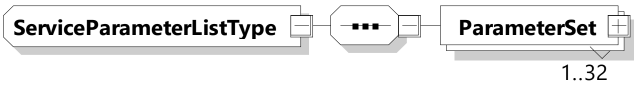

Figure 118 — Schema diagram – ServiceParameterListType

The elements of this message are used according to Table 115.

Table 115 — Semantics and type definition for ServiceParameterListType

Element Name	Type	Semantics
ParameterSet	complexType: ParameterSetType refer to 8.3.5.3.22	Defines parameters for a specific serviceID received from the SECC in the ServiceDiscoveryRes message.

8.3.5.3.22 ParameterSetType

[V2G20-1332] The EVCC and the SECC shall implement this type as defined in Table 116 and Figure 119.

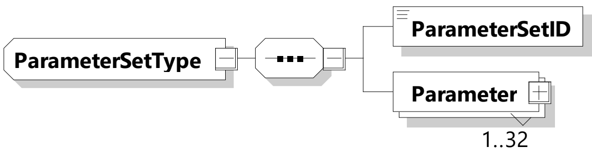

Figure 119 — Schema diagram – ParameterSetType

The elements of this message are used according to Table 116.

Table 116 — Semantics and type definition for ParameterSetType

Element Name	Type	Semantics
ParameterSetID	simpleType: serviceIDType refer to Annex A for the type definition	This element is used to select a specific parameter set for a specific ServiceID when selecting a service using the ServiceSelectionReq message.
Parameter	complexType: ParameterType refer to 8.3.5.3.23	This element is used by the SECC to indicate which service specific parameters can be selected for a certain service using the ParameterSetID. The number of Parameter elements is limited to 32.

8.3.5.3.23 ParameterType

[V2G20-1333] The EVCC and the SECC shall implement this type as defined in Table 117 and Figure 120.

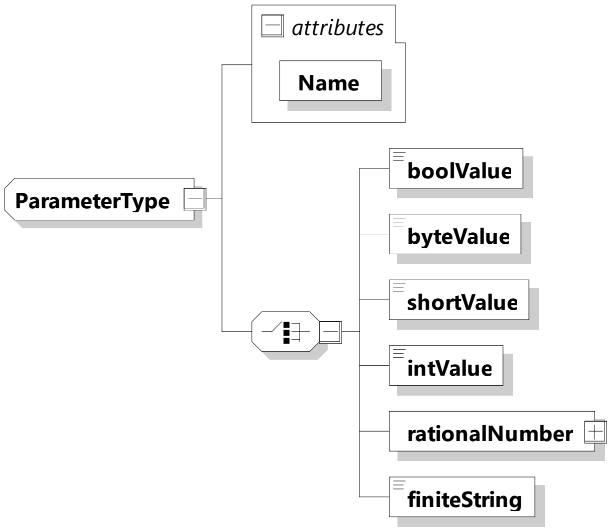

Figure 120 — Schema diagram – ParameterType

The elements of this message are used according to Table 117.

Table 117 — Semantics and type definition for ParameterType

Element Name	Type	Semantics
Name	simpleType: nameType refer to Annex A for the type definition	This element is used to indicate the name of the parameter.
One of the elements: – boolValue – byteValue – shortValue – intValue – rationalNumber – finiteString	xs:boolean (boolValue) xs:byte (byteValue) xs:short (shortValue) xs:int (intValue) nameType (finiteString) refer to Annex A for the type definition RationalNumberType (rationalNumber) refer to 8.3.5.3.8	This element is used to indicate the value for the parameter indicated by the element Name. A choice of 6 different element types. Only one for each parameter can be selected.

8.3.5.3.24 SelectedServiceListType

[V2G20-1334] The EVCC and the SECC shall implement this type as defined in Table 118 and Figure 121.

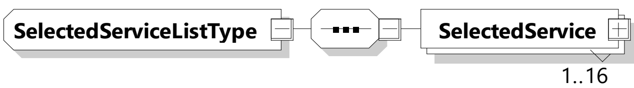

Figure 121 — Schema diagram – SelectedServiceListType

The elements of this message are used according to Table 118.

Table 118 — Semantics and type definition for SelectedServiceListType

Element Name	Type	Semantics
SelectedService	complexType: SelectedServiceType refer to 8.3.5.3.25	This element is used to indicate the selected ServiceID and the associated parameterSet.

8.3.5.3.25 SelectedServiceType

[V2G20-1335] The EVCC and the SECC shall implement this type as defined in Table 119 and Figure 122.

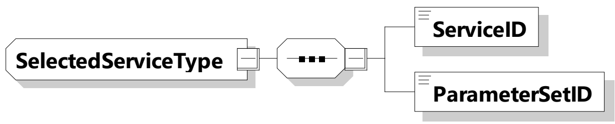

Figure 122 — Schema diagram – SelectedServiceType

The elements of this message are used according to Table 119.

Table 119 — Semantics and type definition for SelectedServiceType

Element Name	Type	Semantics
ServiceID	simpleType: serviceIDType refer to Annex A for the type definition	Unique identifier of the service
ParameterSetID	simpleType: serviceIDType refer to Annex A for the type definition	This element is used to select a specific parameter set for a specific ServiceID when selecting a service using the ServiceSelectionReq message.

8.3.5.3.26 EVSEStatusType

[V2G20-1338] The EVCC and the SECC shall implement this type as defined in Table 120 and Figure 123.

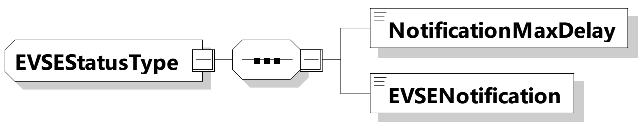

Figure 123 — Schema diagram – EVSEStatusType

The elements of this message are used according to Table 120.

Table 120 — Semantics and type definition for EVSEStatusType

Element Name	Type	Semantics
NotificationMaxDelay	simpleType: xs:unsignedShort	This value is the time in seconds from the point in time this message is sent (relative time) and expected to perform the action immediately. The SECC uses the NotificationMaxDelay element in the EVSEStatus to indicate the time until it expects the EVCC to react on the action request indicated in EVSENotification.
EVSENotification	simpleType: EVSENotificationType Enumeration refer to Annex A for the type definition	This value is used by the SECC to influence the behavior of the EVCC. The EVSENotification contains an action that the SECC wants the EVCC to perform. The requested action is expected by the EVCC until the time provided in NotificationMaxDelay. If the target time is not in the future, the EVCC is expected to perform the action immediately. Requestable actions are: – ScheduleRenegotiation, – ServiceRenegotiation, – MeteringConfirmation, – Pause, – ExitStandby, – Terminate.

8.3.5.3.27 ListOfRootCertificateIDsType

[V2G20-1339] The SECC and the EVCC shall implement this type as defined in Table 121 and Figure 124.

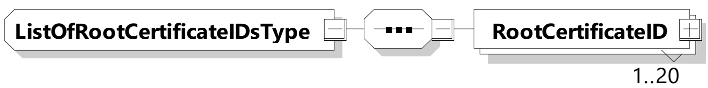

Figure 124 — Schema diagram – ListOfRootCertificateIDsType

The elements of this message are used according to Table 121.

Table 121 — Semantics and type definition for ListOfRootCertificateIDsType

Element Name	Type	Semantics
RootCertificateID	complexType: xmlsig:X509IssuerSerialType	This message element uniquely identifies a V2G root certificate installed in the EVCC.

8.3.5.3.28 DisplayParametersType

[V2G20-1343] The SECC and the EVCC shall implement this type as defined in Table 122 and Figure 125.

NOTE 1 The DisplayParameters have no influence on the energy transfer and are optional for both sides.

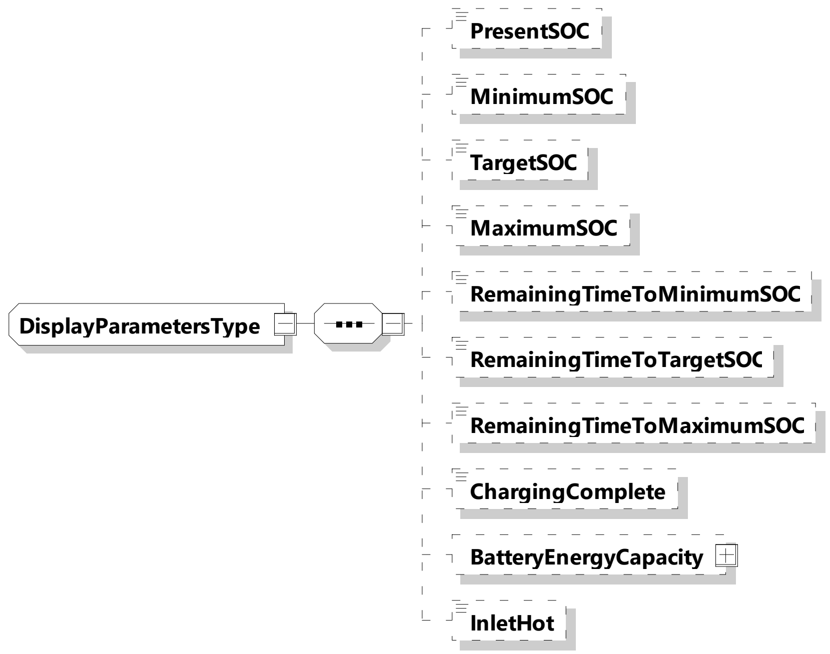

Figure 125 — Schema diagram - DisplayParametersType

The elements of this message are used according to Table 122.

Table 122 — Semantics and type definition for – DisplayParametersType

Element Name	Type	Semantics
PresentSOC	simpleType: percentValueType refer to Annex A for the type definition	Optional: Current State of Charge of the EV battery Mandatory when service parameter MobilityNeedsMode was set to 2
MinimumSOC	simpleType: percentValueType refer to Annex A for the type definition	Optional: Minimum State of Charge EV needs after charging of the EV battery the EV to keep throughout the charging session. Mandatory when service parameter MobilityNeedsMode was set to 2
TargetSOC	simpleType: percentValueType refer to Annex A for the type definition	Optional: Target State of Charge of the EV battery EV needs after charging Mandatory when service parameter MobilityNeedsMode was set to 2
MaximumSOC	simpleType: percentValueType refer to Annex A for the type definition	Optional: The SOC at which the EV will prohibit any further charging.

Element Name	Type	Semantics
RemainingTimeToMinimumSOC	simpleType: xs:unsignedInt	Optional: This element is used to indicate the remaining time it takes to reach the minimum SOC. It is communicated as the offset in seconds from the point in time of sending this message, which is marked in the TimeStamp element of the common V2G message header.
RemainingTimeToTargetSOC	simpleType: xs:unsignedInt	Optional: This element is used to indicate the remaining time it takes to reach the TargetSOC. It is communicated as the offset in seconds from the point in time of sending this message, which is marked in the TimeStamp element of the common V2G message header.
RemainingTimeToMaximumSOC	simpleType: xs:unsignedInt	Optional: This element is used to indicate the remaining time it takes to reach the maximum SOC. It is communicated as the offset in seconds from the point in time of sending this message, which is marked in the TimeStamp element of the common V2G message header.
ChargingComplete	simpleType: xs:boolean	Optional: Indication if the charging is complete from EV point of view (value = 1).
BatteryEnergyCapacity	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: It is the calculated amount of electrical Energy in Wh stored in the battery when the displayed SOC equals 100 %.
InletHot	simpleType: xs:boolean	Optional: Inlet temperature too high to accept specific operating condition.

[V2G20-1340] When EVCC selects the service parameter "MobilityNeedsMode" set to 2, ChargeLoopReq shall contain a DisplayParameter with PresentSOC, TargetSOC, and MinimumSOC.

NOTE 2 For given TargetSOC, MinimumSOC and MaximumSOC values the EV in dynamic control mode will continuously estimate the amount of energy needed to reach those states. See the definition of EVTargetEnergyRequest, EVMinimumEnergyRequest, and EVMaximumEnergyRequest for more details.

8.3.5.3.29 ServiceIDListType

[V2G20-1575] The EVCC and the SECC shall implement this type as defined in Table 123 and Figure 126.

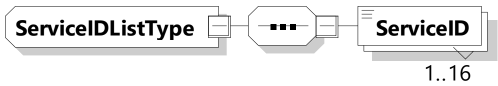

Figure 126 — Schema diagram – ServiceIDListType

The elements of this message are used according to Table 123.

Table 123 — Semantics and type definition for ServiceIDListType

Element Name	Type	Semantics
ServiceID	simpleType: serviceIDType refer to Annex A for the type definition	This element identifies a service which has been offered by the SECC in the ServiceDiscoveryRes message.

8.3.5.3.30 EMAIDListType

[V2G20-1588] The SECC and the EVCC shall implement this type as defined in Figure 127 and Table 124.

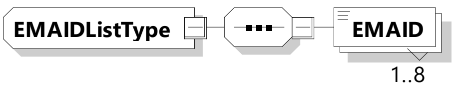

Figure 127 — Schema diagram – EMAIDListType

The elements of this message are used according to Table 124.

Table 124 — Semantics and type definition for EMAIDListType

Element Name	Type	Semantics
EMAID	simpleType identifierType refer to Annex A for the type definition	Contains the EMAID as defined in C.1.

8.3.5.3.31 EIM_AReqAuthorizationModeType

This type contains all elements of the AuthorizationReq message that are only required in case the authorization mode EIM is chosen.

[V2G20-1875] The SECC and the EVCC shall implement this type as defined in Table 125 and Figure 128.

Figure 128 — Schema diagram – EIM_AReqAuthorizationModeType

Table 125 — Semantics and type definition for EIM_AReqAuthorizationModeType

Element name	Type	Semantics
n/a	n/a	n/a

NOTE This parameter does not contain any elements.

8.3.5.3.32 PnC_AReqAuthorizationModeType

This type contains all elements of the AuthorizationReq message that are only required in case the authorization mode PnC is chosen.

[V2G20-1782] The SECC and the EVCC shall implement this type as defined in Table 126 and Figure 129.

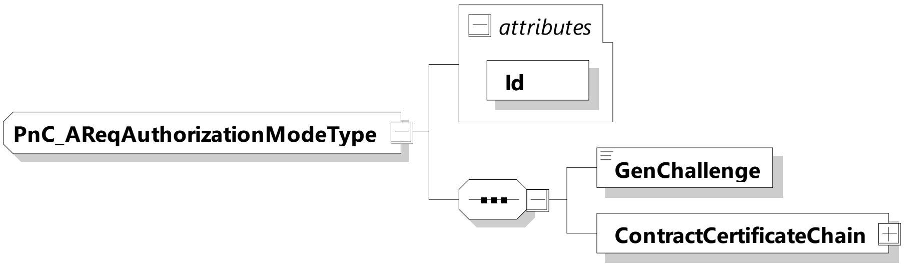

Figure 129 — Schema diagram – PnC_AReqAuthorizationModeType

The elements of this message are used according to Table 126.

Table 126 — Semantics and type definition for PnC_AReqAuthorizationModeType

Element name	Type	Semantics
Id	simpleType: xs:ID	This attribute is used for referencing the GenChallenge element in the signature header.
GenChallenge	simpleType: genChallengeType	The challenge sent by the SECC in AuthorizationSetupRes message. This element contains the generated random number.
ContractCertificateChain	complexType: ContractCertificateChainType refer 8.3.5.3.5	The certificate chain to be used for authorization of the service(s) being utilized at the EVSE.

[V2G20-2110] The GenChallenge field shall be exactly 128 bits long, as stated in [V2G20-697].

[V2G20-2107] The EVCC shall send one (1) contract certificate chain in the element ContractCertificateChain.

NOTE 1 In case EVCC contains more than one (1) contract certificate chains, it is up to the EVCC how to determine the contract certificate chain to be provided.

NOTE 2 One methodology to determine the ontract certificate chain to be provided is the use of the SupportedProviders list. Refer to 8.3.5.3.35 for more details.

8.3.5.3.33 EIM_ASResAuthorizationModeType

[V2G20-1876] The SECC and the EVCC shall implement this type as defined in Figure 130 and Table 127.

Figure 130 — Schema diagram – EIM_ASResAuthorizationModeType

Table 127 — Semantics and type definition for EIM_ASResAuthorizationModeType

Element name	Type	Semantics
n/a	n/a	n/a

NOTE This parameter does not contain any elements.

8.3.5.3.34 PnC_ASResAuthorizationModeType

[V2G20-1877] The SECC and the EVCC shall implement this type as defined in Table 128 and Figure 131.

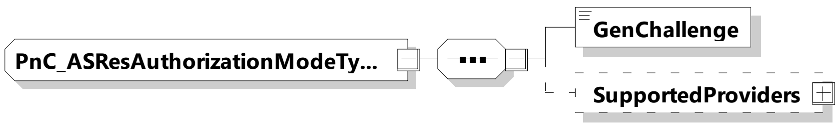

Figure 131 — Schema diagram – PnC_ASResAuthorizationModeType

The elements of this message are used according to Table 128.

Table 128 — Semantics and type definition for PnC_ASResAuthorizationModeType

Element name	Type	Semantics
GenChallenge	simpleType: genChallengeType refer to Annex A for the type definition	The challenge sent by the SECC in AuthorizationSetupRes message. This element contains the generated random number.
SupportedProviders	complexType: SupportedProvidersListType refer to 8.3.5.3.35 for the type definition	Optional: Provides a list of eMSPs supported by the SECC. This parameter allows EVCC to filter the list of contract certificates to be utilized during the payment authorization. Refer to 8.3.4.3.3.1 for further details.

[V2G20-697] The GenChallenge field shall be exactly 128 bits long.

[V2G20-2108] GenChallenge shall be generated per random number generation requirements as defined by 7.3.7.

[V2G20-698] The entropy of the GenChallenge field shall be at least 120 bits.

8.3.5.3.35 SupportedProvidersListType

[V2G20-2703] The SECC and the EVCC shall implement this type as defined in Figure 132 and Table 129.

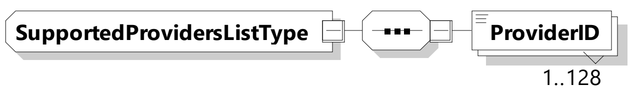

Figure 132 — Schema diagram – SupportedProvidersListType

The elements of this message are used according to Table 129.

Table 129 — Semantics and type definition for SupportedProvidersListType

Element name	Type	Semantics
ProviderID	simpleType: nameType refer to Annex A for the type definition	Alphanumeric characters referring to the eMSP. These are nominally the country code and the provider ID similar to those included in the EMAID. Refer to C.1 for more details of EMAID. The list shall be in ascending order with 0-9 followed by A-Z. All characters shall be in uppercase only. The SupportedProviders includes at least 1 Provider and may include up to 128 providers.

8.3.5.3.36 SignedMeteringDataType

[V2G20-2647] The SECC and the EVCC shall implement this type as defined in Figure 133 and Table 130.

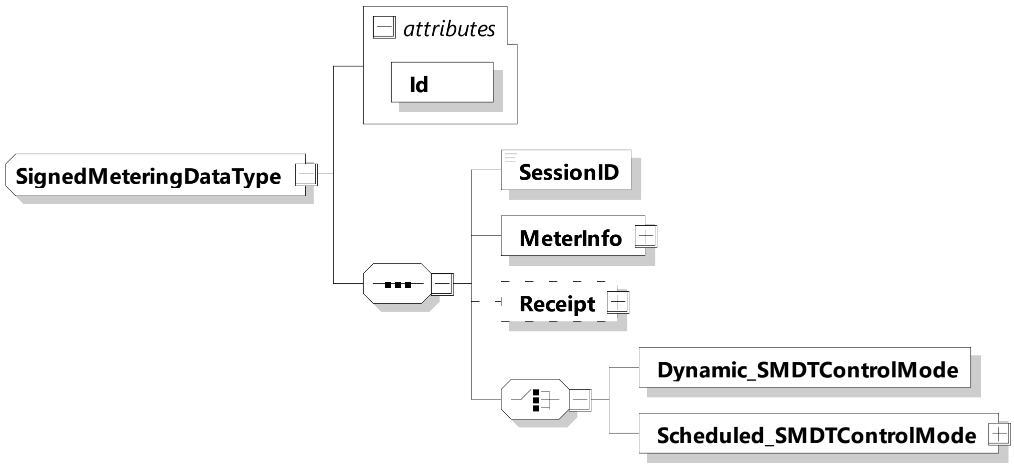

Figure 133 — Schema diagram – SignedMeteringDataType

The elements of this message are used according to Table 130.

Table 130 — Semantics and type definition for SignedMeteringDataType

Element name	Type	Semantics
SessionID	simpleType: sessionIDType refer to Annex A for the type definition	This message element is used by EVCC and SECC for uniquely identifying a V2G communication session. This element is identical with the one included in the message header. It is placed in the body in addition to be able to apply a signature to it (complete body is signed).

Element name	Type	Semantics
MeterInfo	complexType: MeterInfoType refer to 8.3.5.3.6 for the type definition	Includes the energy charged during this service session and other meter relevant data.
Receipt	complexType: ReceiptType refer to 8.3.5.3.59 for the type definition	Optional: Includes the receipt as provided by the SECC.
Dynamic_SMDTControlMode	complexType: Dynamic_SMDTControlModeType refer to 8.3.5.3.37 for the type definition	Contains all elements that are only required in case the dynamic control mode was chosen.
Scheduled_SMDTControlMode	complexType: Scheduled_SMDTControlModeType refer to 8.3.5.3.38 for the type definition	Contains all elements that are only required in case the scheduled control mode was chosen.

8.3.5.3.37 Dynamic_SMDTControlMode

This type contains all elements of the SignedMeteringDataType that are only required in case the dynamic control mode is chosen.

[V2G20-1875] The SECC and the EVCC shall implement this type as defined in Figure 134 and Table 131.

Figure 134 — Schema diagram – Dynamic_SMDTControlMode

Table 131 — Semantics and type definition for Dynamic_SMDTControlMode

Element name	Type	Semantics
n/a	n/a	n/a

NOTE This parameter does not contain any elements.

8.3.5.3.38 Scheduled_SMDTControlMode

This type contains all elements of the SignedMeteringDataType that are only required in case the scheduled control mode is chosen.

[V2G20-1875] The SECC and the EVCC shall implement this type as defined in Figure 135 and Table 132.

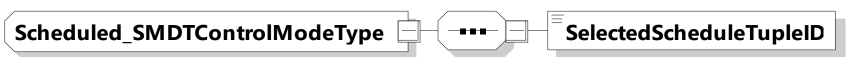

Figure 135 — Schema diagram – Scheduled_SMDTControlMode

The elements of this message are used according to Table 132.

Table 132 — Semantics and type definition for Scheduled_SMDTControlMode

Element name	Type	Semantics
SelectedScheduleTupleID	simpleType: numericIDType refer to Annex A for the type definition	This element includes the ScheduleTupleID the EV chose for this session.

8.3.5.3.39 SignedInstallationDataType

[V2G20-1878] The SECC and the EVCC shall implement this type as defined in Figure 136 and Table 133.

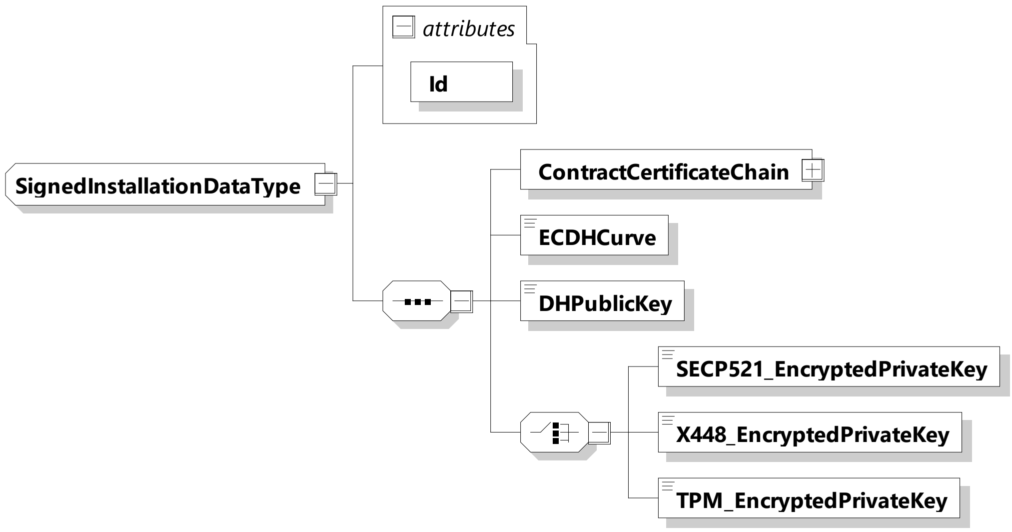

Figure 136 — Schema diagram – SignedInstallationDataType

The elements of this message are used according to Table 133.

Table 133 — Semantics and type definition for SignedInstallationDataType

Element name	Type	Semantics
ContractCertificateChain	complexType: ContractCertificateChainType refer 8.3.5.3.5	The certificate chain to be used for authorization of the service(s) being utilized at the EVSE.
ECDHCurve	simpleType: ECDHCurveType refer to Annex A for the type definition	Parameter indicating which (elliptic curve) EC the eMSP wants to use for generating the session key (seed) used to encrypt the contract certificate private key.
DHPublicKey	simpleType: dhPublicKeyType refer to Annex A for the type definition	Diffie Hellman public key from the SA for generating the session key at the EVCC in order to encrypt the contract signature private key at the SA and decrypt it a the EVCC. Refer to 7.9.2.5.4 for details.

Element name	Type	Semantics
SECP521_EncryptedPrivateKey	simpleType: secp521_EncryptedPrivateKeyType refer to Annex A for the type definition	The 521-bit private key that belongs to the new contract certificate encrypted for EVCCs without the TPM. This is secret data and therefore shall be encrypted. Refer to 7.9.2.5.2 and its subclauses for details.
X448_EncryptedPrivateKey	simpleType: x448_EncryptedPrivateKeyType refer to Annex A for the type definition	The 448-bit private key that belongs to the new contract certificate encrypted for EVCCs without the TPM. This is secret data and therefore shall be encrypted. Refer to 7.9.2.5 and its subclauses for details.
TPM_EncryptedPrivateKey	simpleType: tpm_EncryptedPrivateKeyType refer to Annex A for the type definition	The private key that belongs to the new contract certificate encrypted for EVCCs with the TPM. This is secret data and therefore shall be encrypted. Refer to 7.9.2.5.3 and its subclauses for details.

NOTE Details of various elements in SignedInstallationDataType can be found in 7.9.2.5 and 8.3.4.3.9 and their subclauses.

8.3.5.3.40 ChargingScheduleType

[V2G20-1879] The SECC and the EVCC shall implement this type as defined in Figure 137 and Table 134.

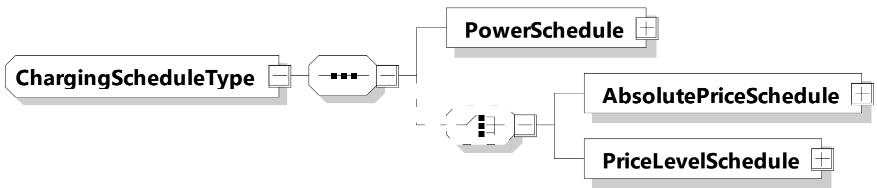

Figure 137 — Schema diagram – ChargingScheduleType

The elements of this message are used according to Table 134.

Table 134 — Semantics and type definition for ChargingScheduleType

Element name	Type	Semantics
PowerSchedule	complexType: PowerScheduleType refer to 8.3.5.3.18	Encapsulating element describing all relevant details for one PowerSchedule as defined by the secondary actor or the SECC. It is used to communicate the hard limit of charging power
AbsolutePriceSchedule	complexType: AbsolutePriceScheduleType PriceScheduleType refer to 8.3.5.3.49	Optional: Absolute Price schedule for this session

Element name	Type	Semantics
PriceLevelSchedule	complexType: PriceLevelScheduleType PriceScheduleType refer to 8.3.5.3.62	Optional: PriceLevel schedule for this session

[V2G20-2176] The SECC shall either provide an AbsolutePriceSchedule or a PriceLevelSchedule as part of the ChargingScheduleType.

8.3.5.3.41 EVEnergyOfferType

The EVEnergyOffer extends the scheduled control mode by adding the possibility for EVCCs to offer energy (discharging capabilities) for a certain price per kWh to the infrastructure. This offer can be included by the SECC into the DischargingSchedule and be offered back to the EVCC again. The EVPowerSchedule describes the maximum flexibility to feedback energy regarding time and power, the energy amount is limited and defined by the absolute value of EVTargetEnergyRequest. A negative value indicates the willingness to discharge under specific conditions, a positive value indicates that the EV currently is not able to offer energy to discharge.

[V2G20-1881] The SECC and the EVCC shall implement this type as defined in Figure 138 and Table 135.

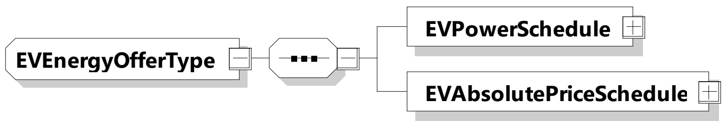

Figure 138 — Schema diagram – EVEnergyOfferType

The elements of this message are used according to Table 135.

Table 135 — Semantics and type definition for EVEnergyOfferType

Element name	Type	Semantics
EVPowerSchedule	complexType: EVPowerScheduleType refer to 8.3.5.3.42	Encapsulating element describing all relevant details for one EVPowerSchedule as defined by the EVCC. It is used to communicate the hard limit of discharging power.
EVAbsolutePriceSchedule	complexType: EVAbsolutePriceScheduleType refer to 8.3.5.3.45	Optional: EVAbsolutePriceSchedule for this session. It is used to communicate the Price for which the EV offers to discharge its stored energy.

8.3.5.3.42 EVPowerScheduleType

[V2G20-1882] The SECC and the EVCC shall implement this type as defined in Figure 139 and Table 136.

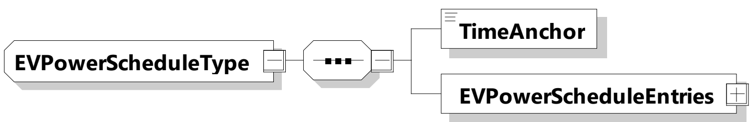

Figure 139 — Schema diagram – EVPowerScheduleType

The elements of this message are used according to Table 136.

Table 136 — Semantics and type definition for EVPowerScheduleType

Element name	Type	Semantics
TimeAnchor	simpleType: xs:unsignedLong	The time that defines the starting point for the EVEnergyOffer. The value is encoded at microseconds resolution in SECC time, a concept that is defined in 8.3.3.3.
EVPowerScheduleEntries	complexType: EVPowerScheduleEntryListType refer to 8.3.5.3.43	List of EVPowerScheduleEntry elements. The number of EVPowerScheduleEntry elements is limited by the parameter MaximumSupportingPoints (according to [V2G20-1056]).

[V2G20-2138] The TimeAnchor of an EVPowerScheduleType element shall define the point in time when the first EVPowerScheduleEntry element starts to be active.

8.3.5.3.43 EVPowerScheduleEntryListType

[V2G20-1883] The SECC and the EVCC shall implement this type as defined in Figure 140 and Table 137.

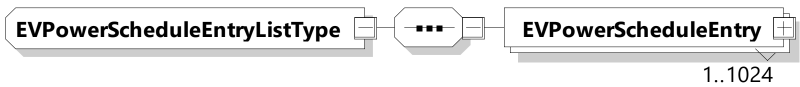

Figure 140 — Schema diagram – EVPowerScheduleEntryListType

The elements of this message are used according to Table 137.

Table 137 — Semantics and type definition for EVPowerScheduleEntryListType

Element name	Type	Semantics
EVPowerScheduleEntry	complexType: EVPowerScheduleEntryType refer to 8.3.5.3.44	List of EVPowerScheduleEntry elements. The number of EVPowerScheduleEntry elements is limited by the parameter MaximumSupportingPoints (according to [V2G20-1056]).

8.3.5.3.44 EVPowerScheduleEntryType

[V2G20-1884] The SECC and the EVCC shall implement this type as defined in Figure 141 and Table 138.

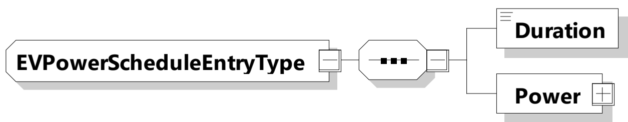

Figure 141 — Schema diagram – EVPowerScheduleEntryType

The elements of this message are used according to Table 138.

Table 138 — Semantics and type definition for EVPowerScheduleEntryType

Element name	Type	Semantics
Duration	simpleType: xs:unsignedInt	Duration of the EVPowerScheduleEntries. The amount of seconds that define the duration of the given EVPowerScheduleEntries.
Power	complexType: RationalNumberType refer to 8.3.5.3.8	Defines maximum amount of power for the duration of this EVPowerScheduleEntry to be discharged from the EV battery through EVSE power outlet. Negative values are used for discharging.

NOTE The precise physical meaning of Power is related to the chosen charging technology. Refer to 8.3.5.3.8.

[V2G20-2139] The duration element shall define the active period of time (s) for the respective parent element of EVPowerScheduleEntryType.

[V2G20-2140] If AC was selected as energy transfer mode, the Power element shall define the maximum amount of apparent power to be drawn from the EV when the element of type EVPowerScheduleEntryType is active. The power can freely be drawn from any phase, but the real AC polyphase discharging behavior should always respect the maximum allowed asymmetry at the grid connection point as defined by the local grid code.

[V2G20-2141] The power elements in the EVPowerScheduleType shall always be smaller or equal to zero as they communicate the discharging capabilities of the EV.

8.3.5.3.45 EVAbsolutePriceScheduleType

[V2G20-1885] The SECC and the EVCC shall implement this type as defined in Figure 142 and Table 139.

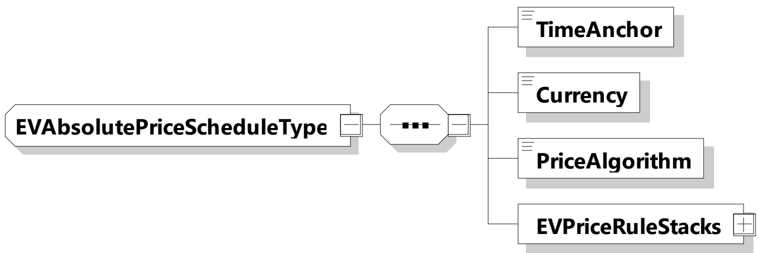

Figure 142 — Schema diagram – EVAbsolutePriceScheduleType

The elements of this message are used according to Table 139.

Table 139 — Semantics and type definition for EVAbsolutePriceScheduleType

Element name	Type	Semantics
TimeAnchor	complexType: xs:unsignedLong	The time that defines the starting point for the EVEnergyOffer. The value is encoded at microseconds resolution in SECC time, a concept that is defined in 8.3.3.3
Currency	complexType: currencyType refer to Annex A for the type definition	Currency for the pricing according to ISO 4217

Element name	Type	Semantics
PriceAlgorithm	simpleType: identifierType refer to Annex A for the type definition	A string in URN notation which shall uniquely identify an algorithm that defines how to compute an energy fee sum for a specific power profile based on the EnergyFee information from the EVPriceRule elements
EVPriceRuleStacks	complexType: EVPriceRuleStackListType Refer to 8.3.5.3.46	A set of pricing rules for energy costs

[V2G20-2142] Only one currency shall be used for all EVAbsolutePriceSchedules.

NOTE 1 It is possible that the currency in EVAbsolutePriceSchedules can be different from the currency in AbsolutePriceSchedule.

NOTE 2 It is up to the SECC to decide on how to react to currencies that are unexpected.

[V2G20-2143] The number of EVPriceRules shall not exceed value of MaximumSupportingPoints.

NOTE 3 For requirements and informative text regarding PriceAlgorithm, please refer to 8.3.5.3.49.1 PriceAlgorithm identifiers.

8.3.5.3.46 EVPriceRuleStackListType

[V2G20-1886] The SECC and the EVCC shall implement this type as defined in Figure 143 and Table 140.

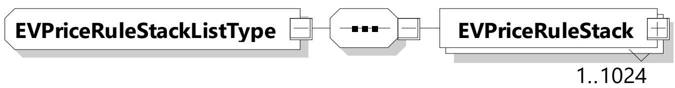

Figure 143 — Schema diagram – EVPriceRuleStackListType

The elements of this message are used according to Table 140.

Table 140 — Semantics and type definition for EVPriceRuleStackListType

Element name	Type	Semantics
EVPriceRuleStack	complexType: EVPriceRuleStackType refer to 8.3.5.3.47	Contains the EVPriceRules.

[V2G20-2144] An EVPriceRuleStack shall become active when the previous EVPriceRuleStack has expired, or when it is the first EVPriceRuleStack in the EVAbsolutePriceSchedule.

[V2G20-2145] An EVPriceRuleStack expires when the current time is greater than the starting time of the EVPriceRuleStack plus the Duration.

[V2G20-2146] The first EVPriceRule in the EVPriceRuleStack shall start at the same time as the parent EVAbsolutePriceScheduleType's TimeAnchor.

8.3.5.3.47 EVPriceRuleStackType

[V2G20-2147] The SECC and the EVCC shall implement this type as defined in Figure 144 and Table 141.

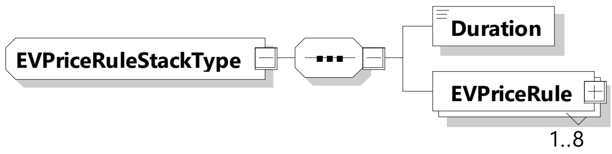

Figure 144 — Schema diagram – EVPriceRuleStackType

The elements of this message are used according to Table 141.

Table 141 — Semantics and type definition for EVPriceRuleStackType

Element name	Type	Semantics
Duration	simpleType: xs:unsignedInt	Duration of the stack of price rules. The amount of seconds that define the duration of the given EVPriceRuleStack.
EVPriceRule	complexType: EVPriceRuleType refer to 8.3.5.3.48	A set of pricing rules for energy costs.

[V2G20-2148] Overlapping (stacked) EVPriceRules shall all start at the same time.

NOTE EVPriceRulesStack allows for EVPriceRules with different PowerRangeStart values.

8.3.5.3.48 EVPriceRuleType

[V2G20-2149] The SECC and the EVCC shall implement this type as defined in Figure 145 and Table 142.

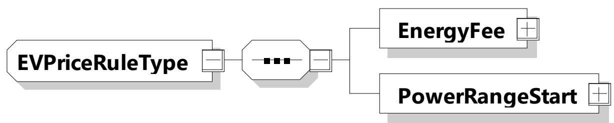

Figure 145 — Schema diagram – EVPriceRuleType

The elements of this message are used according to Table 142.

Table 142 — Semantics and type definition for EVPriceRuleType

Element name	Type	Semantics
EnergyFee	complexType: RationalNumberType refer to 8.3.5.3.8	Cost per kWh NOTE If the energy is free, then this value is zero.
PowerRangeStart	complexType: RationalNumberType refer to 8.3.5.3.8	The EnergyFee applies between this value and the value of the PowerRangeStart of the subsequent EVPriceRule. If the power is below this value, the EnergyFee of the previous EVPriceRule applies.

[V2G20-2150] An EVPriceRule shall become active when the current power is at or below the PowerRangeStart value.

[V2G20-2152] At least one active EVPriceRule shall have a PowerRangeStart with a value of zero.

[V2G20-2153] PowerRangeStart shall be less than or equal to zero.

[V2G20-2154] PowerRangeStart values shall be unique within the same EVPriceRuleStack.

8.3.5.3.49 AbsolutePriceScheduleType

This type represents the actual cost for a charging session. For examples showing the usage, please refer to Annex J.

NOTE IDs within AbsolutePrice are not used in this document, rather, they are provided for informational purposes (e.g. logging).

[V2G20-1887] The SECC and the EVCC shall implement this type as defined in Figure 146 and Table 143.

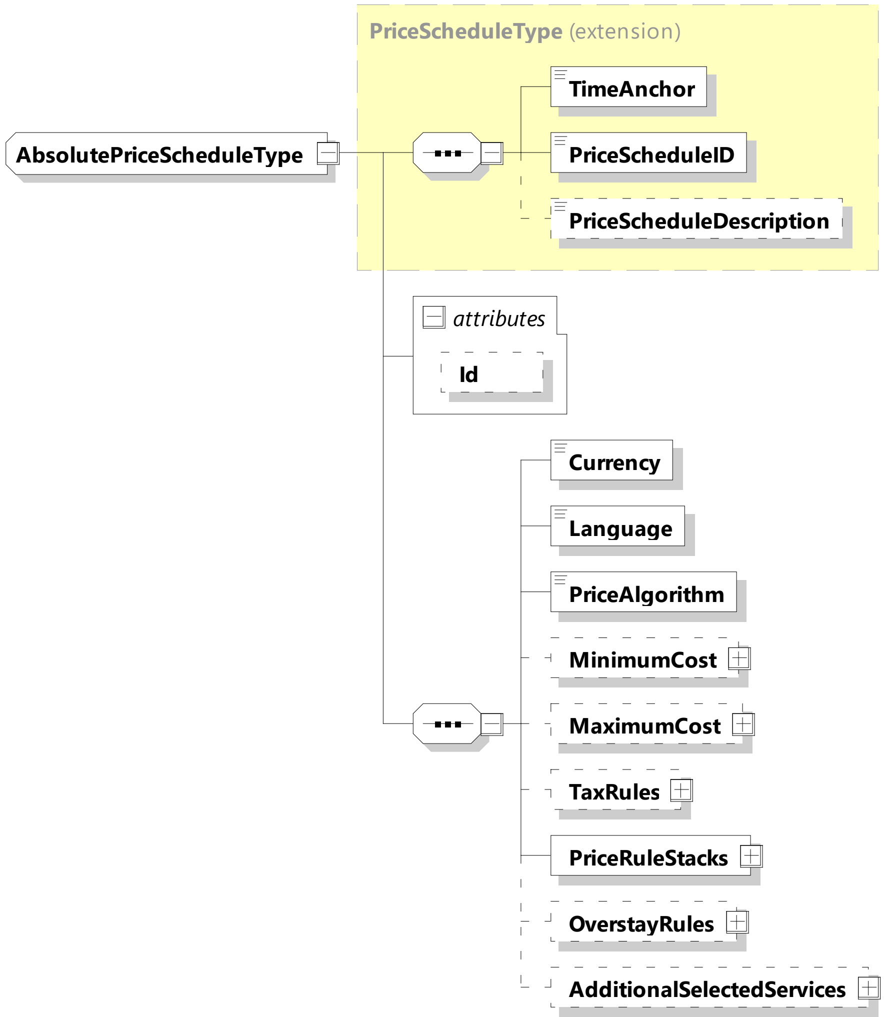

Figure 146 — Schema diagram – AbsolutePriceScheduleType

The elements of this message are used according to Table 143.

Table 143 — Semantics and type definition for AbsolutePriceScheduleType

Element name	Type	Semantics
Id	simpleType: xs:ID	This element is used for referencing the remaining elements of this table in the message header when a signature needs to be applied.

Element name	Type	Semantics
TimeAnchor	simpleType: xs:unsignedLong	The time that defines the starting point for the AbsolutePriceSchedule. The value is encoded at microseconds resolution in SECC time, a concept that is defined in 8.3.3.3
PriceScheduleID	simpleType: numericIDType refer to Annex A for the type definition	Unique ID for the price schedule.
PriceScheduleDescription	simpleType: descriptionType refer to Annex A for the type definition	Optional: Human readable text describing a schedule.
Currency	simpleType: currencyType refer to Annex A for the type definition	Currency for the pricing according to ISO 4217
Language	simpleType: languageType refer to Annex A for the type definition	String that indicates what language is used for the human readable strings in the price schedule. Based on ISO 639.
PriceAlgorithm	simpleType: identifierType refer to Annex A for the type definition	A string in URN notation which shall uniquely identify an algorithm that defines how to compute an energy fee sum for a specific power profile based on the EnergyFee information from the PriceRule elements
MinimumCost	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: Minimum amount to be billed for the overall charging session (e.g. including energy, parking, and overstay)
MaximumCost	complexType: RationalNumberType refer to 8.3.5.3.8	Optional: Maximum amount to be billed for the overall charging session (e.g. including energy, parking, and overstay)
TaxRules	complexType: TaxRuleListType refer to 8.3.5.3.50	Optional: Describes the applicable tax rule(s) for this price schedule
PriceRuleStacks	complexType: PriceRuleStackListType refer to 8.3.5.3.52	A set of pricing rules for parking and energy costs
OverstayRules	complexType OverstayRuleListType refer to 8.3.5.3.55	Optional: A set of overstay rules that allows for escalation of charges after the overstay is triggered

Element name	Type	Semantics
AdditonalSelectedServices	complexType AdditionalServicesListType refer to 8.3.5.3.57	Optional: A set of price rules for optional services (e.g. valet, carwash)

[V2G20-1888] Only one currency shall be used for all AbsolutePriceSchedules.

NOTE 1 It is possible that the currency in EVAbsolutePriceSchedules can be different from the currency in AbsolutePriceSchedule.

NOTE 2 It is up to the SECC and EVCC to decide on how to handle currencies that are unexpected.

[V2G20-1889] The number of PriceRules shall not exceed the value of MaximumSupportingPoints.

[V2G20-1890] If MaximumCost is specified, then the element defines the maximum that may be billed.

[V2G20-1891] If MinimumCost is specified, then the element defines the minimum that may be billed.

[V2G20-2640] AdditionalSelectedServices shall not be included in the calculation of MaximumCost or MinimumCost.

[V2G20-1892] Each PriceScheduleID shall be unique.

[V2G20-1893] When a fee parameter is not sent, it shall mean this fee does not apply.

NOTE 3 This applies to all fee parameters: MaximumCost, MinimumCost, EnergyFee, ParkingFee and ServiceFee.

8.3.5.3.49.1 PriceAlgorithm identifiers

This document defines three identifiers for the PriceAlgorithm element:

– urn:iso:std:iso:15118:-20:PriceAlgorithm:1-Power;

– urn:iso:std:iso:15118:-20:PriceAlgorithm:2-PeakPower;

– urn:iso:std:iso:15118:-20:PriceAlgorithm:3-StackedEnergy.

Their definition is based on the following, generic observations, which are illustrated in Figure 147, Figure 148 and Figure 149. In Figure 147, the example is for a set of PriceRule elements and their associated EnergyFee definitions. In the first hour after midnight (00:00) a PowerSchedule limit of 15 kW is in place and only one PriceRule exists. In the next hour (01:00 to 02:00) the PowerSchedule limit has increased to 30 kW and there are three different PriceRule entries, which are stacked based on their PowerRangeStart value. The pictured EnergyFee values are just examples and not relevant for the definition of the PriceAlgorithm choices.

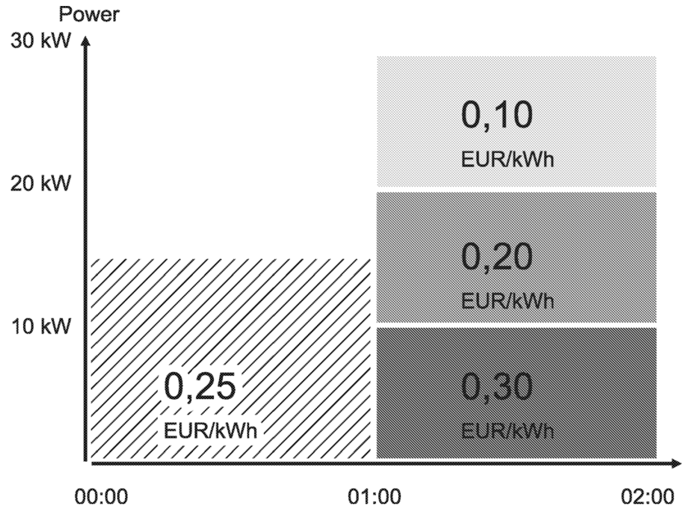

Figure 147 — Example for a set of PriceRule elements and their associated EnergyFee

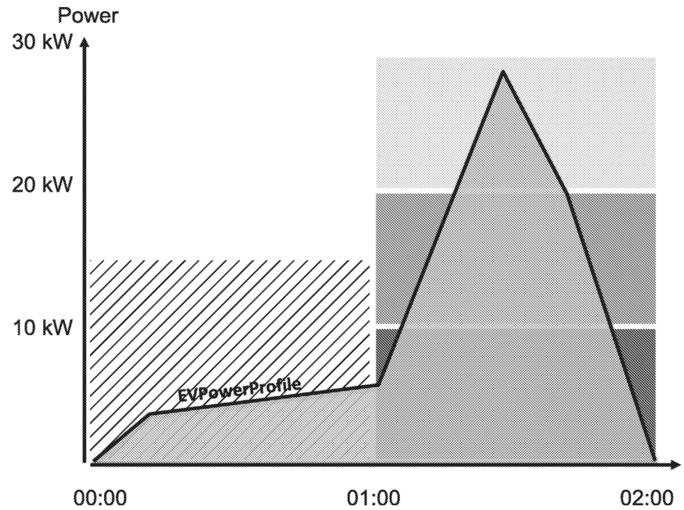

Figure 148 — Example for boundaries of different PriceRules with an EVPowerProfile

Given a hypothetical EVPowerProfile (Figure 148) each intersection of its curve with the boundaries of the different PriceRules will result in an area that defines a particular energy content.

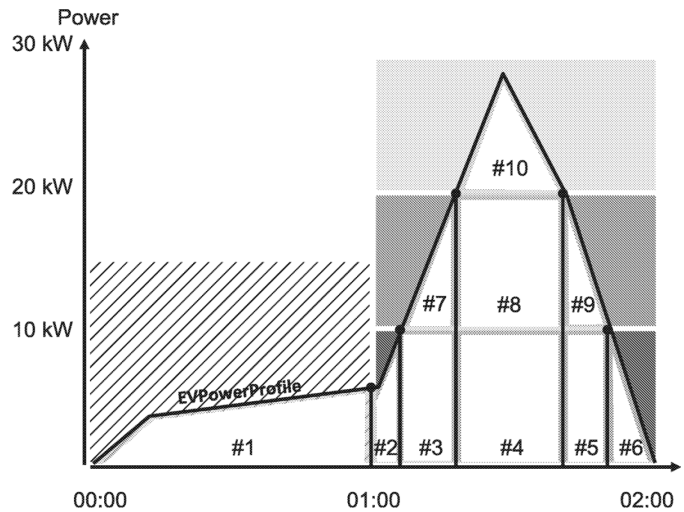

Figure 149 — Example with for the PriceAlgorithm with different PriceRules

In this example (Figure 149), one will end up with 10 distinct areas. While those areas are always the same each, PriceAlgorithm will apply a different rule which determines the price (EnergyFee) that each of the 10 energy content areas need to be multiplied with in order to compute and predict a resulting energy fee sum.

Figure 150, Figure 151 and Figure 152 below show how the three PriceAlgorithm options would be applied. Figure 150 shows "1-Power", Figure 151 shows "2-PeakPower" and Figure 152 shows the "3-StackedEnergy" PriceAlgorithm. The energy content of each of the 10 areas is multiplied with the EnergyFee of the PriceRule with the corresponding color and then a sum is computed.

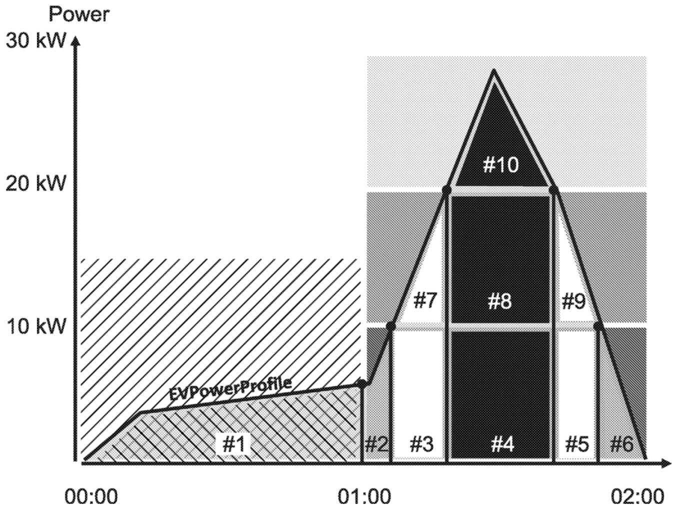

Figure 150 — Applying the PriceAlgorithm for 1-Power

[V2G20-1894] The PriceAlgorithm identifier "urn:iso:std:iso:15118:-20:PriceAlgorithm:1-Power" shall identify an algorithm that follows the principles illustrated in Figure 150. For every given point in time the EnergyFee shall be applied which is defined in the PriceRule for the present power which is applied by the chosen charging technology.

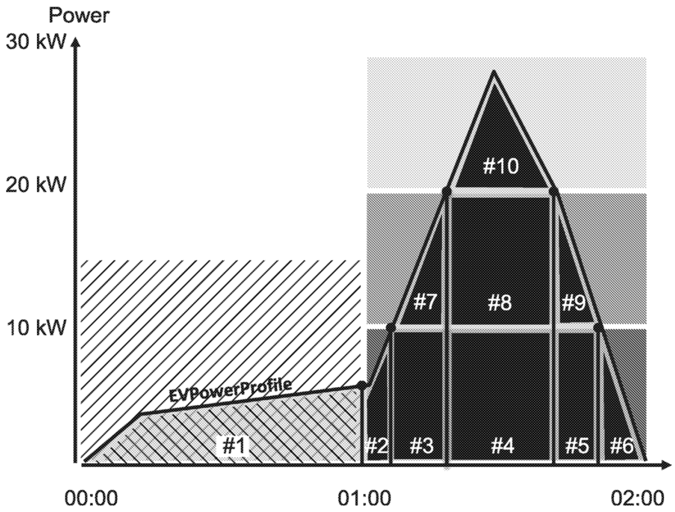

Figure 151 — Applying the PriceAlgorithm for 2-PeakPower

[V2G20-1895] The PriceAlgorithm identifier "urn:iso:std:iso:15118:-20:PriceAlgorithm:2-PeakPower" shall identify an algorithm that follows the principles illustrated in Figure 151. For every time span with one or multiple stacked PriceRules, all energy within that time span shall be priced according to the EnergyFee defined in the PriceRule that matches the highest peak power applied by the chosen charging technology during that time span.

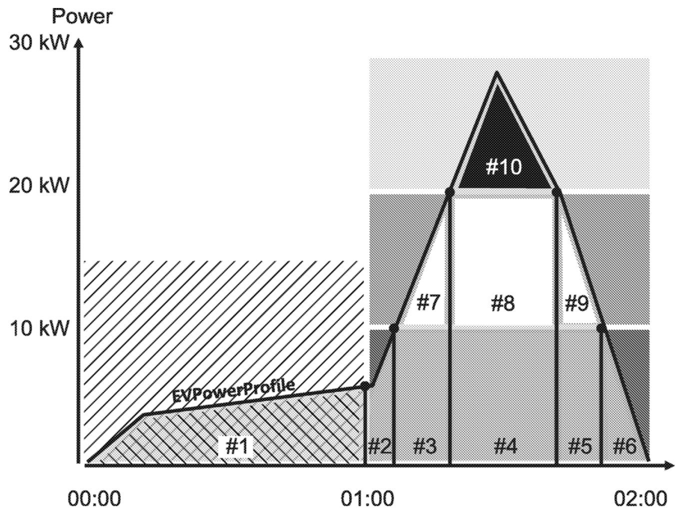

Figure 152 — Applying the PriceAlgorithm for 3-StackedEnergy

[V2G20-1896] The PriceAlgorithm identifier "urn:iso:std:iso:15118:-20:PriceAlgorithm:3-StackedEnergy" shall identify an algorithm that follows the principles illustrated in Figure 152. Energy shall be priced based on the EnergyFee defined in the PriceRules to the individual power bands and the amount of energy that was transferred within each corresponding power band.

NOTE 1 During the actual charging loop the present power level which is relevant for the PriceAlgorithm can be observed by the EVCC in different ways. For AC charging it is provided by the SECC via the EVSEPresentActivePower element. For DC charging it needs be computed based on the EVSEPresentVoltage and EVSEPresentCurrent elements.

NOTE 2 The use of unique URN based identifiers for the PriceAlgorithm allows an interoperable and well defined support of future proprietary use cases like fleet applications or special algorithms which can possibly be mandated by individual national regulation.

8.3.5.3.50 TaxRuleListType

This type represents the list of tax rules to be applied to a charging session.

[V2G20-1894] The SECC and the EVCC shall implement this type as defined in Figure 153 and Table 144.

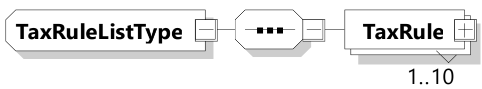

Figure 153 — Schema diagram – TaxRuleListType

The elements of this message are used according to Table 144.

Table 144 — Semantics and type definition for TaxRuleListType

Element name	Type	Semantics
TaxRule	complexType: TaxRuleType refer to 8.3.5.3.51	Individual entry of a list of tax rules

8.3.5.3.51 TaxRuleType

This type represents the tax rule to be applied to a charging session.

[V2G20-1898] The SECC and the EVCC shall implement this type as defined in Figure 154 and Table 145.

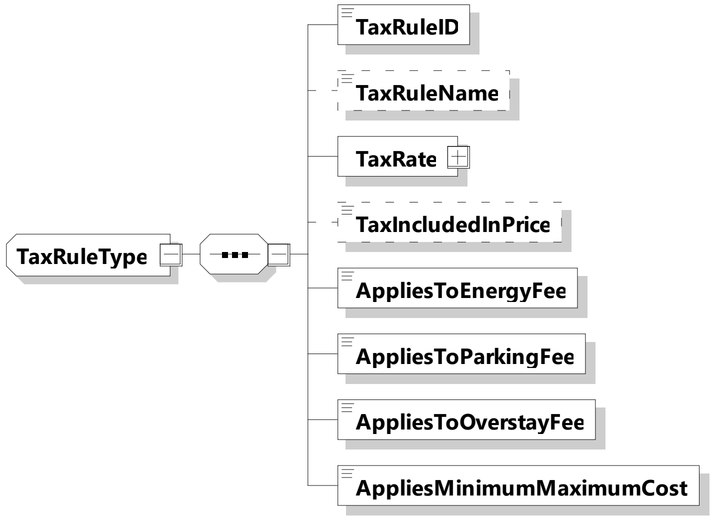

Figure 154 — Schema diagram – TaxRuleType

The elements of this message are used according to Table 145.

Table 145 — Semantics and type definition for TaxRuleType

Element name	Type	Semantics
TaxRuleID	simpleType: numericIDType refer to Annex A for the type definition	Identifier for the tax rule.
TaxRuleName	simpleType: nameType refer to Annex A for the type definition	Optional: Human readable string to identify the tax rule.
TaxRate	complexType: RationalNumberType Refer to 8.3.5.3.8	Percentage of the total amount of applying fee (energy, parking, overstay, MinimumCost and/or MaximumCost)
TaxIncludedInPrice	simpleType: xs:boolean	Optional: Indicates whether the tax is included in any price or not.
AppliesToEnergyFee	simpleType: xs:boolean	Indicates whether this tax applies to Energy Fees.
AppliesToParkingFee	simpleType: xs:boolean	Indicates whether this tax applies to Parking Fees.
AppliesToOverstayFee	simpleType: xs:boolean	Indicates whether this tax applies to Overstay Fees.

Element name	Type	Semantics
AppliesToMinimumMaximumCost	simpleType: xs:boolean	Indicates whether this tax applies to MinimumCost and/or MaximumCost.

[V2G20-1899] Each TaxRuleID shall be unique in AbsolutePrice.

8.3.5.3.52 PriceRuleStackListType

This type contains all of the pricing rules that will apply to a charging session.

[V2G20-2155] The SECC and the EVCC shall implement this type as defined in Figure 155 and Table 146.

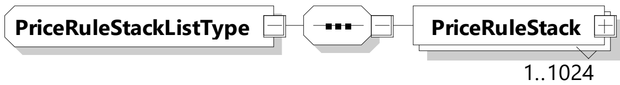

Figure 155 — Schema diagram – PriceRuleStackListType

The elements of this message are used according to Table 146.

Table 146 — Semantics and type definition for PriceRuleStackListType

Element name	Type	Semantics
PriceRuleStack	complexType: PriceRuleStackType Refer to 8.3.5.3.53	Contains the PriceRules. The number of PriceRuleStack elements is limited by the parameter MaximumSupportingPoints (according to [V2G20-1056]).

8.3.5.3.53 PriceRuleStackType

This type represents the collection of pricing rules that will apply to a charging session. This element defines how long the various price rules will be active.

[V2G20-1900] The SECC and the EVCC shall implement this type as defined in Figure 156 and Table 147.

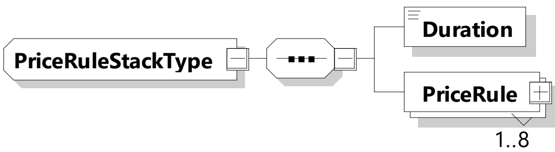

Figure 156 — Schema diagram – PriceRuleStackType

The elements of this message are used according to Table 147.

To enable error handling for a V2G communication session setup the EVCC monitors the time between reception of SECC discovery response, SessionSetupRes and PowerDeliveryRes, respectively. This allows the EVCC to decide about a successful or failed charging session after the defined timeouts.

The monitoring of a V2G communication session is based on the following timing concepts:

– message timing: monitors the timing between the request and the response of a Request-Response-Pair. This allows, for example, the EVCC application to initiate the error handling if no response is sent;
– sequence timing: monitors the timing of subsequent request-response-pairs. This allows, for example, the SECC application to initiate the error handling if an expected next request message was not sent;
– ongoing timing: monitors the timing in case of sending a request-response-pair repeatedly based on the parameter EVSEProcessing equal to "Ongoing". This allows, for example, the EVCC to initiate the error handling in case the SECC exceeds the processing time;
– communication setup timing: monitors the time from the moment of an established data link until the session setup message. It allows deciding if the communication setup was successful within a defined time.

The timers are compared to predefined time values as decision criterion. The EVCC and the SECC distinguish between two categories:

– timeout: if the specified time is exceeded the related error handling is initiated;
– performance time: if the specified time is exceeded the performance requirement is not fulfilled.

NOTE 1 While exceeding a timeout always causes an error handling, the performance time does not necessarily cause error handling if not defined differently by requirements. Depending on the system behavior (e.g. transmission time) no error can occur if the corresponding communication partner does not detect a timeout but the probability for causing a timeout is high.

NOTE 2 Timeouts are observed at the application layer where they are also enforced (according to requirement [V2G20-1966] for example). It is important for each sending party to keep in mind that all processing (e.g. TLS encryption) and transmission delays between both application layer sides will be part of the observed timing on the receiving side. See Figure 212 for a detailed illustration.

NOTE 3 This document supports PLC and WiFi communication. Both technologies come with different average and worst case latencies as well as congestion and retransmissions characteristics. For example, in WiFi networks real world round trip latencies can reach up to 25 ms to 100 ms or even more. As mentioned before it is important that those latencies are taken into account when deciding on the implementation of a message timing strategy in the application layer of the sending side. The timing requirements of Table 215 have been defined with the assumption that round-trip end-to-end transmission delays will never exceed 250 ms.

8.5.3 DC service

The monitoring of a DC V2G communication session includes the following timing concepts:

– DC_CableCheck timing: The monitoring of the cable check is carried out by the EV using the V2G_EVCC_DC_CableCheck_Timer. It is started when the EV requests the EVSE to start the Cable Check, and ends when the EVSE has finished the cable check, or when the V2G_EVCC_DC_CableCheck_Timer expires;
– DC_PreCharge timing: The monitoring of the pre charge is carried out by the EV using the V2G_EVCC_DC_PreCharge_Timer. It is started when the EV starts the pre charging by sending the first DC_PreChargeReq message, and ends when the pre charging has finished, indicated by the EV determining that the EVSE output voltage, as measured inside the EV, has sufficiently been adjusted to the EV RESS voltage, or when the V2G_EVCC_DC_PreCharge_Timer expires.

8.5.4 Message sequence and communication session

8.5.4.1 Common

Message timers, sequence timers, timeouts, and performance times are defined for EVCC and SECC separately and are summarized in Table 214. Timeouts and performance times are parameterized for messages separately to describe different processing times. Table 215 defines the common values for each V2G message type.

[V2G20-434] The EVCC shall implement the EVCC specific timeouts and performance times defined in Table 214 and Table 215.

[V2G20-435] The SECC shall implement the SECC specific timeouts and performance times defined in Table 214 and Table 215.

Table 214 — Common EVCC and SECC timers, timeouts, performance times

Name	Type	Applicable for	
		EVCC	SECC
V2G_EVCC_Msg_Timer	Message timer in the EVCC	x	
V2G_SECC_Msg_Timer	Message timer in the SECC		x
V2G_EVCC_Sequence_Timer	Sequence timer in the EVCC	x	
V2G_SECC_Sequence_Timer	Sequence timer in the SECC		x
V2G_EVCC_Ongoing_Timer	Ongoing timer in the EVCC	x	
V2G_SECC_Ongoing_Timer	Ongoing timer in the SECC		x
V2G_EVCC_Msg_Timeout (MessageType)	Timeout for the message timer The value is defined depending on the parameter MessageType as defined in Table 215.	x	
V2G_SECC_Msg_Performance_Time (MessageType)	Performance time for the message timer The value is defined depending on the parameter MessageType as defined in Table 215.		x
V2G_EVCC_Sequence_Performance_Time	Performance time for the sequence timer as defined in Table 215.	x	
V2G_SECC_Sequence_Timeout	Timeout for the sequence timer as defined in Table 215.		x
V2G_EVCC_Ongoing_Timeout	Timeout for ongoing timer	x	
V2G_SECC_Ongoing_Performance_Time	Performance time for ongoing timer		x

Table 215 — Common EVCC and SECC message sequence and session timing parameter values

Name	MessageType	Value [s]
V2G_EVCC_Msg_Timeout(MessageType)	SupportedAppProtocolReq	2
	SessionSetupReq	2
	VehicleCheckInReq	2

Name	MessageType	Value [s]
	VehicleCheckOutReq	2
	AuthorizationSetupReq	2
	AuthorizationReq	2
	CertificateInstallationReq	5
	ServiceDiscoveryReq	2
	ServiceDetailReq	5
	ServiceSelectionReq	2
	ChargeParameterDiscoveryReq	2
	ScheduleExchangeReq	2
	PowerDeliveryReq	2
	MeteringConfirmationReq	2
	SessionStopReq	2
V2G_SECC_Msg_Performance_Time(MessageType)	SupportedAppProtocolRes	1,5
	SessionSetupRes	1,5
	VehicleCheckInRes	1,5
	VehicleCheckOutRes	1,5
	AuthorizationSetupRes	1,5
	AuthorizationRes	1,5
	CertificateInstallationRes	4,5
	ServiceDiscoveryRes	1,5
	ServiceDetailRes	4,5
	ServiceSelectionRes	1,5
	ChargeParameterDiscoveryRes	1,5
	ScheduleExchangeRes	1,5
	PowerDeliveryRes	1,5
	MeteringConfirmationRes	1,5
	SessionStopRes	1,5
V2G_EVCC_Sequence_Performance_Time	(all other messages)	40
V2G_SECC_Sequence_Timeout	(all other messages)	60
V2G_EVCC_Ongoing_Timeout	Response messages with parameter EVSEProcessing equal to "Ongoing"	60
V2G_SECC_Ongoing_Timeout	Request messages with parameter EVProcessing equal to "Ongoing"	60

Name	MessageType	Value [s]
V2G_EVCC_Ongoing_Performance_Time	Request messages with parameter EVProcessing equal to "Ongoing"	55
V2G_SECC_Ongoing_Performance_Time	Response messages with parameter EVSEProcessing equal to "Ongoing"	55

Figure 212 illustrates how the common message timers, sequence timers, timeouts, and performance times are applied in the EVCC and the SECC.

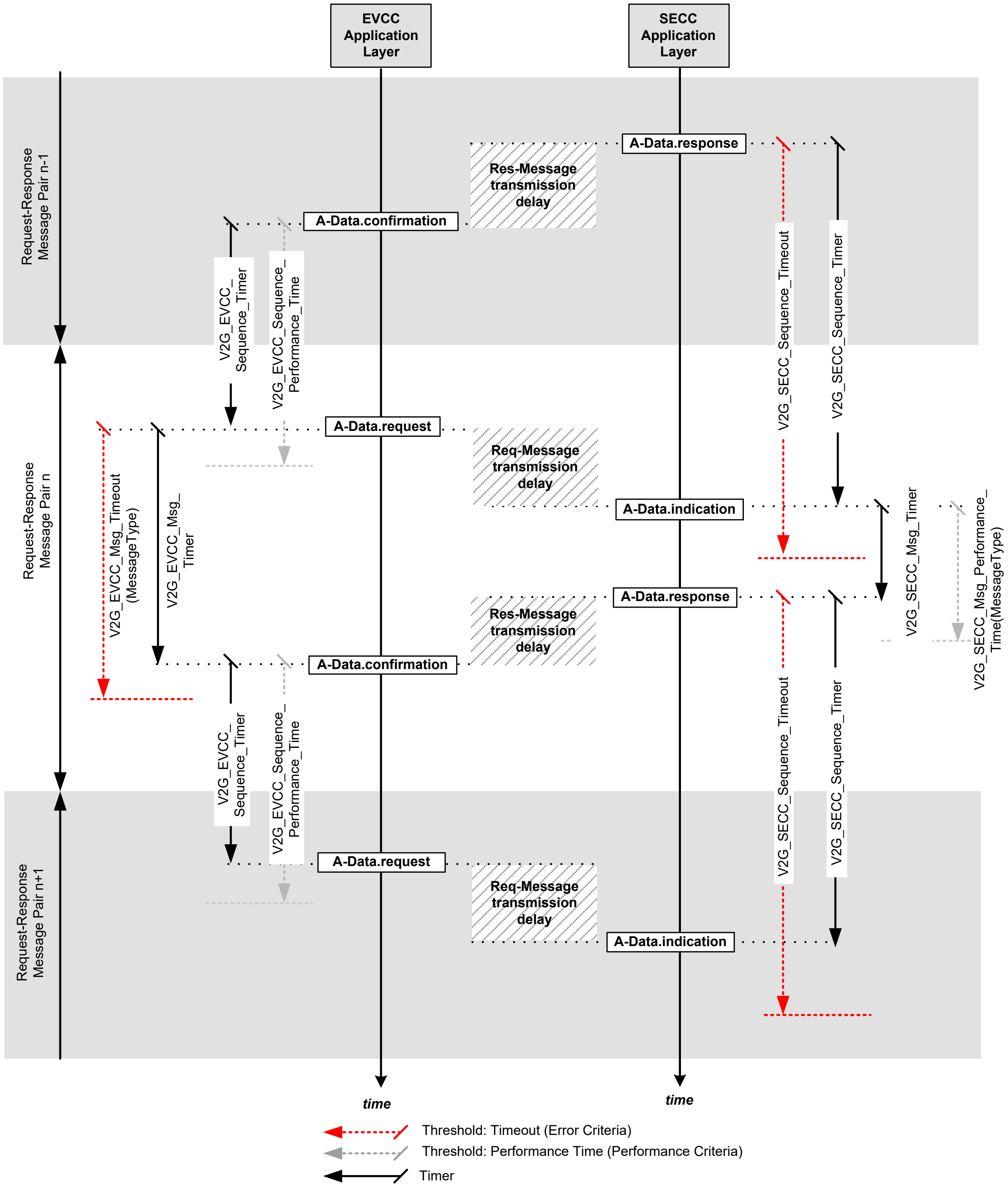

Figure 212 — Message sequence and session timing

8.5.4.1.1 EVCC timing for request-response message pairs

[V2G20-436] The EVCC shall set the timeout V2G_EVCC_Msg_Timeout depending on the value MessageType as defined in Table 215, reset the V2G_EVCC_Msg_Timer and start monitoring the V2G_EVCC_Msg_Timer when it sends a request message.

NOTE 1 In this document sending a request message is described by A-Data.request.

[V2G20-437] The EVCC shall wait for the response message corresponding to the request message sent before.

[V2G20-438] The EVCC shall stop waiting for the response message and stop monitoring the V2G_EVCC_Msg_Timer when V2G_EVCC_Msg_Timer is equal or larger than V2G_EVCC_Msg_Timeout(MessageType) and no response message was received. It shall then apply the error handling as defined in 8.6.

NOTE 2 In this document receiving a response message is described by A-DATA.confirmation.

[V2G20-439] The EVCC shall stop waiting for the response message and stop monitoring the V2G_EVCC_Msg_Timer when V2G_EVCC_Msg_Timer is smaller than V2G_EVCC_Msg_Timeout(MessageType) and it received a response message. It shall then process the response message as defined in 8.6.

NOTE 3 In this document receiving a response message is described by A-Data.confirmation.

[V2G20-440] The EVCC shall ignore any message that is not a valid response message.

8.5.4.1.2 SECC timing for response-request message sequence

[V2G20-441] The SECC shall set the timeout V2G_SECC_Sequence_Timeout to the value as defined in Table 215, reset the V2G_SECC_Sequence_Timer and start monitoring the V2G_SECC_Sequence_Timer when it sends a response message.

NOTE 4 In this document sending a response message is described by A-Data.response.

[V2G20-442] The SECC shall wait for a request message.

[V2G20-443] The SECC shall stop waiting for a request message and stop monitoring the V2G_SECC_Sequence_Timer when V2G_SECC_Sequence_Timer is equal or larger than V2G_SECC_Sequence_Timeout and no request message was received. It shall then stop the V2G communication session.

NOTE 5 In this document receiving a request message is described by A-Data.indication. A-Data.indication (A_Msg="message name") signalizes the successful reception of a valid request message for the V2G message that is given by A_Msg where "Valid message" means that all mandatory elements are filled in so that it can be deserialized.

[V2G20-444] The SECC shall stop waiting for a request message and stop monitoring the V2G_SECC_Sequence_Timer when V2G_SECC_Sequence_Timer is smaller than V2G_SECC_Sequence_Timeout and it received a request message. It shall then process the response message as defined in 8.6.

NOTE 6 In this document receiving a request message is described by A-Data.indication.

[V2G20-445] The SECC shall ignore any message that is not a valid request message.

8.5.4.2 AC specific message sequence and session timing

[V2G20-1499] The EVCC shall implement the EVCC specific AC timeouts and performance times defined in Table 216.

[V2G20-1500] The SECC shall implement the SECC specific AC timeouts and performance times defined in Table 216.

Table 216 — AC EVCC and SECC timeouts and performance times

Name	MessageType	Value [s]
V2G_EVCC_Msg_Timeout(MessageType)	AC_ChargeLoopReq	0,5
V2G_SECC_Msg_Performance_Time(MessageType)	AC_ChargeLoopReq	0,25
V2G_EVCC_Sequence_Performance_Time	AC_ChargeLoopReq	0,25
V2G_SECC_Sequence_Timeout	AC_ChargeLoopRes	0,5

8.5.4.3 DC specific message sequence and session timing

[V2G20-1501] The EVCC shall implement the EVCC specific DC timeouts and performance times defined in Table 217.

[V2G20-1502] The SECC shall implement the SECC specific DC timeouts and performance times defined in Table 217.

Table 217 — DC EVCC and SECC timeouts and performance times

Name	MessageType	Value [s]
V2G_EVCC_Msg_Timeout(MessageType)	DC_CableCheckReq	2
	DC_PreChargeReq	2
	DC_ChargeLoopReq	0.5
	DC_WeldingDetectionReq	2
V2G_SECC_Msg_Performance_Time(MessageType)	DC_CableCheckReq	1,5
	DC_PreChargeReq	1,5
	DC_ChargeLoopReq	0,25
	DC_WeldingDetectionReq	1,5
V2G_EVCC_Sequence_Performance_Time	DC_ChargeLoopReq	0,25
V2G_SECC_Sequence_Timeout	DC_ChargeLoopRes	0,5

8.5.4.4 WPT specific message sequence and session timing

[V2G20-5069] The EVCC shall implement the EVCC specific WPT timeouts and performance times defined in Table 218.

[V2G20-5070] The SECC shall implement the SECC specific WPT timeouts and performance times defined in Table 218.

Table 218 — WPT EVCC and SECC timeouts and performance times

Name	MessageType	Value [s]
V2G_EVCC_Msg_Timeout(MessageType)	WPT_FinePositioningSetupReq	2
	WPT_FinePositioningReq	2
	WPT_PairingReq	2
	WPT_AlignmentCheckReq	2

	WPT_ChargeLoopReq	0,5
V2G_SECC_Msg_Performance_Time(MessageType)	WPT_FinePositioningSetupReq	1,5
	WPT_FinePositioningReq	1,5
	WPT_PairingReq	1,5
	WPT_AlignmentCheckReq	1,5
	WPT_ChargeLoopReq	0,25
V2G_EVCC_Sequence_Performance_Time	WPT_ChargeLoopReq	0,25
V2G_SECC_Sequence_Timeout	WPT_ChargeLoopRes	0,5

8.5.4.5 ACDP specific message sequence and session timing

[V2G20-4020] The EVCC shall implement the EVCC specific ACDP timeouts and performance times defined in Table 219.

[V2G20-4021] The SECC shall implement the SECC specific ACDP timeouts and performance times defined in Table 219.

Table 219 — ACDP EVCC and SECC timeouts and performance times

Name	MessageType	Value [s]
V2G_EVCC_Msg_Timeout(MessageType)	ACDP_VehiclePositioningReq	2
	ACDP_ConnectReq	2
	ACDP_DisconnectReq	2
	ACDP_SystemStatusReq	2
V2G_SECC_Msg_Performance_Time(MessageType)	ACDP_VehiclePositioningReq	1,5
	ACDP_ConnectReq	1,5
	ACDP_DisconnectReq	1,5
	ACDP_SystemStatusReq	1,5

8.5.5 Session setup and ready to charge

8.5.5.1 Common

Timing parameters applicable to the common communication session setup and ready to charge time defined in this document are shown in Table 220. Table 221 defines the values for the related common performance times and the timeouts.

Table 220 — Common EVCC and SECC V2G communication session setup timing parameters

Parameter name	Definition	Implementation	
		EVCC	SECC
V2G_EVCC_CommunicationSetup_Timer	Communication setup timer in the EVCC	x	
V2G_SECC_CommunicationSetup_Timer	Communication setup timer in the SVCC		x

Parameter name	Definition	Implementation	
		EVCC	SECC
V2G_EVCC_CommunicationSetup_Timeout	Timeout for the communication setup timer as defined in Table 221.	x	
V2G_SECC_CommunicationSetup_Performance_ Time	Performance time for the communication setup timer as defined in Table 221.		x

[V2G20-605] The EVCC and SECC shall implement the timing parameter values defined in Table 221.

Table 221 — Common EVCC and SECC message sequence and session timing parameter values

Parameter name	Value [s]	Implementation	
		EVCC	SECC
V2G_SECC_CommunicationSetup_Performance_Time	18		x
V2G_EVCC_CommunicationSetup_Timeout	20	x	

Figure 213 illustrates how the timing parameters defined in Table 220 are applied.

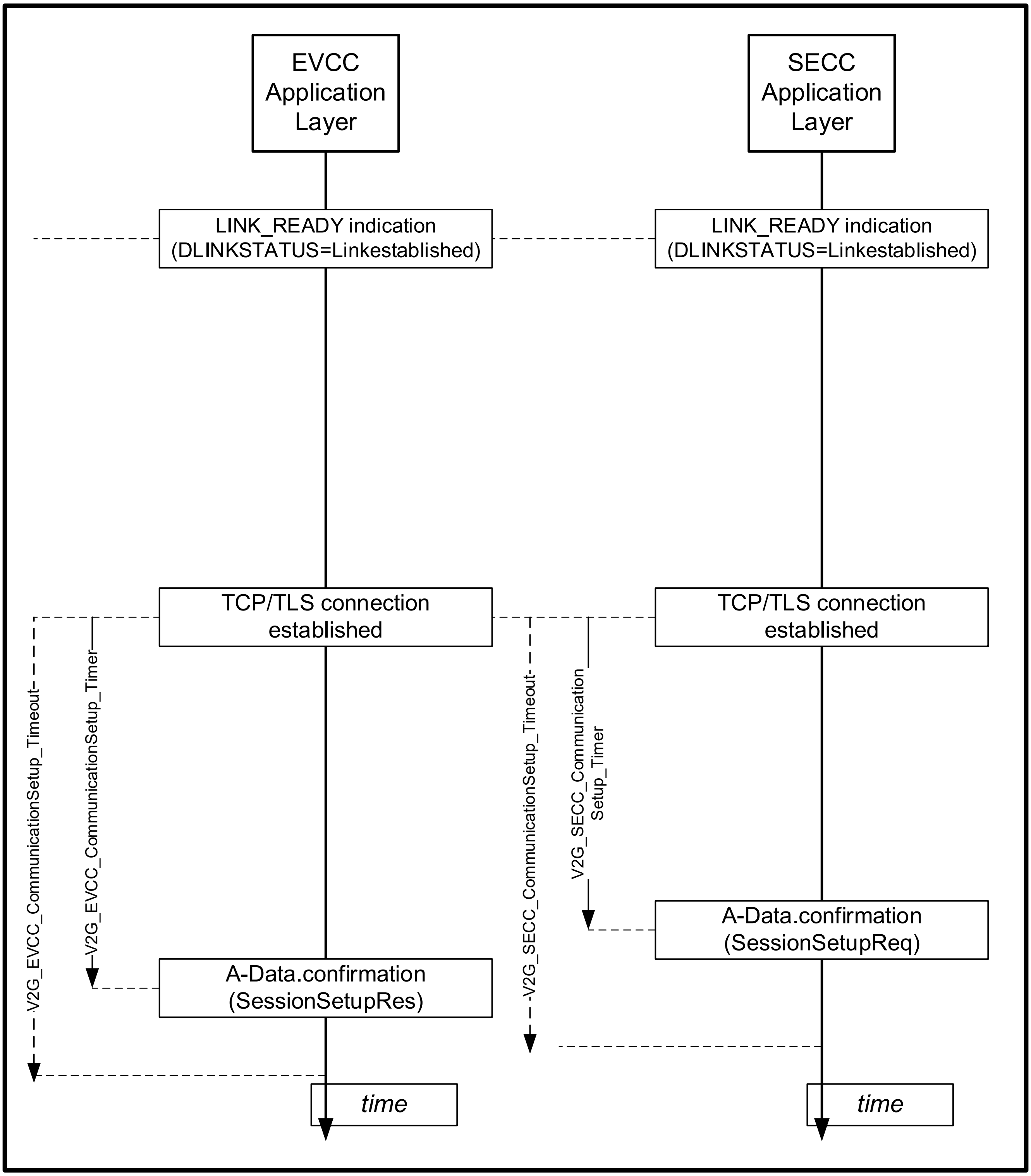

Figure 213 — CommunicationSetup timing

8.5.5.1.1 EVCC Timing for communication session setup

[V2G20-2295] If the EVCC receives an SDP response message with parameter DiagStatus equal to "Ongoing" for the first time it shall start the timer V2G_EVCC_Ongoing_Timer and wait for parameter DiagStatus equal to "Finished with EVSEID" or parameter DiagStatus equal to "Finished without EVSEID".

[V2G20-2296] If [V2G20-2295] applies, the EVCC shall stop and then restart the SDP process when V2G_EVCC_Ongoing_Timer is equal or larger than V2G_EVCC_Ongoing_Timeout and no parameter EVSEProcessing equal to "Finished" has been received.

[V2G20-2298] The EVCC shall set the timeout V2G_EVCC_CommunicationSetup_Timeout to the value as defined in Table 221, reset the V2G_EVCC_CommunicationSetup_Timer and start monitoring the V2G_EVCC_CommunicationSetup_Timer when a successful TCP/TLS connection has been established.

[V2G20-447] The EVCC shall wait for the SessionSetupRes message.

[V2G20-448] The EVCC shall stop waiting for the SessionSetupRes message and stop monitoring the V2G_EVCC_CommunicationSetup_Timer when V2G_EVCC_CommunicationSetup_Timer is equal or larger than V2G_EVCC_CommunicationSetup_Timeout and no SessionSetupRes message was received. It shall then stop the V2G communication session.

NOTE 1 In this document receiving the response message "SessionSetupRes" is described by A-Data.confirmation (SessionSetupRes).

[V2G20-449] The EVCC shall stop waiting for the SessionSetupRes message and stop monitoring the V2G_EVCC_CommunicationSetup_Timer when V2G_EVCC_CommunicationSetup_Timer is smaller than V2G_EVCC_CommunicationSetup_Timeout and a SessionSetupRes message was received. It shall then process the response message as defined in 8.6.

NOTE 2 In this document receiving the response message "SessionSetupRes" is described by A-Data.confirmation (SessionSetupRes).

8.5.5.1.2 SECC Timing for communication session setup

[V2G20-2303] The SECC shall set the timeout V2G_SECC_CommunicationSetup_Performance_Time the value as defined in Table 221, reset the V2G_SECC_CommunicationSetup_Timer and start monitoring the V2G_SECC_CommunicationSetup_Timer when a successful TCP/TLS connection has been established.

[V2G20-715] The SECC shall wait for the SessionSetupReq message.

[V2G20-2300] The SECC shall stop waiting for SessionSetupReq and stop monitoring the V2G_SECC_CommunicationSetup_Timer when V2G_SECC_CommunicationSetup_Timer is smaller than V2G_SECC_CommunicationSetup_Performance_Time and a SessionSetupRes message was received. It shall then process the response message as defined in 8.6.

[V2G20-716] The SECC shall stop waiting for SessionSetupReq and stop monitoring the V2G_SECC_CommunicationSetup_Timer when V2G_SECC_CommunicationSetup_Timer is equal or larger than V2G_SECC_CommunicationSetup_Performance_Time and no SessionSetupRes message was sent. It shall then apply [V2G20-034].

NOTE In this document sending the response message "SessionSetupRes" is described by A-Data.response(SessionSetupRes).

8.5.5.1.3 EVCC Timing for EVSEProcessing parameter

[V2G20-710] If the EVCC receives a V2G response message with parameter EVSEProcessing equal to "Ongoing" for the first time in a response message it shall start the timer V2G_EVCC_Ongoing_Timer and wait for parameter EVSEProcessing equal to "Finished".

[V2G20-2101] If [V2G20-710] applies, the EVCC may stop the V2G communication session when V2G_EVCC_Ongoing_Timer is equal or larger than V2G_EVCC_Ongoing_Timeout and no parameter EVSEProcessing equal to "Finished" has been received.

8.5.5.1.4 SECC Timing for EVSEProcessing parameter

[V2G20-712] If the SECC sends a V2G response message with parameter EVSEProcessing equal to "Ongoing" for the first time in a response message it shall start the timer V2G_SECC_Ongoing_Timer.

[V2G20-2102] If [V2G20-712] applies, the SECC may send ResponseCode equal to "FAILED" when V2G_SECC_Ongoing_Timer is equal or larger than V2G_SECC_Ongoing_Performance_Time and no parameter EVSEProcessing equal to "Finished" has been sent. The SECC shall stop the V2G communication session.

8.5.5.2 DC specific timings

Timing parameters applicable to the DC communication session setup and ready to charge time defined in this document are shown in Table 222. Table 223 defines the values for the related DC performance times and the timeouts.

Table 222 — DC specific EVCC and SECC V2G communication session setup timing parameters

Parameter name	Definition	Implementation	
		EVCC	SECC
V2G_EVCC_DC_CableCheck_Timer	Cable check timer in the EVCC	x	
V2G_SECC_DC_CableCheck_Timer	Cable check timer in the SECC		x
V2G_EVCC_DC_PreCharge_Timer	DC_PreCharge timer in the EVCC	x	
V2G_SECC_DC_PreCharge_Timer	DC_PreCharge timer in the SECC		x
V2G_EVCC_DC_CableCheck_Timeout	Timeout for the DC_CableCheck timer as defined in Table 223.	x	
V2G_SECC_DC_CableCheck_Performance_Time	Performance time for the DC_CableCheck timer as defined in Table 223.		x
V2G_EVCC_DC_PreCharge_Timeout	Timeout for the DC_PreCharge timer as defined in Table 223.	x	
V2G_SECC_DC_PreCharge_Performance_Time	Performance time for the DC_PreCharge timer as defined in Table 223.		x

[V2G20-1774] The EVCC and SECC shall implement the DC timing parameter values defined in Table 223.

Table 223 — DC specific EVCC and SECC message sequence and session timing parameter values

Parameter name	Value [s]	Implementation	
		EVCC	SECC
V2G_SECC_DC_CableCheck_Performance_Time	38		x
V2G_EVCC_DC_CableCheck_Timeout	40	x	
V2G_SECC_DC_PreCharge_Performance_Time	9		x
V2G_EVCC_DC_PreCharge_Timeout	10	x	

8.5.5.2.1 EVCC Timing for cable check

[V2G20-700] The EVCC shall set the timeout V2G_EVCC_DC_CableCheck_Timeout to the value as defined in Table 223, reset the V2G_EVCC_DC_CableCheck_Timer and start monitoring the V2G_EVCC_DC_CableCheck_Timer when sending the message DC_CableCheckReq for the first time in a charging session.

NOTE 1 In this document, sending a request message is described by A-Data.request.

[V2G20-1397] The EVCC shall wait for the cable check of the EVSE to finish indicated by the reception of a DC_CableCheckRes with ResponseCode equal to "OK" and EVSEProcessing equal to "Finished".

NOTE 2 In this document, receiving a response message is described by A-Data.confirmation.

[V2G20-1398] The EVCC shall stop waiting for the Cable check of the EVSE to finish and stop monitoring the V2G_EVCC_DC_CableCheck_Timer when V2G_EVCC_DC_CableCheck_Timer is equal or larger than V2G_EVCC_DC_CableCheck_Timeout. It shall then apply the error handling as defined in 8.6.

[V2G20-1399] The EVCC shall stop waiting for the cable check of the EVSE to finish and stop monitoring the V2G_EVCC_DC_CableCheck_Timer if V2G_EVCC_DC_CableCheck_Timer is smaller than V2G_EVCC_DC_CableCheck_Timeout and a DC_CableCheckRes message with ResponseCode equal to "OK" and EVSEProcessing equal to "Finished" was received. It shall then process the response message as defined in 8.6.

NOTE 3 In this document, receiving a response message is described by A-Data.confirmation.

8.5.5.2.2 EVCC Timing for pre charging

[V2G20-704] The EVCC shall set the timeout V2G_EVCC_DC_PreCharge_Timeout to the value as defined in Table 223, reset the V2G_EVCC_DC_PreCharge_Timer and start monitoring the V2G_EVCC_DC_PreCharge_Timer when sending the message DC_PreChargeReq for the first time in a charging session.

NOTE 4 In this document, sending a request message is described by A-Data.request.

[V2G20-1400] The EVCC shall wait for the pre charging to finish, indicated by the EV determining that the EVSE output voltage, as measured inside the EV, has sufficiently been adjusted to the EV RESS voltage.

[V2G20-706] The EVCC shall stop waiting for the pre charging to finish and stop monitoring the V2G_EVCC_DC_PreCharge_Timer when V2G_EVCC_DC_PreCharge_Timer is equal or larger than V2G_EVCC_DC_PreCharge_Timeout. It shall then apply the error handling as defined in 8.6.

[V2G20-1401] The EVCC shall stop monitoring the V2G_EVCC_DC_PreCharge_Timer when V2G_EVCC_DC_PreCharge_Timer is smaller than V2G_EVCC_DC_PreCharge_Timeout and pre charging has finished, indicated by the EV determining that the EVSE output voltage, as measured inside the EV, has sufficiently been adjusted to the EV RESS voltage. It shall then process the response message as defined in 8.6.

8.5.5.3 WPT specific timings

Operational timing requirements are given in IEC 61980-2.

8.5.6 V2G message synchronization for AC and DC with IEC 61851-1 signalling

8.5.6.1 Overview

ISO 15118 based energy transfer control extends PWM signal over a control pilot wire according to IEC 61851-1:2017, Annex A. For this, the messaging on application layer is synchronized with the CP states defined in IEC 61851-1:2017, Annex A.

ISO 15118 based messaging is able to manage the AC and DC charging process for a complete charging session from the beginning to the end in 5 % or 100 % duty cycle case.

In this subclauses, terms and definitions in requirements are applied as defined in ISO 15118-3 and IEC 61851-1:2017, Annex A.

Figure 214 shows an example for AC energy transfer with BC and HLC-C in relation to service session and phases of data link setup, V2G setup and V2G power transfer loop during a V2G communication session.

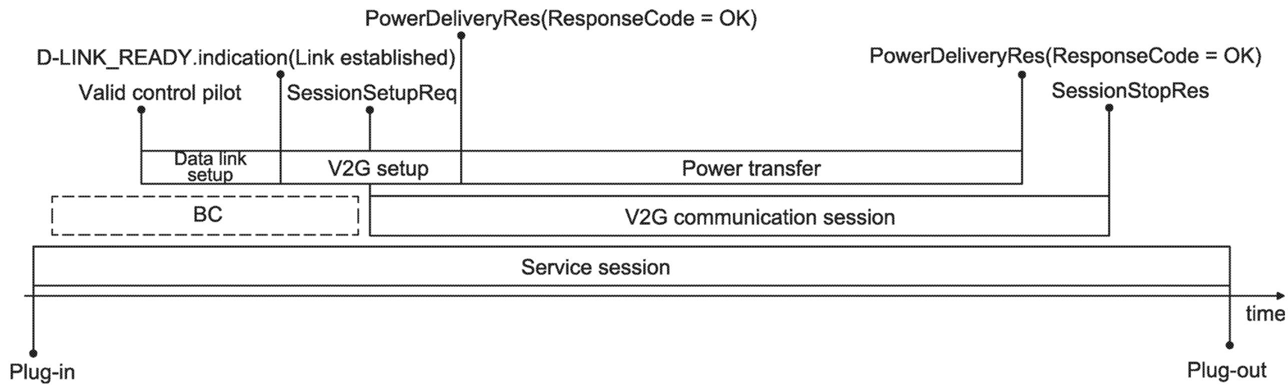

Figure 214 — AC example for BC and HLC-C energy transfer in relation to service session

The V2G power transfer loop is defined as V2G messaging phase for controlling the energy transfer process by the ISO 15118 series in normal operation. The power transfer phase under control of the ISO 15118 series is defined as high level communication charging (HLC-C).

The entry and exit conditions for the V2G power transfer loop are as follows:

- entry condition: PowerDeliveryRes message with parameter ResponseCode equal to OK after PowerDeliveryReq with ChargeProgress equals "Start";
- exit condition: PowerDeliveryRes with parameter ResponseCode equal to OK after PowerDeliveryReq with ChargeProgress equals "Stop".

Besides the requirements for request-response message pairs and request-response message sequences as defined in 8.6.4, the EVCC and the SECC shall conform to the requirements as defined in 8.5.6.2, 8.5.6.3, and 8.5.6.4.

8.5.6.2 Common requirements for AC and DC

The following requirements apply for the EVCC.

[V2G20-1615] The EVCC shall use the message PowerDeliveryReq with parameter ChargeProgress set to "Start" to start HLC-C based energy transfer.

[V2G20-1616] If [V2G20-1615] applies and the PowerDeliveryRes message has been received with ResponseCode set to OK, the EV shall apply the energy transfer limits according to the V2G messaging.

[V2G20-1310] The EV shall stop the energy transfer (no current drawn by EV) before sending the message PowerDeliveryReq with the parameter ChargeProgress set to "Stop".

[V2G20-1311] The EVCC shall use the message PowerDeliveryReq with parameter ChargeProgress set to "Stop" to stop HLC-C based energy transfer.

In general, the ISO 15118 series is based on the requirements as defined in IEC 61851-1.

[V2G20-843] An ISO 15118-enabled EV shall conform to IEC 61851-1.

[V2G20-731] If [V2G20-1463] applies, the EVCC can establish a new or resumed V2G communication session by applying [V2G20-014].

[V2G20-2200] If [V2G20-728] applies, the EVCC can establish a new V2G communication session by applying [V2G20-014].

Since the V2G session was terminated due to an error, the EVCC has to mitigate the conditions that resulted in the error before establishing a new V2G communication session.

The following requirements apply for the SECC.

[V2G20-1404] ISO 15118-enabled EVSE shall conform to IEC 61851-1.

[V2G20-2030] If [V2G20-1632] applies, the SECC shall be allowed to establish a new or resumed V2G communication session by applying [V2G20-027].

[V2G20-2115] At the point in time when requesting pause by setting the parameter EVSENotification to "Pause" in dynamic control mode, the SECC shall ensure that the applied power level is equal to 0 kW (no current drawn by EV or EVSE).

The following applies for the EVCC during a standby period.

[V2G20-1390] When the EVCC wants to enter a standby period, after having received the ScheduleExchangeRes with parameter "EVSEProcessing" set to "Finished", the EVCC shall send a PowerDeliveryReq with parameter ChargeProgress set to "Standby", while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Performance_Time.

[V2G20-2113] After receiving the PowerDeliveryRes with ResponseCode equal to "OK" as a response to a previous PowerDeliveryReq message with ChargeProgress equal to "Standby", the EVCC shall send a ChargeLoopReq while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Performance_Time.

[V2G20-2116] At the point in time when requesting standby by PowerDeliveryReq with parameter ChargeProgress= "Standby", the EVCC shall ensure that the applied power level is equal to zero kW (no current drawn by EV or EVSE).

[V2G20-2117] After receiving a PowerDeliveryRes with ResponseCode equal to "OK" as a response to the previous PowerDeliveryReq message with ChargeProgress equal to "Standby", the EVCC shall not process any elements in the ChargeLoopRes except EVSEStatus, ResponseCode, MeterInfo and Receipt, until the next PowerDeliveryReq message with ChargeProgress set to "Start", "Stop", or "Renegotiate" is send.

During standby both EV and EVSE contactors are to stay closed. The standby concept is solely used as communication state within this document and therefore is not reflected in related documents, e.g. IEC 61851-1 or IEC 61851-23.

[V2G20-1391] When the EVCC wants to enter a standby period, after having received ChargeLoopRes, the EVCC shall send a PowerDeliveryReq with parameter ChargeProgress set to "Standby", while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Performance_Time.

[V2G20-1393] When the EVCC wants to exit a standby period, after having received ChargeLoopRes, the EVCC shall send a PowerDeliveryReq with parameter ChargeProgress set to "Start", "Stop", or "Renegotiate", while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Performance_Time.

[V2G20-1394] When the EVCC wants to enter a standby period, after having received MeteringConfirmationRes, the EVCC shall send a PowerDeliveryReq with parameter ChargeProgress set to "Standby", while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Performance_Time.

[V2G20-1395] When the EVCC wants to exit a standby period, after having received MeteringConfirmationRes, the EVCC shall send a PowerDeliveryReq with parameter ChargeProgress set to "Start", "Stop", or "Renegotiate", while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Performance_Time.

The following applies for the SECC during a standby period.

[V2G20-1385] The SECC shall enter standby, after having received PowerDeliveryReq with ChargeProgress set to "Standby", by sending a PowerDeliveryRes with ResponseCode is set to "OK" while V2G_SECC_Sequence_Timer is smaller than V2G_SECC_Sequence _Timeout.

[V2G20-2118] After sending a PowerDeliveryRes with ResponseCode equal to "OK" as a response to the previous PowerDeliveryReq message with ChargeProgress equal to "Standby", the SECC shall not process any elements in the ChargeLoopReq except MeterInfoRequested and DisplayParameters, until the next PowerDeliveryReq message with ChargeProgress set to "Start", "Stop", or "Renegotiate" is received.

During standby both EV and EVSE contactors are to stay closed. The standby concept is solely used as communication state within this document and therefore is not reflected in related documents, e.g. IEC 61851-1 or IEC 61851-23.

[V2G20-1386] The SECC shall exit standby, after having received PowerDeliveryReq with ChargeProgress set to "Start", "Stop", or "Renegotiate", by sending a PowerDeliveryRes with ResponseCode set to "OK", while V2G_SECC_Sequence_Timer is smaller than V2G_SECC_Sequence_Timeout.

[V2G20-1700] The message PowerDeliveryRes shall contain the ResponseCode "WARNING_StandbyNotAllowed" if the EVCC has sent a PowerDeliveryReq containing the parameter ChargeProgress set to "Standby" and the SECC does not allow for a standby period. All mandatory parameters shall be filled in with arbitrary XSD conform values. All values shall be chosen in a way that results in a minimal size of the response message.

The following applies if the EVCC is unable to enter a standby period.

[V2G20-1396] After receiving the message "PowerDeliveryRes" with ResponseCode equal to "WARNING_StandbyNotAllowed", the EVCC shall ignore all other parameters of the response message and shall do one (1) of the following.

- Proceed with the initially agreed EVPowerProfile or in dynamic control mode follow the demand of the SECC within V2G_EVCC_Sequence_Performance_Timeout according to Table 215. The next allowed message shall be the ChargeLoopReq message. Refer to 8.3.1 and 8.3.4.1 for details of requesting standby of the V2G session.

OR

- Request schedule renegotiation within V2G_EVCC_Sequence_Performance_Timeout according to Table 215. Refer to Annex F for further details.

OR

- Stop the V2G communication according to requirements defined 7.4.

The following applies if the SECC declines a request to enter a standby period from the EVCC.

[V2G20-1387] After receiving the PowerDeliveryReq with ChargeProgress set to "Standby", the SECC shall send a PowerDeliveryRes with ResponseCode set to "WARNING_StandbyNotAllowed" while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Timeout, if the SECC does not allow a standby period. All mandatory parameters shall be filled in with arbitrary XSD conform values. All values shall be chosen in a way that results in a minimal size of the response message.

NOTE After declining a standby request from the EVCC, the SECC expects the EVCC to proceed according to [V2G20-1396].

8.5.6.3 AC specific requirements

The following requirements apply for the EVCC.

[V2G20-930] The EVCC shall measure a PWM of 5 % duty cycle before sending the message ChargeParameterDiscoveryReq.

NOTE 1 In the case of AC energy transfer, the HLC can be started with using a 5 % PWM or a 100 % PWM (static current).

[V2G20-1617] The EVCC shall signal CP State B before sending the first PowerDeliveryReq with ChargeProgress equals "Start" within V2G communication session.

[V2G20-847] The EVCC shall signal CP State C or D no later than 250 ms after sending the first PowerDeliveryReq with ChargeProgress equals "Start" within V2G communication session.

[V2G20-848] The EVCC shall signal CP State B no later than 250 ms after sending PowerDeliveryReq with ChargeProgress equals "Stop".

[V2G20-849] If [V2G20-847] applies and no error has been identified, the EVCC shall not change the CP State until [V2G20-848] applies.

NOTE 2 In case of schedule renegotiation, the contactor stays closed to allow energy transfer based on the existing energy transfer limits during schedule renegotiation.

The following applies for the EVCC during a standby period.

[V2G20-2120] After receiving a PowerDeliveryRes with ResponseCode equal to "OK" as a response to the previous PowerDeliveryReq message with ChargeProgress equal to "Standby", and if EVCC selected the service parameter MobilityNeedsMode set to 2, the EVCC shall process the elements TargetSOC, MinimumSOC, DepartureTime and AckMaxDelay in the AC_ChargeLoopRes messages.

The following requirements apply for the SECC supply equipment. The SECC contactor is closed before sending the PowerDeliverRes message.

[V2G20-858] After receiving a PowerDeliveryReq with parameter ChargeProgress equal to "Start" the SECC shall monitor the Contactor status and start the V2G_SECC_Msg_Performance_Timer.

[V2G20-1317] In case of High-level communication based energy transfer, the SECC shall not close the Contactor before receiving PowerDeliveryReq(ChargeProgress = Start).

[V2G20-860] If no error is detected, the SECC shall close the Contactor no later than 3s after measuring CP State C or D.

[V2G20-861] The SECC shall respond with PowerDeliveryRes containing "ResponseCode = OK" within V2G_SECC_Msg_Performance_Time according to Table 215, if the SECC measures a closed Contactor and [V2G20-858] has applied before.

[V2G20-862] The SECC shall respond with PowerDeliveryRes containing "ResponseCode = FAILED_ContactorError" if V2G_SECC_Msg_Timer is equal or larger than V2G_SECC_Msg_Performance of "PowerDelivery" according to Table 215 and SECC measures opened Contactor.

[V2G20-863] After receiving a PowerDeliveryReq with parameter ChargeProgress equal to "Stop", the SECC shall monitor the Contactor status and start the V2G_SECC_Msg_Performance_Timer.

[V2G20-864] The SECC shall respond with PowerDeliveryRes containing "ResponseCode = OK" within V2G_SECC_Msg_Performance_Time according to Table 215, if the SECC measures an open Contactor and [V2G20-863] has applied before.

NOTE 3 The ISO 15118 series does not require a specific implementation for controlling the contactor in an EVSE. Control of the power contactor depends on the EVSE architecture.

8.5.6.4 DC specific requirements

The following requirements apply for the EVCC.

[V2G20-912] After receiving a ScheduleExchangeRes with EVSEProcessing set to "Finished" and before sending a DC_CableCheckReq message, the EVCC shall change to CP State C or D as defined in IEC 61851-1.

[V2G20-913] After sending a PowerDeliveryReq message with ChargeProgress equal to "Stop" and receiving the corresponding PowerDeliveryRes message, the EVCC shall change to CP State B as defined in IEC 61851-1 before sending the next request message.

[V2G20-914] If [V2G20-912] applies and no error has been identified, the EVCC shall not change the CP State until [V2G20-913] applies.

The following applies for the EVCC during a standby period.

[V2G20-1380] When the EVCC wants to enter a standby period, after having received DC_PreChargeRes with ResponseCode equal to "OK" and if the EVCC determines that the EVSE output voltage, as measured inside the EV, has sufficiently been adjusted to the EV RESS voltage, the EVCC shall send a PowerDeliveryReq with ChargeProgress equal to "Standby, while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Performance_Time.

[V2G20-2119] After receiving a PowerDeliveryRes with ResponseCode equal to "OK" as a response to the previous PowerDeliveryReq message with ChargeProgress equal to "Standby", and if EVCC selected the service parameter MobilityNeedsMode set to 2, the EVCC shall process the elements TargetSOC, MinimumSOC, DepartureTime and AckMaxDelay in the DC_ChargeLoopRes messages.

[V2G20-1381] When the EVCC wants to enter a standby period, after having received DC_ChargeLoopRes, the EVCC shall send a PowerDeliveryReq with parameter ChargeProgress equal to "Standby", while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Performance_Time.

[V2G20-1383] When the EVCC wants to exit a standby period, after having received DC_ChargeLoopRes, the EVCC shall send a PowerDeliveryReq while V2G_EVCC_Sequence_Timer is smaller than V2G_EVCC_Sequence_Performance_Time according to Table 215.

NOTE The SECC handles the DC_PreChargeReq as described in [V2G20-858], [V2G20-861], [V2G20-862].

The following requirements apply for the SECC.

[V2G20-1408] The SECC shall apply a CP duty cycle of 5% from start of data link setup until end of V2G communication session.

The SECC waits for positive authorization and availability of energy before entering the charging loop.

[V2G20-916] After receiving a DC_CableCheckReq message, the SECC shall monitor the CP State and start the V2G_SECC_Msg_Performance_Timer.

[V2G20-917] The SECC shall respond with DC_CableCheckRes containing "ResponseCode = OK" within V2G_SECC_Msg_Performance_Time according to Table 215, if the SECC measures a CP State C or D.

[V2G20-918] The SECC shall respond with DC_CableCheckRes containing "ResponseCode = "FAILED" within V2G_SECC_Msg_Performance_Time according to Table 215, if the SECC measures no CP State C or D.

[V2G20-919] After sending a PowerDeliveryRes message with parameter ResponseCode set to "OK" in response to a PowerDeliveryReq with parameter ChargeProgress set to "Stop", the SECC shall try to measure CP state B.

[V2G20-920] After receiving a DC_WeldingDetectionReq message or a SessionStopReq message, the SECC shall wait for a maximum of V2G_SECC_Msg_Performance_Time to measure CP state B.

[V2G20-921] After receiving a DC_WeldingDetectionReq message or a SessionStopReq message, the SECC shall send the corresponding response message with parameter ResponseCode set to "OK" within V2G_SECC_Msg_Performance_Time according to Table 103, if CP State B was measured.

[V2G20-922] After receiving a DC_WeldingDetectionReq message or a SessionStopReq message, the SECC shall send the corresponding response message with parameter ResponseCode set to "FAILED" within V2G_SECC_Msg_Performance_Time according to Table 103, if CP State B was not measured.

8.5.6.5 AC and DC reverse power transfer specific requirements

8.5.6.5.1 General

In the discharging services, EVCC and SECC shall align to IEC 61851-1 signalling in the same way as in the charging services. Until the discharging service is added in IEC 61851-1, it is defined as an extension within this document. Basic status definitions between charging and discharging are similar, but trigger conditions for charging and discharging loops are defined for reverse power transfer system. It shall be given by HLC control under the condition of IEC 61851-1 signalling agreement.

8.5.6.5.2 Common BPT specific requirements for AC and DC

Most of common requirements defined in 8.5.6.2 shall be applied with following exceptions.

[V2G20-1138] ISO 15118 enabled EVCC shall confirm to IEC 61851-1 in case of charging. Also ISO 15118 enabled EVCC shall similarly work as defined in IEC 61851-1 in case of discharging.

[V2G20-1409] ISO 15118 enabled SECC shall confirm to IEC 61851-1 in case of charging. Also ISO 15118 enabled SECC shall similarly work as the EVSE defined in IEC 61851-1 in case of discharging.

8.5.6.6 ACDP specific requirements

The following requirements apply for the EVCC.

[V2G20-4022] After receiving a ACDP_ConnectRes message with "ResponseCode = OK" and before sending a DC_CableCheckReq message, the EVCC shall change to CP State C or State D as defined in IEC 61851-1.

[V2G20-4023] If [V2G20-4022] applies and no error has been identified, the EV shall not change the CP State until [V2G20-913] applies.

The following requirements apply for the SECC.

[V2G20-4024] The EVSE shall apply a CP duty cycle of 100 % from start of Data Link Setup until end of V2G communication session as defined in IEC 61851-23-1.

NOTE 1 For ACDP noise robustness CP line uses only a DC signal. PWM is not allowed.

NOTE 2 The following DC specific requirements apply: [V2G20-917].

8.5.7 V2G message synchronization with IEC 61980-2 signalling for WPT

8.5.7.1 Overview

In general, the ISO 15118 series is based on the requirements as defined in IEC 61851-1.

[V2G20-5071] An ISO 15118-enabled EV shall conform to IEC 61980-2 and ISO 19363:2020.

[V2G20-5072] During HLC-C, the EV shall apply all energy transfer limits as negotiated by the ISO 15118 series in addition to the limits defined by IEC 61980-2.

[V2G20-5073] ISO 15118-enabled EVSE shall conform to IEC 61980-2.

8.6 Message sequencing and error handling

8.6.1 Overview

In a V2G communication session the EVCC and the SECC exchange request-response message pairs based on a predefined sequence. In this document this is referenced as request-response message sequence. These request-response message sequences allow both sides to synchronize the process in any situation and to control the communication between two V2G entities.

The basic error handling concept is based on application level timers for request-response message sequences. This enables a V2G entity to manage any processing error and communication error on application level after waiting for the expected behavior until a specified timeout. Therefore, a V2G entity can decide on a successful processing after waiting for the positive result until a timeout. In case there is no error in the communication layers below the application layer, the application has the option to terminate the TCP communication in case of application error. A terminated TCP connection before a SessionStopRes is always interpreted as an error by the EVCC or the SECC.

Besides the timeout concept defined in this document, it does not limit additional error detection mechanisms as long as such mechanisms do not lead to incompatibility.

Within a charging session, an EVCC can establish a new V2G communication session after an error by applying the V2G communication state processing as described in 7.4. In this case, the EVCC and the SECC start the communication in the same way as for the first V2G session setup.

In general, the processing time of a request message in an SECC is limited by the performance requirements defined in 8.5.4. If an SECC requires more processing time it may apply the parameter EVSEProcessing (set to Ongoing) to avoid a timeout at the EVCC by explicitly indicating that the processing is not finished yet and the EVCC waits for finishing the processing before proceeding with the message sequence. The EVCC will then repeat the request message periodically until the EVSE signals the finishing of the processing by the parameter EVSEProcessing (set to finished).

8.6.2 Basic definitions for error handling

The basic error handling for a request-response-message pair and a request-response message sequence is based on the ResponseCode included in the response message of the SECC. Depending on the value in the ResponseCode the EVCC decides if it can proceed with the standard request-response message sequence or if it handles an error.

In this document, the ResponseCode as defined in Annex A is interpreted by the EVCC as follows.

– OK:

  Any value starting with "OK" or "OK_" indicates a positive response. Detailed information may be provided by OK_<additional info>. This information may be used to differentiate the reaction on the positive response.

– FAILED:

  Any value starting with "FAILED" or "FAILED_" indicates a negative response. Detailed information may be provided by FAILED_<additional info>. This information may be used to differentiate the reaction on the negative response. This response is considered a fatal error. The SECC and EVCC will terminate the communication session after sending/receiving this response code.

– WARNING:

  Any value starting with "WARNING_" indicates a response that requires an adjustment or reaction by the EVCC, otherwise a subsequent negative response is to be expected in the next message pair. Detailed information is to be provided by WARNING_<additional info>. This information is to be used to differentiate the reaction on the response. This response is not considered a fatal error. The SECC and EVCC will not terminate the communication session after sending/receiving this response. They can continue the communication if proper remedial actions (as defined for that particular response) are taken by the EVCC.

The processing of the remaining parameters in a response message in the EVCC depends on the parameter ResponseCode. If the ResponseCode contains a value starting with OK indicating that no error was detected, the EVCC can process other parameters of the response message mandatorily or optionally as defined in 8.3.5.7.1. If the ResponseCode contains a value starting with FAILED indicating that an error was detected, the EVCC is expected to ignore other parameters of the response message. If the ResponseCode contains a value starting with WARNING indicating that something is not as expected, the EVCC may either process the parameters or ignore them depending on the specifications defined for that particular message.

As long as the message content of the received request message is valid and the request message is accepted in the current SECC state (sequencing), the ResponseCode is set to OK. This is also true for response messages with EVSEProcessing set to Ongoing. With the ResponseCode set to OK the SECC indicates that the request message is accepted and valid, but cannot be finally answered at this moment and more time is needed to send the valid values in the response message. When the valid values are sent in the response message, the SECC indicates this with the parameter EVSEProcessing (set to Finished) and the EVCC can proceed with the sequence as expected for the first request message before receiving the parameter EVSEProcessing (set to Ongoing). If the SECC sets the ResponseCode to FAILED it identified an error in processing or an invalid message and the standard error handling applies.

8.6.3 ResponseCode handling

8.6.3.1 Common requirements

Besides the requirements for request-response message pairs and request-response message sequences as defined in 8.6.4 the EVCC and the SECC shall conform to the following requirements.

[V2G20-734] If the ResponseCode contains a value starting with OK, the EVCC shall process the other parameters of the response message as defined in 8.6.

[V2G20-735] If the ResponseCode contains a value starting with FAILED, the EVCC shall ignore the other parameters of the response message.

[V2G20-736] If the SECC sends a ResponseCode containing a value starting with FAILED all mandatory parameters shall be filled in with arbitrary XSD conform values. Additionally, all values shall be chosen in a way that results in a minimal size of the response message.

[V2G20-2201] If the ResponseCode contains a value starting with WARNING, the EVCC shall take appropriate actions as defined for that particular WARNING response for that particular message.

NOTE 1 Each WARNING ResponseCode can have different reaction requirements for the EVCC. In some cases the EVCC can be required to ignore the other parameters of the response message and take a pre-defined action while in other cases the EVCC can be required to take an action based on the parameters received in the message but act in a different way than when it receives OK ResponseCode for the same message.

[V2G20-2202] If the SECC sends a ResponseCode containing a value starting with WARNING all mandatory parameters shall be filled as defined for that particular WARNING response for that particular message.

NOTE 2 Each WARNING ResponseCode can have a different requirements for the parameters. In some cases the SECC can be required to send default/arbitrary XSD conform values while in others the SECC can be required to send some specific values so that the EVCC can take an action based on the parameters received in the message.

In general each response message can contain two types of ResponseCode values "OK" or "FAILED". In addition some response messages can further contain the ResponseCode value "WARNING_" (see Table 224).

[V2G20-457] A response message shall contain the ResponseCode "OK" in the "ResponseCode" attribute if the processing of the request message was successful. If later on a specific positive "ResponseCode" is defined for a dedicated situation, this ResponseCode shall be used.

[V2G20-458] A response message shall contain the ResponseCode "FAILED" in the "ResponseCode" attribute if the processing of the request message was not successful and no specific "ResponseCodeType" is defined for the concrete error case.

[V2G20-459] The response message shall contain the ResponseCode "FAILED_SequenceError" if the SECC has received an unexpected request message.

[V2G20-460] The response message shall contain the ResponseCode "FAILED_UnknownSession" if the SessionID in any request message except SessionSetupReq is not equal to the SessionID value stored for the currently active V2G communication session.

[V2G20-461] The response message shall contain the ResponseCode "FAILED_SignatureError" if the validation of the Security element in the message header failed.

[V2G20-462] The message "SessionSetupRes" shall contain the specific ResponseCode "OK_NewSessionEstablished" if processing of the SessionSetupReq message was successful and a different SessionID is contained in the response message than the SessionID in the request message.

[V2G20-463] The message "SessionSetupRes" shall contain the specific ResponseCode "OK_OldSessionJoined" if processing of the SessionSetupReq message was successful and the same SessionID as used in the request message is contained in the response message.

[V2G20-464] The message "ServiceDetailRes" shall contain the ResponseCode "FAILED_ServiceIDInvalid" if the ServiceID contained in the ServiceDetailReq message was not part of the offered EnergyTransferServiceList or VASList during ServiceDiscovery.

[V2G20-467] The message "ServiceSelectionRes" shall contain the ResponseCode "FAILED_ServiceSelectionInvalid" if the SelectedServiceList contained in the ServiceSelectionReq message contains a ServiceID which was not contained in the offered EnergyTransferServiceList or VASList of ServiceDiscoveryRes.

[V2G20-1618] The message "ServiceSelectionRes" shall contain the ResponseCode "FAILED_NoEnergyTransferServiceSelected" if the SelectedServiceList contained in the ServiceSelectionReq message does not contain a ServiceID value equal to one which was offered by the SECC with the parameter ChargeService or BPT service in the ServiceDiscoveryRes.

[V2G20-2203] The message "CertificateInstallationRes" shall contain the ResponseCode "WARNING_CertificateExpired" if the SECC (or SA) deduces that one (or more) of the certificates in the OEMProvisioningCertificate chain it received in the CertificateInstallationReq message has expired. This includes any cross certificates (if applicable).

[V2G20-2204] The message CertificateInstallationRes shall contain the ResponseCode "WARNING_CertificateNotYetValid" if the SECC (or SA) deduces that one (or more) of the certificates in the OEMProvisioningCertificate chain it received in the CertificateInstallationReq message is not yet valid. This includes any cross certificates (if applicable).

[V2G20-1412] The message "CertificateInstallationRes" shall contain the ResponseCode "WARNING_CertificateRevoked" if the SECC (or SA) deduces (using OCSP responses or CRL) that one (or more) of the certificates in the OEMProvisioningCertificate chain it received in CertificateInstallationReq message is revoked. This includes any cross certificates (if applicable).

[V2G20-2205] The message "CertificateInstallationRes" shall contain the ResponseCode "WARNING_CertificateValidationError" if certificate validation of one (or more) of the certificates in the OEMProvisioningCertificate chain it received in the CertificateInstallationReq message fails due to any issue other than those identified by [V2G20-2203], [V2G20-1412] and [V2G20-2204]. This includes any cross certificates (if applicable).

NOTE 3 ResponseCode "WARNING_CertificateValidationError" for message "CertificateInstallationRes" is a catch-all response code for cases that are not covered by the specific certificate validation error response codes defined in other requirements for "CertificateInstallationRes" message. This includes cases where certificate validation fails due to a mismatch in the curves in the OEM provisioning certificate received in "CertificateInstallationReq" message and the curves the SECC/SA is allowed to process per the configuration parameter as defined by [V2G20-2320]. For example, in case the SECC/SA is unable to validate the OEM provisioning certificate because the SECC/SA cannot process the curves in the OEM provisioning certificate (due to changes to configuration parameter as defined by [V2G20-2320]), the SECC will send the ResponseCode "WARNING_CertificateValidationError".

[V2G20-2206] The message "CertificateInstallationRes" shall contain the ResponseCode "WARNING_NoContractMatchingPCIDFound" if the SA(s) indicate to the SECC that after a complete search, no contract could be found for the PCID contained in the OEM provisioning certificate received in the OEMProvisioningCertificate chain contained in the CertificateInstallationReq message.

[V2G20-2207] The message "CertificateInstallationRes" shall contain the ResponseCode "WARNING_NoCertificateAvailable" if a new contract certificate cannot be retrieved from secondary actor within V2G_SECC_Msg_Performance_Time according to Table 215.

NOTE 4 This response code indicates a timeout. The timeout could be due to many reasons including communication errors, taking longer than the timeout allows to retrieve the contract certificate from the secondary actor, etc.

[V2G20-2208] The message CertificateInstallationRes shall contain the ResponseCode "WARNING_eMSPUnknown" if the SECC does not recognize any one (or more) of the eMSPs indicated by the EMAID included in the PrioritizedEMAIDs list it received in CertificateInstallationReq message.

[V2G20-2209] The message "AuthorizationRes" shall contain the ResponseCode "WARNING_AuthorizationSelectionInvalid" if the SelectedAuthorizationService contained in the AuthorizationReq message was not part of the offered AuthoritzationServices of AuthorizationSetupRes message.

[V2G20-2210] The message AuthorizationRes shall contain the ResponseCode "WARNING_ContractCancelled" if the SECC received the information (potentially from SA; it is not in the scope of this document to define how the SECC received this information) that the contract associated with the ContractCertificate it received in AuthorizationReq message, in attribute ContractSignatureCertChain, has been cancelled.

NOTE 5 This response code is related to the actual customer contract between the SA and the customer. It is not related to the contract certificate itself. For example, the contract certificate can still be valid while the customer possibly cancelled the actual contract with the eMSP. In that case, at the SA's discretion, this response code is expected to be utilized.

[V2G20-2211] The message AuthorizationRes shall contain the ResponseCode "WARNING_eMSPUnknown" if the SECC does not recognize the eMSP indicated by the EMAID included in the ContractCertificate it received in AuthorizationReq message, in attribute ContractSignatureCertChain.

[V2G20-2212] The message "AuthorizationRes" shall contain the ResponseCode "WARNING_CertificateExpired" if the SECC (or SA) deduces that one (or more) of the certificates in the ContractCertificate chain it received in the AuthorizationReq message, in attribute ContractSignatureCertChain, is expired. This includes any cross certificates (if applicable).

[V2G20-2213] The message "AuthorizationRes" shall contain the ResponseCode "WARNING_CertificateNotYetValid" if the SECC (or SA) deduces that one (or more) of the certificates in the ContractCertificate chain it received in the AuthorizationReq